AIS AUTOMATION
SOFTWARE SYSTEMS S&T GROUP

User Documentation

ToolCommander® – Manual

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Kontron AIS GmbH

Contents

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Introduction

1 Introduction

This document describes the operation of a machine control that is based on ToolCommander, a control application framework for machine control. The information contained in this manual explains the operation of all framework functionality, parts of a ToolCommander based application might use slightly different logic or graphical representation. For details please refer to the manual that comes with the specific application.

1.1 Abbreviations

Abbreviation	Meaning
AMHS	Automated Material Handling System
CJ	Control Job
EC	Equipment Constant
FA	Factory Automation (Host Interface)
GUI	Graphical User Interface
I/O	Input/ Output
LL	Load Lock
LP	Loadport
MSDE	Microsoft Data Engine
PJ	Process Job
PM	Process Module
PMC	Process Module Controller
PPM	Process Program Management
RaP	Recipe and Parameters (SEMI E139)
SECS	SEMI Equipment Communications Standard
SEMI®	Semiconductor Equipment and Materials International
SQL	Structured Query Language

TC	ToolCommander
TM	Transport Module
TMC	Transport Module Controller
XML	Extensible Markup Language

Table 1 - Abbreviations

2

General Application Operation

2 General Application Operation

The ToolCommander usually consists of two applications: a visualization and a control executable. The tasks of the software are structured in the following way:

Visualization	Control
• machine state visualization	• connection to the process and transfer modules
• device parameter management	• control interface to all components
• flow and module recipe management	• recipe execution
• protocol, trend, message display	• protocol, trend, message data storage

Table 2 - Software Task Structure

2.1 User Interface Elements

This chapter gives a general overview of the operation of the application. Operation of the system by keyboard and mouse is in a manner known from Microsoft Windows®. The user interface employs input elements (number and text input fields, tables etc.) according to the Windows® standard. For further information see the Windows® manual.

2.1.1 Buttons

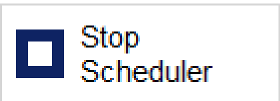


Figure 1 - Button Stop Scheduler

Buttons are controls that work like a push button and execute some program function when being pressed.

Examples

Press the button *Cancel* to cancel the actual dialog.

Press the button *Save* to save the actual configuration.

Press the button *Print* to print the actual dialog.

Some buttons display their selection status using a blue salience. If the button function is active the salience is displayed, otherwise it is hidden.

2.1.2 Input and Output Frames

Figure 2 - Input Frame

A frame is a rectangular window used to input or output text or figures (e.g. when specifying a setpoint of a recipe). To change the input value the frame needs to be selected by clicking on it.

2.1.3 Lists

Type	Name	Operation Mode	Occupation	Reserved Material
Module Slot	Handling Slot 1	Automatic	Empty	-none-
Module Slot	Handling Slot 2	Automatic	Empty	-none-

Figure 3 - Listbox

Lists are used to visualize and modify structured data. Information is displayed in form of a multi column list (table). List box entries can be selected by clicking on the according line.

2.1.4 Combo-Boxes

Combo-boxes are used to select a value from several options provided. By clicking on the arrow all possible values are shown. By clicking on a list entry you can activate a value. Now it appears in the upper part of the combo-box which is always visible. That means it is selected. The lower part is automatically closed.

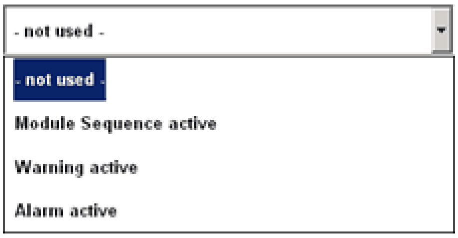


Figure 4 - Combo-box

2.1.5 Check-Boxes

A check-box works like an on/off switch. A checked box ☑ is activated whereas a non-checked box ☐ is deactivated. It can be operated by clicking the box or the assigned explanation text.

2.1.6 Tab Control

Tab control elements are used to switch the display between several user dialogs.

Figure 5 - Tab control

Selection of a particular dialog is done by clicking the assigned tab.

2.1.7 Treeview

Treeview controls show hierarchical relationships between arbitrary elements and allow selection of one particular of these elements.

Pressing ⊞ symbol beside a group name opens that group. Now the elements of the group are displayed. By pressing ⊟ you can close the group.

Tree view entries can be selected by clicking on them.

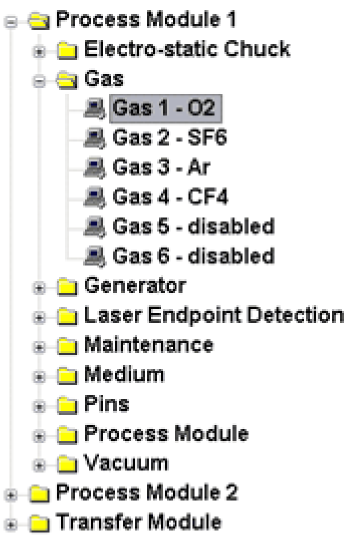


Figure 6 - Tree View

2.2 Touch Screen Support

If the touch screen support is activated one of the following windows will be opened after an edit-frame is clicked. The type of window opened depends on the data type of the edit-frame clicked (text, number etc.)

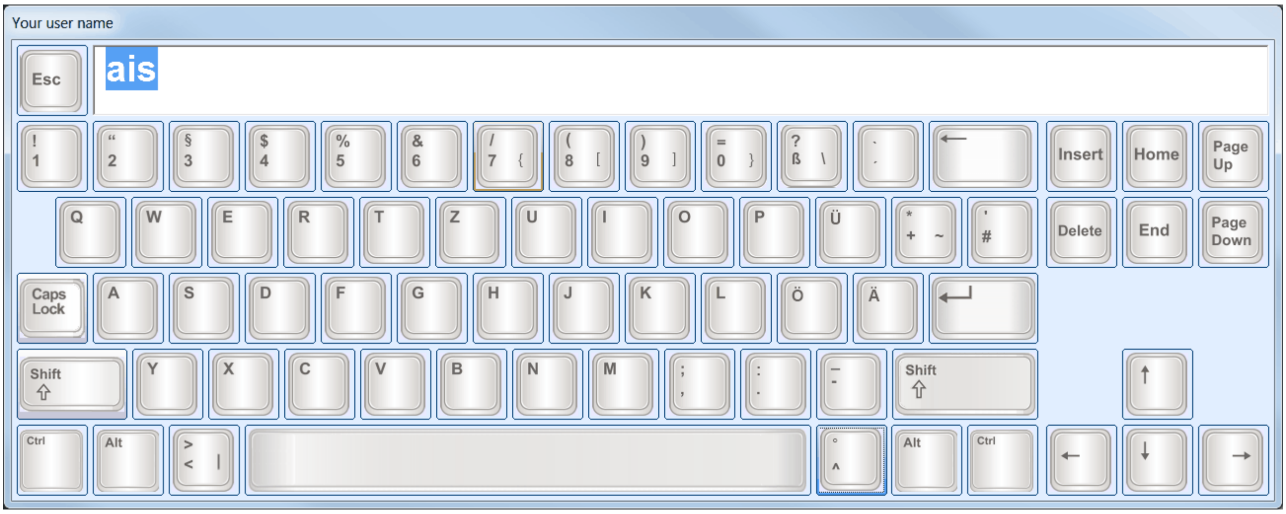


Figure 7 - Touch Screen Text Input Tool

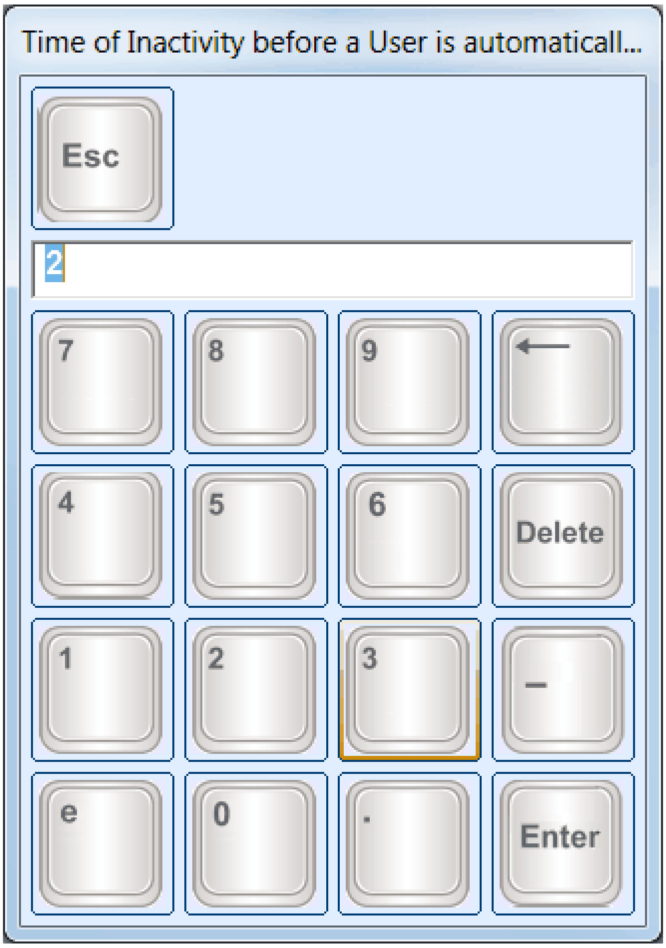


Figure 8 - Touch Screen Number Input Tool

The input elements have the following functionality:

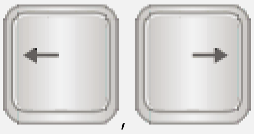
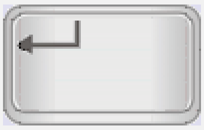
Button	Function
	enter the according character
	activate / deactivate capital letters (shift) for the next action
	activate/ deactivate capital letters (caps lock)
	erase characters left (backspace) resp. right (Delete) of the cursor
	move the cursor
	cancel the input
	accept the input
	activate special characters

Table 3 - Touch Screen Buttons

2.3 Screen Layout

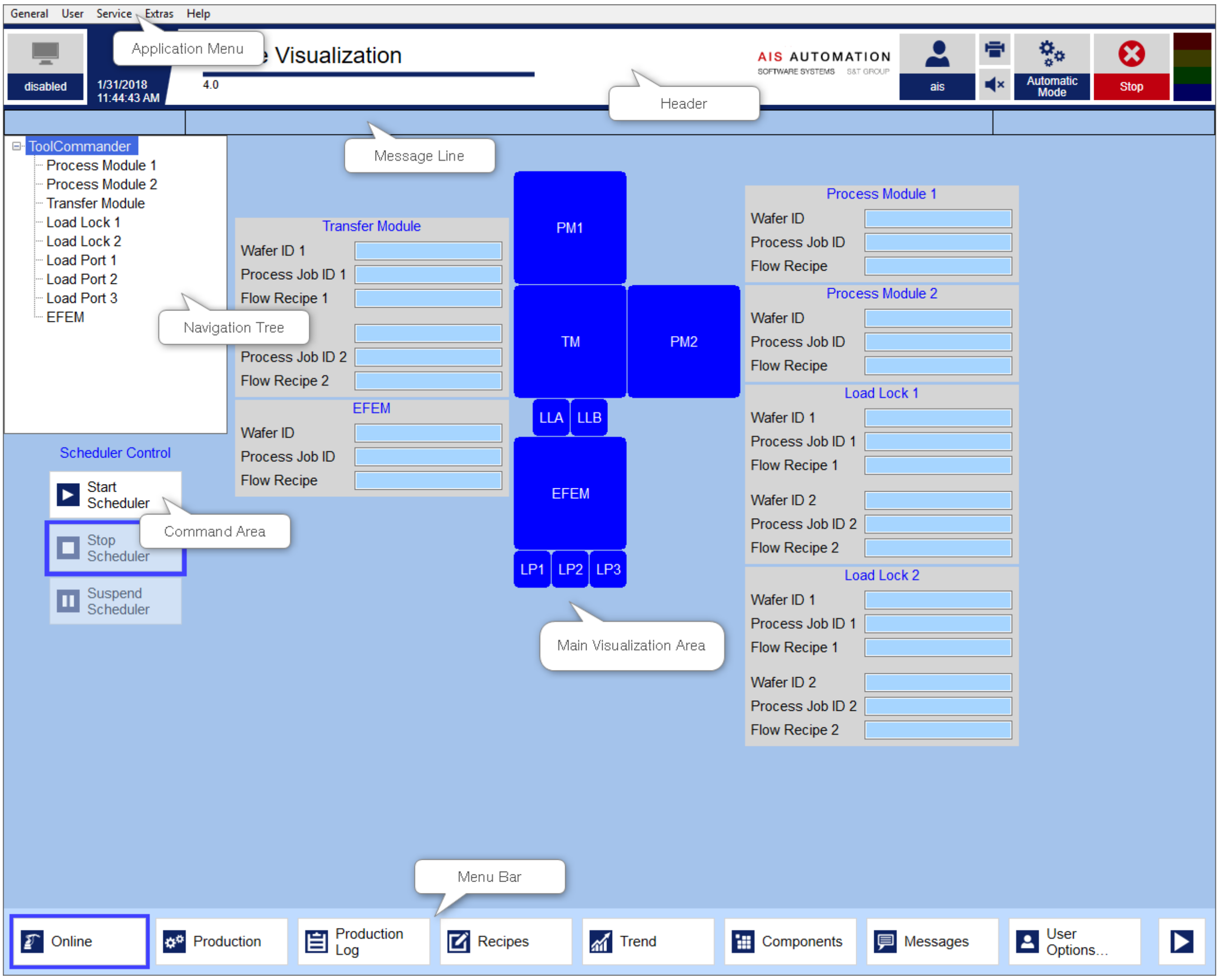


Figure 9 - Screen Layout

2.4 Header

Figure 10 - Header

Host connection state

The header shows the current communication state as well as the current control state on the left header button.

The color of the header button indicates the **communication state**. The following communication states are available:

- **Connected**: The equipment is connected with the host interface. The monitor icon in the button is shown in green color.
- **Connecting**: The equipment is attempting to connect to the host interface. The monitor icon in the button is shown in yellow color.
- **Connection error**: An error occurs while the equipment is connecting to the host interface. The monitor icon in the button is shown in red color.
- **Disabled**: In this case, the connection is disabled or not installed. The control state is not defined. The monitor icon in the button is shown in gray color.

The label of the header button indicates the current **control state** of the ToolCommander. The following control states are available according to SEMI E30.

- **Online Remote**: The hosts controls the equipment.
- **Online Local**: The host has full access to information of the equipment, but the control is limited.
- **Host Offline**: The equipment allows no host control. The operator intends to go online.
- **Equipment Offline**: The equipment allows no host control. The system awaits operation to attempt to go online.
- **Attempt Online**: The equipment allows no host control. Equipment has responded to an operator instruction to attempt to go online.

Please consider the label **Disabled** is used if the communication state is Disabled.

Switch to automatic mode

If you confirm this action in the dialog which is opened after this button was pressed then all modules, components and I/O channels are set to automatic mode.

Technological stop

The stop button can be invisible depending on the configuration of the application.

NOTICE
Pressing the Stop button causes a stop of logical processes inside the control system to cancel processing of substrates. For emergency situations, use the hardware EMO button to completely stop the whole plant.

User

Click the header button **User** to open the log on window.
Enter user name and password and press the button **Log On**.
Press the button **Log Off** to log off the current logged on user.

Only one user can be logged in at once. When using the function Log on while still a user is logged in, the user will be logged off automatically.
The name of the current user is shown in the label of header button.

Print screenshot

Press the button **Print screenshot** in the header to print a screenshot of the main screen of the application. The screen shot is printed on the default printer set in the Windows operating system. Open dialogs and messages will not be printed. Dependent on the configuration of the application the function can be deactivated. In that case, the button is not visible.

Light Tower

The light tower works corresponding to the hardware light tower. It shows the current machine states. The logic for the activation of the single lights and buzzers depends on the concrete logic of the ToolCommander based project/ machine control.

2.5 Message Line

Part of the ToolCommander is a message system. The last message is shown in the message line. It can be used as a quick information source in special system conditions.

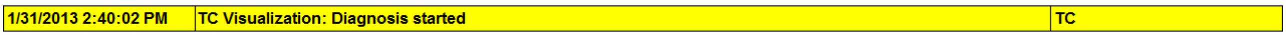


Figure 11 - Message Line

2.6 Main Visualization Area

In this area the screens selected by menu operating are shown. All the module overviews, input masks, diagnosis tools etc. are displayed in this area. It is used for multiple purposes as working and display area.

2.7 Command Area

This section contains the different command buttons, depending on the selected function and module.

2.8 Menu Bar

This area includes buttons which can be used to switch the main area and the display in the navigation area (changeover between functional modules). This menu is always visible and can be operated at any time. The currently selected function is indicated by a blue salience around the assigned button.

The menu consists of two parts which can be switched with the buttons ▶ and ◀.

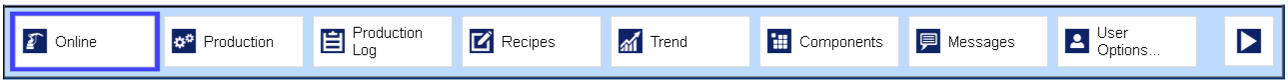


Figure 12 - Menu Bar Left

Figure 13 - Menu Bar Right

Online: The online visualization shows the state of all units of the machine or module (depending on the selection in the navigation tree).

Production: Provides access to the ToolCommander Production Editor where jobs and material can be administrated, and the machine status is displayed.

Production Log: Provides access to the job and wafer history.

Recipe: This selection contains all function to create, edit and delete flow and module recipes.

Trend: Access to the trending module of the selected chamber/ module.

Components: The component overview.

Messages: The message system provides a visualization that shows a list of all active messages and another list of all archived messages. Messages provide different recovery options that can be viewed and executed at the messages screen.
If the messages function is not currently selected and there are active or not yet acknowledged alarms active in the system the button **Messages** will have a red salience. If warning messages (and not alarms) are active that salience will be yellow.

User options: The user option dialog allows user logon and logoff as well as user administration.

I/O Diagnosis: Using the I/O diagnosis, selected binary and analog I/O channels can be switched manually. Use this menu entry to get access to this function.

Language: This menu item enables selecting different languages.

Settings: This item provides access to the adjustment of common application settings.

Information: Use this menu entry to display information about program version etc.

Logbook: The Log Book allows writing and reading some notes.

Help: Opens the online help. Depending on the settings this button might be hidden.

System Diagnosis: The system diagnosis is a service tool that allows viewing and modifying internal software states. Access to it must be restricted to experienced and trained users. Safety interlocks are not active when modifying internal software states using the diagnosis.

Exit Program: Exits the program.

2.9 Application Menu

The Menu offers a central access to following functionality. Depending on the settings this menu might be hidden.

2.9.1 Menu General

Function	Description
Online Visualization	Corresponds with the menu button *Online*.
Production	Corresponds with the menu button *Production* (see chapter Production).
Production Log	Corresponds with the menu button *Production Log* (see chapter Production Log).
Recipes	Corresponds with the menu button *Recipes* (see chapter Flow Recipes).
Trend	Corresponds with the menu button *Trend* (see chapter Trend).
Components	Corresponds with the menu button *Components* (see chapter Components).
Messages	opens the following sub-menu (see chapter Message System):
Alarms	Selects the menu button *Messages* and the sub-dialog *Alarms* of the message view (see chapter Alarms View).
Archive	Selects the menu button *Messages* and the sub-dialog *Archive* of the message view (see chapter Message Archive View).
Alarm List Popup...	Opens a popup window which displays all currently active alarms (see chapter Alarm List Popup).

Exit	Corresponds with the menu button *Exit Program* (see chapter Exit Program).

Table 4 - Application Menu General

2.9.2 Menu User

Function	Description
Log On...	Opens the log on window where the user may enter his/ her user name and password (see chapter *Log On/ Log Off*).
Log Off	Logs the current user off. This item is disabled if there is no user logged on (see chapter *Log On/ Log Off*).
Password...	Is used for changing the password of the user currently logged on. This item is disabled if there is no user logged on (see chapter *Change Password*).
Administration...	Opens the user administration dialog. This item is disabled if the current user does not have the user administration right (see chapter *User Manager*).

Table 5 - Application Menu User

2.9.3 Menu Service

Function	Description
Language...	Corresponds with the menu button *Language* (see chapter Language Selection).
Application Settings	Corresponds with the menu button *Settings* (see chapter Application Settings).
I/O Diagnosis	Corresponds with the menu button *I/O Diagnosis* (see chapter I/O Diagnosis).
Minimize Window	Minimizes the application window. Depending on the settings this item might be disabled if the user does not have the right *Windows Access*.
VAC Diagnosis...	Opens the VAC Diagnosis. If the user does not have the according right the menu item is disabled (see chapter VAC Diagnosis).

Table 6 - Application Menu Service

2.9.4 Menu Extras

Function	Description
Log Book	Corresponds with the menu button *Log Book*.
Change Date/ Time...	Opens the Windows dialog for setting the date and time of the computer. Depending on the settings this item might be disabled if the user does not have the right *Windows Access*.
Set Auto Mode	If the user confirms this action in the dialog which is opened after this item was selected then all modules, components and I/O channels are set to automatic mode.

Table 7 - Application Menu Extras

2.9.5 Menu Help

Function	Description
Program Information...	Corresponds with the menu button *Information*.
Documentation ...	Opens the operation manual. Depending on the application settings this menu item might be disabled.

Table 8 - Application Menu Help

2.10 Navigation Tree

The system consists of different modules. Several program functions are provided for single modules, e.g.

- Online Visualization
- Recipe Editors
- I/O Diagnosis
- Online and Offline Trend.

In the navigation tree you can select the module to be effected by one of these functions.

Additionally to the modules you can select the root item to apply actions to the entire machine.

2.11 Visualization of Operating and Process States

Component Animation

Components within the visualization will be displayed according to their state. These states are listed in the table below.

Aggregate	Gray	Yellow	Green	Red
Devices that can be switched on/ off	off	in transition	on	fault
Devices that may open/ close	closed	in transition	open	fault

Table 9 - State Colors

Substrate Processing Status

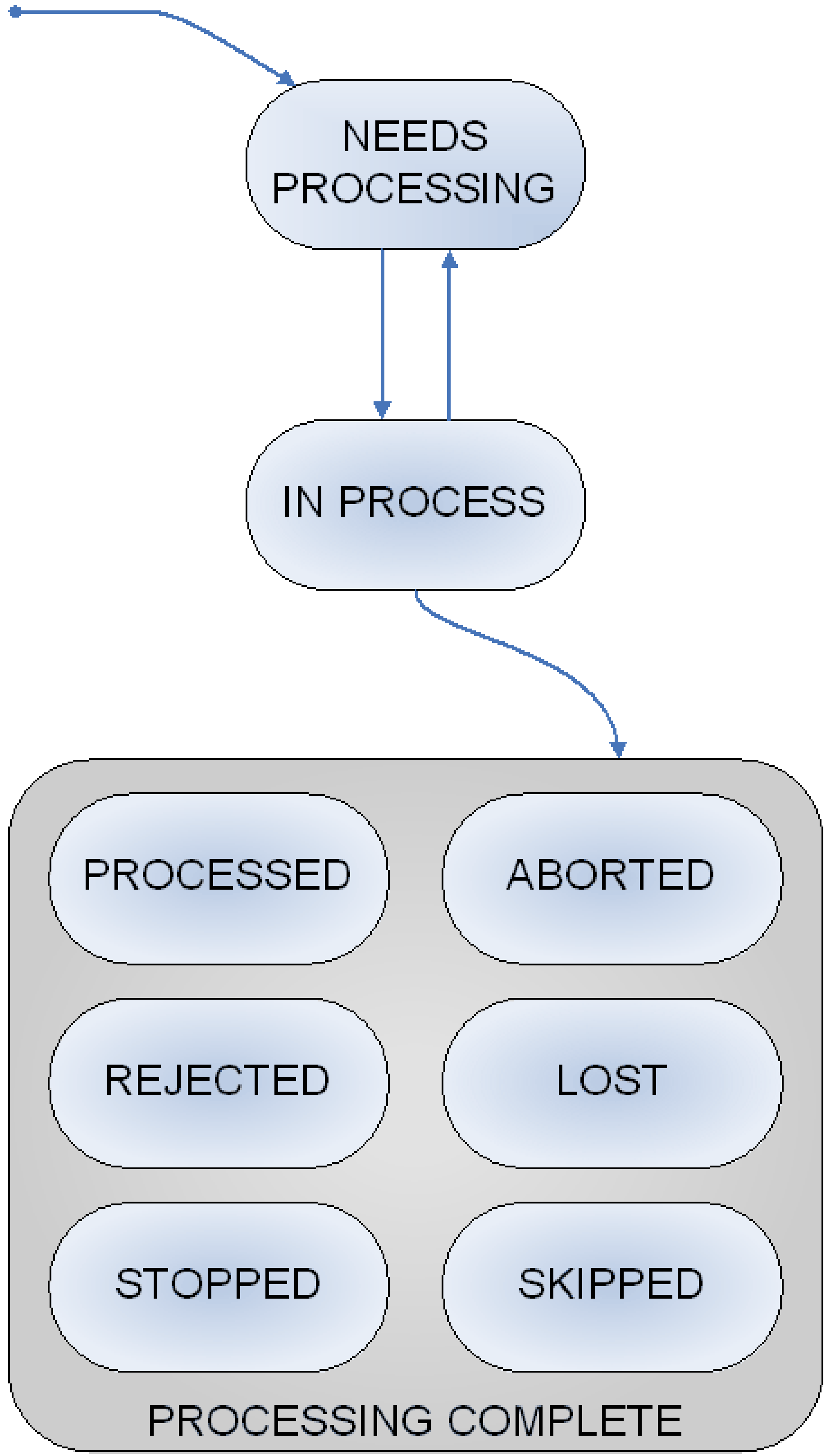


Figure 14 - Substrate Processing Status

The process state for a substrate will be displayed as a symbol at current location. Every process state is related to a unique color and pattern listed in the table below.

Status	Color, Salience, Pattern
Needs processing	light pink striped
In process	white filled
Processed	dark green filled
Rejected	orange filled
Stopped	light blue filled
Aborted	red filled
Lost	red/yellow striped
Skipped	blue filled

Table 10 - Substrate Process Colors

2.12 Screens Not Available

If the user selects a screen (menu and navigation combination) that is not available (e.g. recipe editor for a Loadport) the following screen will be displayed:

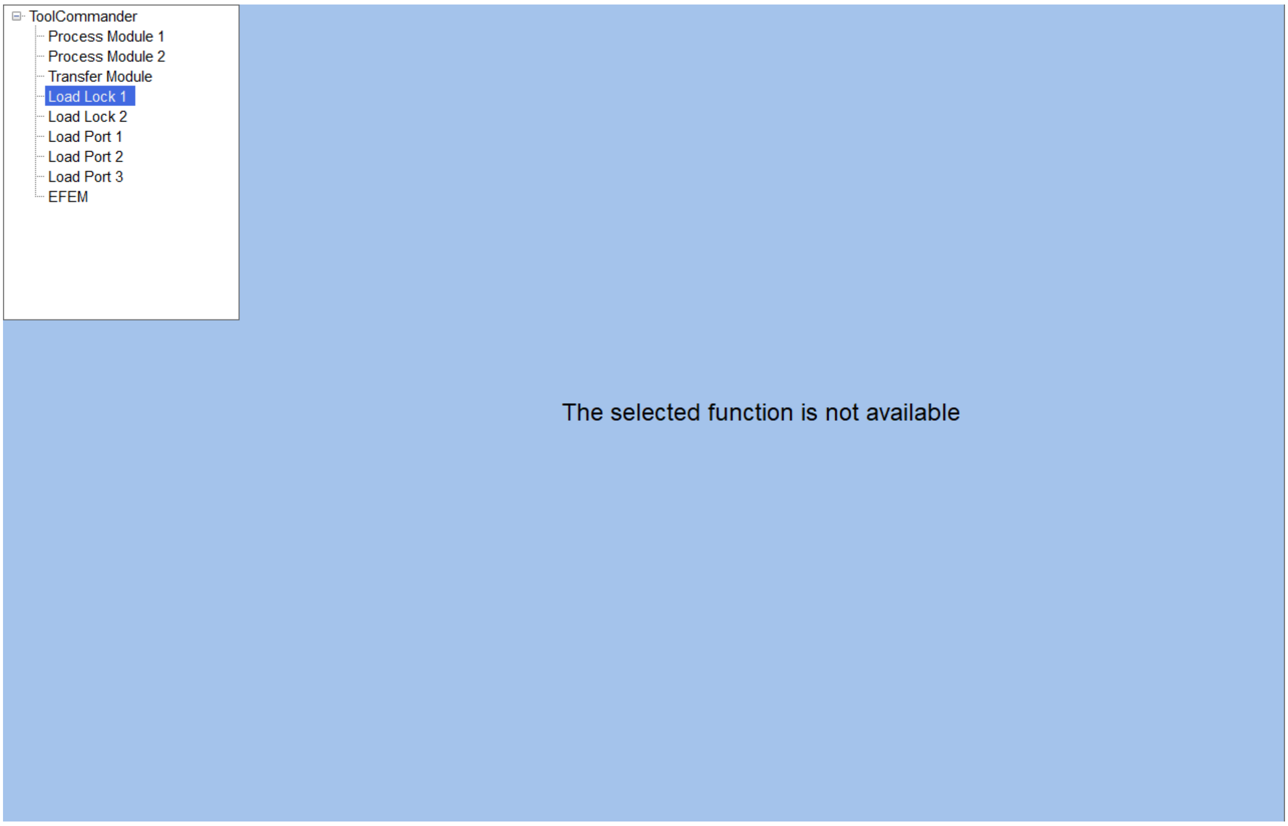


Figure 15 - Not Available Function

2.13 Insufficient User Rights

When the user tries to access a restricted screen (e.g. recipe editor) but does not have the sufficient user right to see its content the following screen will be displayed:

Figure 16 - Insufficient User Rights

3

Production

3 Production

The ToolCommander Production screen allows users to explore and configure the ToolCommander. In addition to this, the dialog can be used for initial operation assistance and system diagnostics.

3.1 Overview

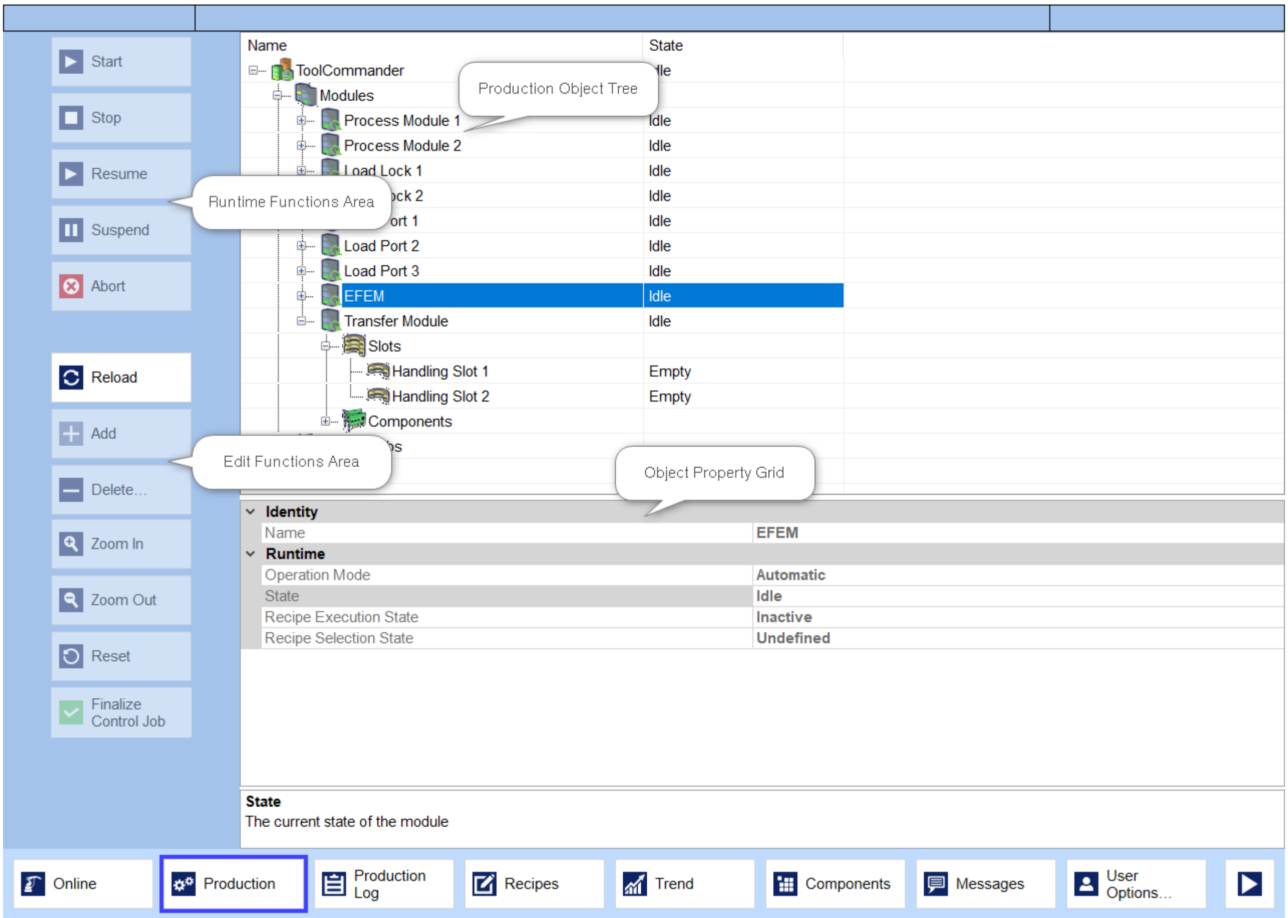


Figure 17 - Production Screen

3.1.1 Production Object Tree

The production object tree allows users to explore the actual state of the ToolCommander. Depending on the selected item in the tree, different functions are available in the edit function area and the runtime function area on the left side of the screen. Additionally, properties of the selected item are displayed in the object property grid.

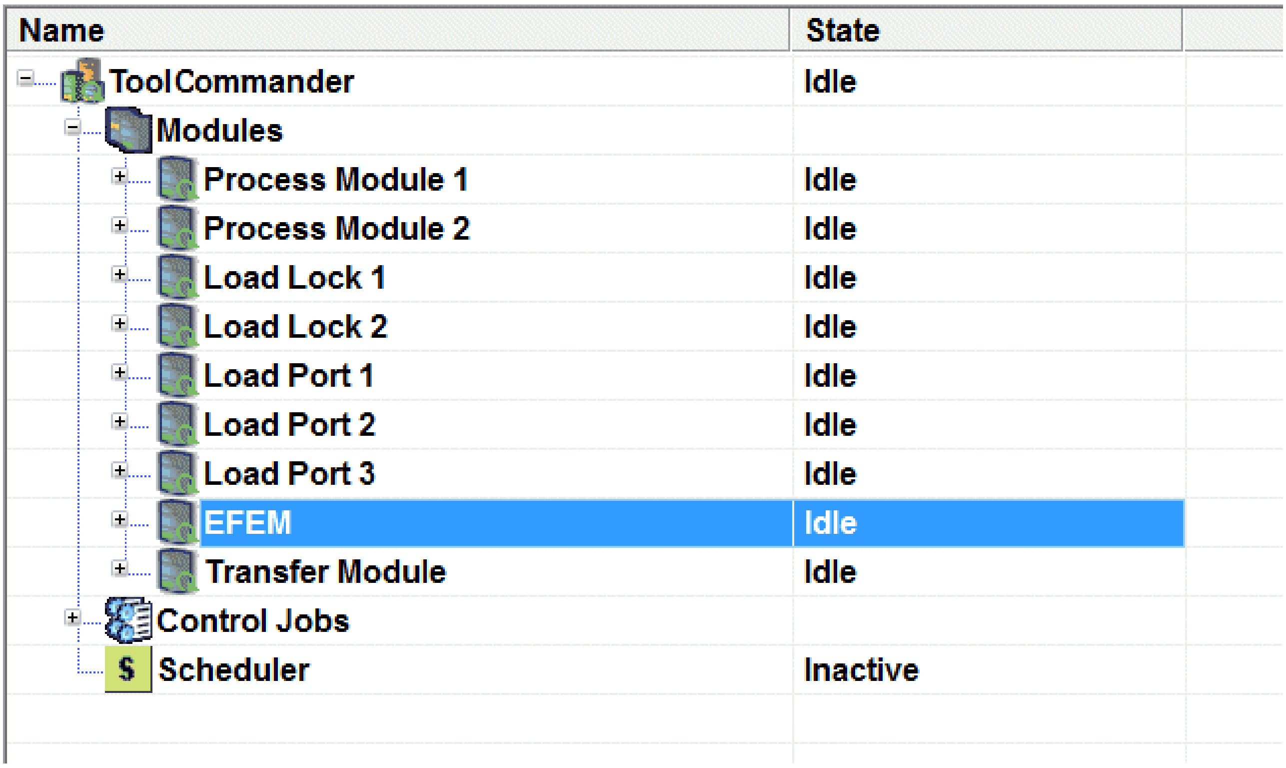


Figure 18 - Production Object Tree

For showing the sub-elements of a tree node, click or press the key Cursor Right (→). For hiding the sub-elements click or press Cursor Left (←).

3.1.2 Object Property Grid

The object property grid allows users to view and edit the properties of each single item in the production object tree. Properties displayed in gray color cannot be changed.

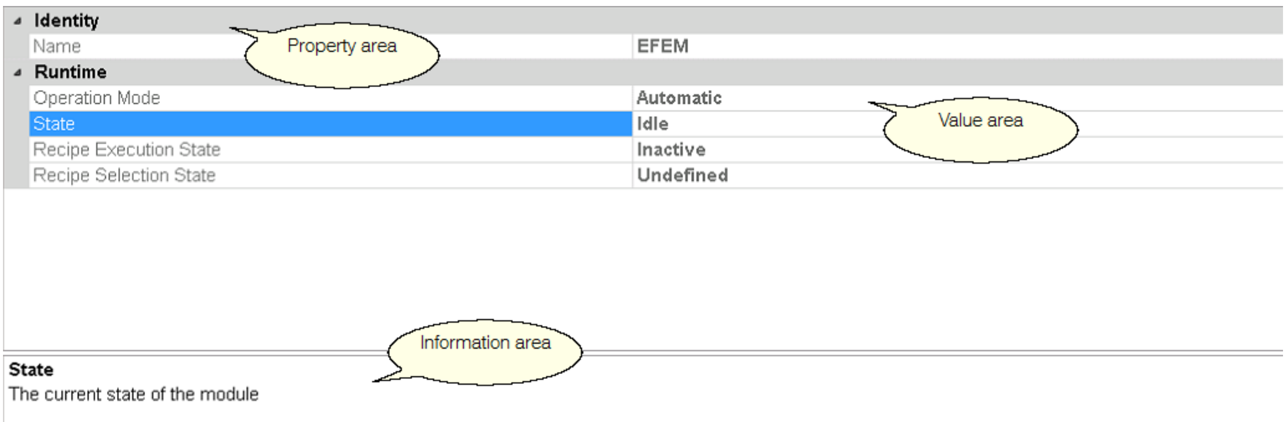


Figure 19 - Object Property Grid with Data of Selected Control Job

The object property grid is divided into three sections:

- Property area
- Value area
- Information area

The input options of the changeable properties differ. Properties like the name of an object can be entered via keyboard. The second input option is a drop-down list in which the user can select an item.

Figure 20 - Free text input for the name of the Control Job

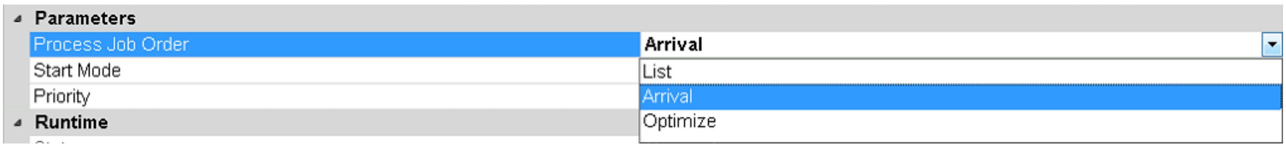


Figure 21 - Combo-box for Selecting the Process Job Order of a Control Job

For a better view, some properties of an element are grouped. The group title depends on the kind of the selected object in the tree.

3.1.3 Runtime Function Area

The runtime function area enables users to start and stop production. Depending on the selection in the object tree, users can start, stop, suspend, resume, and abort the scheduler, control and process jobs.

Scheduler and ToolCommander are always started or stopped at the same time. When one of these is selected the buttons Start and Stop operate all three instances.

Depending on the selection in the object tree, the buttons at the runtime function area are enabled or disabled. For example, a slot cannot be started or stopped.

3.1.4 Edit Function Area

The edit function area provides functionality for manipulations in the production object tree.

Press the button **Reload** to refresh the production object tree.

The second button Add provides functionality for adding material to slots, creating new control jobs or assign new process jobs to existing control jobs (for details about material, slots and jobs see the following chapters). The name of the button depends on the selection in the production object tree.

Press the button **Add** to add a new object to the tree objects currently selected. The name of the button is changed depending on the selected object. You can add other objects to the following tree objects. The button is deactivated if you have selected one of the other tree objects.

- **Slot List (Slots)**: Press the button Add Material(s) to add one material to each slot of the list. By default, the material is name with the name of the module followed by a continuous number.
- **Slot**: Press the button Add Material to add a material to the selected slot. By default, the material name is set to the name of the module followed by a continuous number.
- **Control Job List (Control jobs)**: Press the button **Add Control Job** to add a control job to the control job list. By default, the new control job is named with "CJ" followed by a continuous number.
- **Control Job**: Press the button **Add Process Job** to add a process job to the process job list. By default, the new process job is named with "PJ" followed by a continuous number.
- **Process Job List (Process Jobs)**: Press the button **Add Process Job** to add a process job to the process job list. By default, the new process job is named with "PJ" followed by a continuous number.

Press the button **Delete**..., to remove the material, the control job or the process job currently selected. The button is deactivated if you have selected another object in the tree or the selected object is currently processing. You have to confirm deleting the object to avoid deleting by accident. Press the button **Yes** to confirm otherwise press **No**.

By default, you see the entire object tree with all modules, control jobs and the scheduler. Limit the displayed objects to any branch of the object tree by the zooming function.
Select the required node of the tree and press the button **Zoom in**. The button is deactivated if you have selected the node of the presented tree.
Press the button **Zoom out** to go back to the previous view. The button is deactivated if the entire tree is visible.
A special zoom editor is available for Carriers. All slots and properties of slots of the carrier are presented in a table. Select multiple slots to assign equal properties (i.e. process job) at the same time.

Press the button **Reset** to reset the state of a material to state **Needs Processing**. The button is only activated if you have selected a material in the object tree and the current state is not **Needs Processing**.

Press the **Finalize Control Job** to finalize the settings for the selected control job. The control cannot be started before.

3.2 Equipment

The root of the production tree symbolizes the entire equipment. The following table provides an overview of the properties of that node:

Property name	Description	Editable

Name	The name of the equipment.	No
Active Hours	The duration in hours the equipment spent in production.	No
Finished Material Count	Number of materials processed.	No
Virtual Material Count	Number of material logged on the equipment, but are not physically there. The material has to be fetched to the equipment before processing.	No
Production State	Defines if the equipment is stopped currently or is processing.	No
Flow Recipe Count	The number of flow recipes.	No
Control Job Count	The number of control jobs.	No
Material Count	The number of materials currently present in the machine.	No
Virtual Carrier Count	Carrier which are not at the equipment but announced.	No
Material Throughput	The average material throughput per hour.	No

Table 11 - Equipment Properties

The ToolCommander element in the tree contains the following sub-elements:

- Modules
- Control Jobs
- Scheduler

3.3 Modules

The Modules node contains a list of all physical modules of the tool. Modules can be of different types and have sub nodes. After selecting a module its properties are displayed in the object property grid.

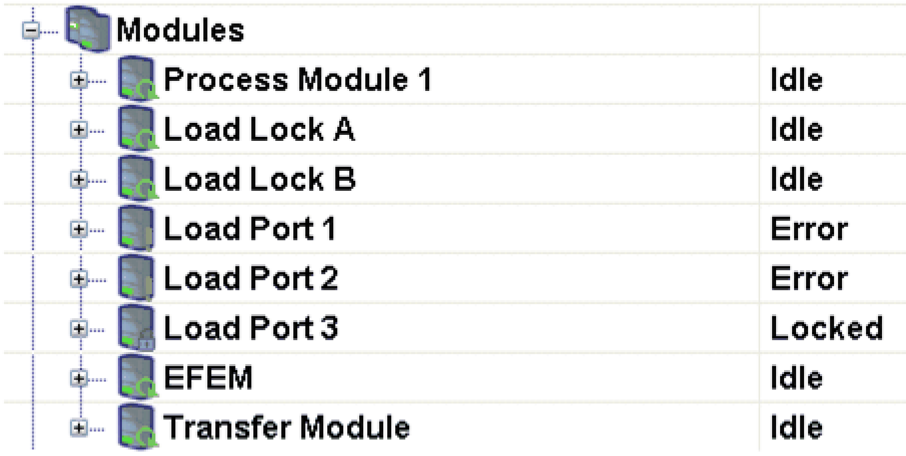
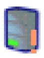


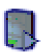
Figure 22 - Module Node in the Production Object Tree

In the production object tree modules can have different symbols. The symbol of a module depends on its current state:

 Idle

 Error

 Locked

 Active

Modules have the following properties:

Property name	Description	Editable
Name	The unique name of the module.	No
Operation Mode	The actual module operation mode (see table Operation Mode).	No
State	The actual module state; possible values are listed in table Module State.	No
Recipe Execution State	The actual recipe execution state of the module (see table Recipe Execution State).	No
Recipe Selection State	The actual recipe selection state of the module (see table Recipe Selection State).	No

Table 12 - Module Properties

The following tables show the values that can be assigned for enumerated properties:

Value	Description
Disabled	The module is disabled. No transports from or to this module are allowed.
Automatic	The module is in automatic mode, the scheduler will use it.
Manual	The Load port is in manual mode, the scheduler will ignore it. Manual transports are allowed.
Not Installed	The module is not installed, the scheduler will ignore it.

Table 13 - Operation Mode

Value	Description
Offline	The module is not available.
Idle	The module is ready for processing.
Active	The module is processing or transporting.
Not Installed	The module is not installed.
Suspended	The module is suspended.
Error	The module is in error mode.
Initializing	The module is initializing.
Shutting Down	The module is shutting down.
Pending To Active	The module is in state change from idle to active.
Locked	The module is temporarily not available.
Pre Conditioning	The module is processing a pre-conditioning recipe.
Post Conditioning	The module is processing a post-conditioning recipe.

Table 14 - Module State

Value	Description

Inactive	The recipe was not yet executed.
Active	A recipe is currently being executed.
Finished	Recipe execution finished successfully.
Failed	Recipe execution failed.

Table 15 - Recipe Execution State

Value	Description
Pending	Recipe selection is pending. The scheduler started a recipe download to the module.
Selected	The recipe was downloaded to the module.
Error	The recipe download failed.
Waiting For Selection	The recipe is currently being downloaded to the module.

Table 16 - Recipe Selection State

The modules child nodes depend on the type of the module. A Process Module contains a number of slots which can get materials. Every slot can get only one material. The slot count varies from module to module and depends on the physical capacity. In addition to the slots, a module can have components.

Figure 23 - Process Module with its Slots and Components

Handling systems like the Transfer Module will be defined as module that contains slots and components.

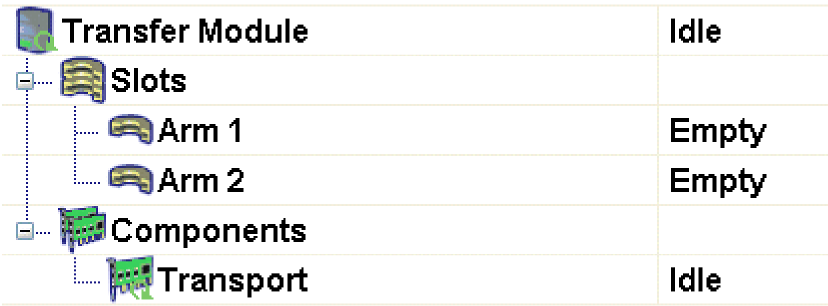


Figure 24 - Transport Module with its Slots and Components

The Components node of transfer modules contains a component named Transport. A Transport component has component parameters that depend on the transfer system. The Transport is being created by the ToolCommander runtime system and is not changeable by the user. Following parameters are available for the Transport component:

Property name	Description	Editable
Name	The unique name of the component	No
Category	The category name of the component.	No
Enforce Half Moves	Indicates if the transport component supports half moves; Possible values are True and False	No
Supports Alignment	Indicates if the transport component supports alignment; Possible values are True and False	No
Supports Exchange	Indicates if the transport component supports material exchanges; Possible values are True and False	No
State	The current state of the module component; Possible values are Offline, Deactivating, Initializing, Error, Activating, Idle, Active, Suspended, Shutting Down, Pending To Active	No
Operation Mode	The current mode of operation of the component; Possible values are Disabled, Automatic, Manual and Not Installed	No

Table 17 - Transport Component

3.4 Loadports

Loadports are a special extension of a module and have the properties described below:

Property name	Description	Editable

Name	The unique name of the module	No
Port ID	The unique numerical port ID of the Loadport	No
Carrier ID Reader Available	Determines if the module supports a carrier ID reader	No
Operation Mode	The actual module operation mode; Possible values are Disabled, Automatic, Manual and Not Installed	No
State	The actual module state; Possible values are Offline, Idle, Active, Suspended, Error, Initializing, Shutting Down, Pending To Active, Locked, Pre Conditioning and Post Conditioning	No
Transfer State	The actual SEMI E87 loadport transfer state. The parameter is only available if a 300mm host interface is installed; Possible values are Out Of Service, Ready To Load, Ready To Unload and Transfer Blocked	No
Access Mode	Loadport access mode to the carrier. The parameter is only available if a 300mm host interface is installed; Possible values are Auto and Manual	No
Carrier Association State	The actual Loadport carrier association state. The parameter is only available if a 300mm host interface is installed; Possible values are Not Associated and Associated	No
Reservation State	The actual SEMI E87 Loadport reservation state. The parameter is only available if a 300mm host interface is installed; Possible values are Reserved and Not Reserved	No

Table 18 - Loadport Properties

Value	Description
Out Of Service	Transfer to/ from this Loadport is disabled.
Ready To Load	The Loadport is available to be loaded with an external carrier or with a carrier that is currently located inside the equipment (i.e. internal buffer).
Ready To Unload	The Loadport is available for unloading of a carrier to a material handling equipment. When the Load Port is being used by the equipment, the state shall transition to Transfer Blocked.

Transfer Blocked	Because of Loadport related activity being performed, transfer is not available to/ from this Loadport at this time.

Table 19 - Transfer State

Value	Description
Auto	Only automated carrier transfers are allowed. The production equipment shall have the capability of generating an alarm if a manual delivery is attempted.
Manual	Only manual carrier transfers are allowed. The production equipment shall have the capability of generating an alarm if an auto delivery is attempted.

Table 20 - Loadport Access Mode

Value	Description
Not Associated	There is no carrier association present for this Loadport
Associated	A Carrier ID has been associated with this Loadport. The Loadport is not available for a new carrier association.

Table 21 - Loadport Carrier Association

Value	Description
Not Reserved	No reservations exist for the Loadport.
Reserved	There is a reservation for future activity at the Loadport. In this state, the Access Mode may not be changed.

Table 22 - Loadport Reservation State

Value	Description
Auto	Only automated carrier transfers are allowed. The production equipment shall have the capability of generating an alarm if a manual delivery is attempted.
Manual	Only manual carrier transfers are allowed. The production equipment shall have the capability of generating an alarm if an auto delivery is attempted.

Table 23 - Carrier Access Mode

Modules of type Loadport usually contain carrier locations.

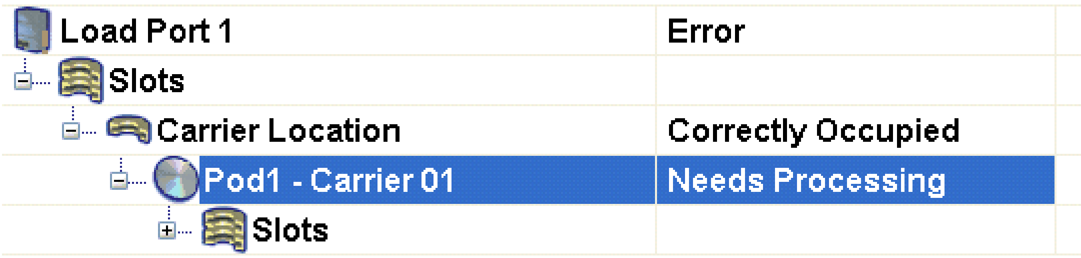


Figure 25 - Loadport with Carrier Locations

At carrier locations the user can insert a carrier. A carrier contains a number of slots.

3.5 Control Jobs

Control jobs provide a supervisory level of control for process jobs on material processing equipment. A detailed description of control jobs can be found in the standard SEMI E94.

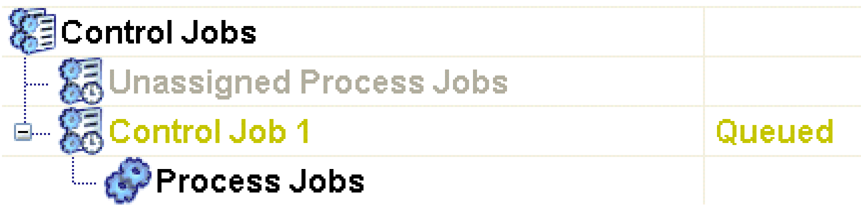


Figure 26 - Control Jobs

By selecting the tree item *Control Jobs* the user can create a new control job by clicking the button Add Control Job in the edit function area. The functionality for deleting a control jobs is also provided.

A control job has the following properties:

Property name	Description	Editable
Name	The name of the control job.	Yes
Author	The name of the user who created the control job.	No
Process Job Order	The order in which the process jobs are processed; Possible values are List, Arrival and Optimize.	Yes
Start Mode	The start mode of the control job; Possible values are Automatic and Manual.	Yes
Priority	The assigned priority of the control job which is taken into account in the choice of the next control job.	Yes

State	The state of the control job; Possible values are Queued, Selected, Waiting For Selection, Executing, Paused and Completed.	No
Ready For Processing	Indicates whether the control job could be started for processing; Possible values are True and False.	No
Start Time	The time when the control job was started.	No
Finish Time	The time when the control job was finished.	No
Expected Start Time	The expected time when the control job will be started.	No
Expected Finish Time	The expected time when the control job will be finished.	No
Creation Time	Date and time when the control job was created.	No
Modification Time	Date and time of the last modification of the control job.	No

Table 24 - Control Job

Value	Description
List	The process jobs will be started in the order of the list/ priority.
Arrival	The process jobs will be started in the order of material arrival.
Optimize	The process jobs will be started in an order which optimizes the equipment throughput.

Table 25 - Process Job Order

Value	Description
Automatic	Automatic start of job if all conditions are fulfilled
Manual	Manual start of job

Table 26 - Start Mode

Value	Description
Queued	The control job was created or deselected

Selected	The control job was selected for execution. Resources will be reserved and are not available for other control jobs. Since the material has not arrived yet it is not possible to start the control job.
Waiting For Start	The material arrived and the control job has to be started manually. The start command is issued either by the operator or the host.
Executing	The control job is running. Process jobs are executed in the configured order.
Paused	The execution is paused, no further process jobs will be started
Completed	All process jobs of the control job were finished successfully, stopped or aborted.

Table 27 - Control Job State

The sub-elements of control jobs are the process jobs.

The node *Unassigned Process Jobs* is used to display process jobs that were created by the host (if a host is connected to the application) before the created and assigned a control job.

3.5.1 Add Control Jobs

In some situations it is needed to manually create new control jobs. For doing this, select the item named Control Jobs in the production object tree.

The button Add changes its caption to Add Control Job. After clicking the button Add Control Job a new control job will be created and displayed as a sub-element of the element called Control Jobs.

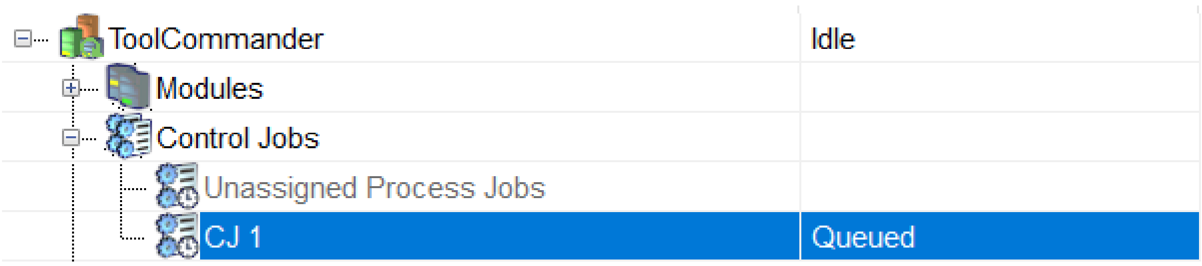


Figure 27 - Newly Created Control Job

For changing its properties the newly created control job was automatically selected after the creation.

3.5.2 Finalize Control Jobs

If you have done all necessary steps to create a valid control job the next step is to finalize it. Only finalized control jobs are considered by the scheduler.

For finalizing a control job, first select it and then click the button Finalize Control Job. If the button is not enabled, one or more of the following conditions are not fulfilled:

- No control job is selected
- The control job is already finalized
- The control job has no process job
- The subordinate process job has no assigned flow recipe

3.5.3 Delete Control Jobs

For deleting the job simply select it and click the button Delete... Before the final deletion of the control job, a dialog will be shown in which the user has to confirm the deletion action.

3.5.4 Start and Stop Control Jobs

Starting a control job can be done after selecting it in the tree-view and pressing the Start button. This action only is enabled if:

- the control job is set to manual start
- the control job was finalized

Control Jobs can be aborted and stopped after they were selected. Details can be found in SEMI E94.

3.6 Process Jobs

Process jobs are described in the standard SEMI E40. They define which materials are to be processed with which flow recipe.

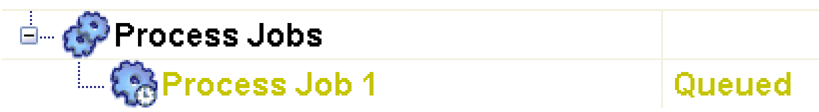


Figure 28 - Process Job

A new process job can be created and applied to a control job by clicking the button Add Process Job. Each control job may contain one or more process jobs.

The following table describes the properties of process jobs.

Property name	Description	Editable
Name	The name of the process job	Yes

Property name	Description	Editable
Author	The name of the user who created the process job	No
Flow Recipe	The flow recipe assigned to the process job	Yes
Start Mode	Determines if the job will be started automatically when all dependencies are fulfilled; Possible values are Automatic and Manual	Yes
Priority	The priority assigned to the process job	Yes
Cycle Count	If the process shall be executed more than once the member of execution cycles can be set.	Yes
State	The state of the process job; Possible values are Queued, Setting Up, Waiting For Start, Processing, Stopping, Stopped, Pausing, Paused, Aborting, Aborted and Process Complete	No
Cycles Executed	The number of executed cycles	No
Quantity	The quantity of materials to be processed by the job	No
Processed Quantity	The number of materials already processed in the process job	No
Start Time	The time when this process job was started	No
Finish Time	The time when this process job was finished	No
Expected Start Time	The expected time when the process job will be started	No
Expected Finish Time	The expected time when the process job will be finished	No
Creation Time	Date and time when the process job was created	No
Modification Time	Date and time when this process job was modified last	No

Table 28 - Process Job Properties

Value	Description
Automatic	Automatic start of job, if all conditions are fulfilled
Manual	Manually start of job

Table 29 - Process Job Start Mode

Value	Description
Arrival	The processing order of materials depends on the arrival of materials at the Load Port
List	The processing requires that the materials arrived at the Load Port are in the same order as defined in the process job. If the materials arrive at the Loadport in an order not equal to the one defined in the process job, the process job would not start processing.
Optimize	The process job optimizes the processing order of the materials.

Table 30 - Process Job Material Order

Value	Description
Queued	The process job is awaiting execution. Depending on the equipment's configuration, one or more process job may be in this state.
Setting Up	The processing resource performs pre-conditioning, awaits material arrival and prepares for material processing
Waiting For Start	Only used in manual start mode. In this state the setup of the process job is complete and the job is waiting for a start command.
Processing	The processing resource is doing the actual material processing using the equipment recipe(s) specified by the process job.
Stopping	The processing resource is performing a stop procedure to terminate processing in an orderly manner. For the processing equipment this may require sending all related materials to its output destination.
Stopped	This is the final state for those jobs that have been in the Stopping state.
Pausing	The processing resource continues to the first safe pausing place and then ceases activity.
Paused	All processing resource activity has ceased. The process job is awaiting a Resume (or Stop or Abort) command.
Aborting	The processing resource is performing an abort or an optional error recovery procedure. The abort procedure will cause immediate termination of the processing. It is the responsibility of the processing resource to cease physical activity as quickly as possible, having archived a safe condition.

Value	Description
Aborted	This is the final state for those jobs that have been in the Aborting state.
Process Complete	The processing resource has completed processing all material specified by the process job. When all material removed from the processing resource, processing resource performs any required post-conditioning. This includes all operations in the processing resource after material departure, which is required by the recipe.

Table 31 - Process Job State

The sub-elements of a process job are the assigned materials and, if the assigned flow recipe contains recipe parameters, the flow recipe structure including parameters.

For assigning material to a process job see chapter Add Material .

3.6.1 Process Job Parameters

After a flow recipe was assigned to a process job the flow recipes parameters (if any) are displayed in a sub-tree of the process job.

When a recipe parameter is selected its value is displayed in the property grid and can be changed there.

3.6.2 Add Process Jobs

Because of the relationship between control jobs and process jobs, you have to select a control job first for adding a new process job. After selecting the control job, the button Add changes its caption to Add Process Job. You can only add process jobs to a control job when the state of the control job is Queued. It isn't possible to add process jobs to a running control job.

After clicking button Add Process Job a new process job is created and assigned to the selected control job. The new process job automatically will be selected after creation. At the bottom object property grid the properties of the new process job can be manipulate.

For finalizing the control job it is needed to assign a flow recipe to the process job. This recipe must be created in the recipe dialog of the application. To assign a flow recipe to a process job change its property called Flow Recipe to an existing recipe.

3.6.3 Delete Process Jobs

For deleting the process job simply select it and click the button Delete…

Deletion of a job only is possible while it was not yet started and if the equipment is not in control state "Online Remote" (controlled by host).

3.6.4 Start and Stop Process Jobs

Starting a process job can be done after selecting it in the tree-view and pressing the Start button. This action only is enabled if:

- the process job is set to manual start
- the control job this process job belongs to already was started
- the process job has material assigned
- the material of the process job already arrived at the machine

To stop or abort a process job is has to be selected and then the according button needs to be pressed. For details see SEMI E40.

3.7 Material

A material has the following properties:

Property name	Description	Editable
Name	The name of the material	Yes
Type	The type of the material. Note: The material type is only editable if there is no further material in the material (carrier) and if it is in the state Needs Processing.	Yes
Acquired ID	The acquired material ID.	No
Assigned Process Job	The process job the material is assigned to	Yes
Comment	A comment for the material	Yes
State	The current state of the material; Possible values are listed in table Material State	No

Property name	Description	Editable
Arrival State	Indicates if the material has been received; Possible values are listed in table Material Arrival State. The parameter is only available if a 300mm host interface is installed.	No
Transport State	The current transport state of the material; Possible values are listed in table Material Transport State	No
Current Position	Indicates where the material currently is located in the machine. Can be used to inform the ToolCommander about material movement.	Yes
Target Position	Indicates the target location of the material	No
Substrate Reading State	The current reading state of the substrate; Possible values are listed in table Substrate Reading State. The parameter is only available if a 300mm host interface is installed and not for material of type carrier.	No
Ready For Processing	Indicates if the carrier may be processed; Possible values are True and False	No
Accessed	Indicates if it was possible to access the carrier. The parameter is only available for material of type carrier.	No
Alignment Angle	The angle the material is currently aligned to.	No
Current Flow Step Name	The name of the currently active flow recipe step for this material.	No
Creation Time	The time when the material was created.	No
Load Time	The time when the material was loaded.	No
Unload Time	The time when the material was unloaded.	No
Carrier ID State	Indicates the current state of the carrier relating to its ID. The parameter is only available for material of type carrier and if a 300mm host interface is installed.	No
Carrier Slot Map State	Indicates the current state of the carrier relating to its slot map. The parameter is only available for material of type carrier and if a 300mm host interface is installed.	No

Property name	Description	Editable
Carrier Accessing State	Indicates the current state of the carrier relating to its accessing state. The parameter is only available for material of type carrier and if a 300mm host interface is installed.	No

Table 32 - Material Properties

Value	Description
Needs Processing	The material is not yet processed (initial state).
In Process	Processing of the material has been started.
Processed	The material was processed.
Aborted	Processing has terminated abnormally.
Stopped	Processing has terminated at completion of certain processing step.
Rejected	Processing of this material was rejected.
Lost	The material has been removed from the equipment by an external agent or it is physically missing.
Skipped	The material was not processed.

Table 33 - Material State

Value	Description
Undefined	The material has been created but no mapping was completed yet.
Arrived	The material arrived at the equipment and is physically present at a specific location.
Not Arrived	The material has not yet arrived at the equipment, but was registered by the host during process job creation.

Table 34 - Material Arrival State

Value	Description
At Source	Material was not yet moved.
At Work	Material was moved from source but has not yet reached the final destination.
At Destination	Material has reached its final destination.

Table 35 - Material Transport State

Value	Description
Not Confirmed	No confirmation was received.
Waiting For Host	Asked the host for confirmation but did not receive an answer yet.
Confirmed	ID was confirmed by the host.
Confirmation Failed	The host did not confirm the ID.

Table 36 - Substrate Reading State

3.7.1 Add Material

In the production view you can create and assign a material to a slot of a module. For doing this select a single slot or the element named Slots and click the button Add Material.

In the configuration file VCTC.ini there is next to other the option "UseMaterialTypeSelectionDialog", which is set to false per default. New materials will be created with the default material type if the option is not enabled. If it is enabled the following dialog when adding material to one or more slots to select the type of the new material if there are multiple material types compatible for the selected slot(s). In the list of shown material types the default material type defined for the selected slot(s) is selected. If multiple slots are selected and there are defined different default material types, no material type is selected in the dialog.

One material will be inserted with the selected type if only one slot was selected. If the entry Slots was selected, then a new material will be created in every available slot with the selected material type.

Depending on the prior selection, the application will create one material or a number of materials with that material type, which was selected in the dialog if shown. Each for any slot in the collection.

For using the material with a specific process job, you have to assign it to the process job. Select the material and change its property Assigned Process Job to an existing process job. A material can only be assigned to one specific process job.

When assigning material to a process job it will also be added to the collection “Materials” of that job.

3.7.2 Delete Material

For removing material from a process job you can use the materials collection below the Materials item of the process job. If you select this material and click the button Delete… it will be removed from the process job. The material itself will not be removed from the equipment. You can also select the material in the slot and change its property Assigned Process Job for removing the material from the process job.

To completely remove a material from the slot, select the material in the slot and press Delete… It is important for a complete deletion of a material that the material in the slot is selected, not the material in the process job (Figure Material Assigned to a Process Job). Deletion of material needs to be confirmed by the user.

Note

Deleting material from a slot will cause the ToolCommander and TMC to treat the according slot as empty. So it could try to place different material to that same slot, even if the slot physically would still be occupied. The user is responsible to make sure the mapping data in ToolCommander corresponds to the physical placement of wafers in the tool.

3.7.3 Move Material

Using this function the user may tell the system that a wafer was moved manually (without using the system components like TMC; e.g. by manually moving a wafer or by using service functions of a robot etc.)

To do so the user needs to select the wafer that was moved in the tree. Then, from the property grid, Current Position needs to be selected. Opening the combo-box assigned to this property will open a tree-view structure of all material positions (slots) in the tool the selected material might be moved to (slot must be compatible with the material type and be empty). Selecting one of the slots displayed will update the data storage of ToolCommander and change the material to the slot selected. Using this function no movement of material will be started by ToolCommander.

Note

Moving material using this function will cause the ToolCommander internal data storage of wafer positions to update to what the user entered. The user is responsible to make sure the mapping data in ToolCommander corresponds to the physical placement of wafers in the tool. Otherwise ToolCommander could try to place a wafer to a slot it believes to be empty but that still might be occupied by another wafer which ToolCommander does not know to be there.

3.8 Scheduler

Another item in the production tree is the Scheduler. The Scheduler handles jobs and controls the production and the flow of any material through the machine.

Figure 29 - Scheduler of the ToolCommander

The following properties are available:

Property name	Description	Editable
Deadlock Prevention Mode	Determines whether deadlocks are prevented; Possible values are listed in table Deadlock Prevention Mode.	No
Deadlock Solving Mode	Determines whether deadlocks are solved automatically if possible; Possible values are listed in table Deadlock Solving Mode.	No
Carrier Mode	The carrier mode determines if the equipment can handle multiple carriers at the same time; Possible values are listed in table Carrier Mode.	No
Substrate Mode	Determines how multiple substrates shall be handled by the scheduler; Possible values listed in table Substrate Mode.	No
Schedule Material Order	Determines whether materials which are still in the equipment or materials from Load Port/ cassette are preferred; possible values are listed in table Schedule Material Order.	No
Exchange Wait Tolerance (ms)	The maximum wait time in milliseconds for a material exchange during transport.	No
Control Job Mode	Defines how the Scheduler handle multiple control jobs.	No
Use Multi Exchange Mode	Defines if multiple exchanges are linked successively.	No
Use Wet Bench Mode	Defines if a recipe with preconditions is to be started before the previous step is finished.	No
State	The actual state of the scheduler; Possible values are listed in table Scheduler State.	No

Table 37 - Scheduler Properties

Value	Description
Off	No deadlock prevention.
Avoid intersections	The scheduler will keep one module free for parking substrates.

Table 38 - Deadlock Prevention Mode

Value	Description
Ignore	Deadlocks are ignored.
Parking	Deadlocks are solved automatically (if possible).

Table 39 - Deadlock Solving Mode

Value	Description
Parallel	Multiple carriers are processed at the same time.
Sequential	A single carrier is processed and the next carrier may start after all materials returned to the carrier.
Sequential Overlapped	A single carrier is processed but the next carrier may start after the last unprocessed material from the first carrier started processing.

Table 40 - Carrier Mode

Value	Description
Parallel	The equipment is handling multiple materials at the same time.
Sequential	Only one material is handled and until it is finished and unloaded no other substrate is loaded.

Table 41 - Substrate Mode

Value	Description
Prefer Active Materials	Prefer active materials during scheduling. Unloading materials has a higher priority than loading new materials.
Prefer Inactive Materials	Prefer inactive materials during scheduling. Loading new materials has a higher priority than unloading materials.

Table 42 - Schedule Material Order

Value	Description
Idle	The scheduler is not working (inactive or no material to be scheduled).

Value	Description
Active	The scheduler is working.
Stopping	The scheduler is being stopped but waiting for the equipment to finish active operations.
Suspending	The scheduler is being suspended. Active Process Jobs will be finished but no new jobs will be started in this state.

Table 43 - Scheduler State

3.8.1 Scheduler Operations

The dialog **Scheduler Operations** shows all operations created by the scheduler. The following figure illustrates the operations viewer by an example.

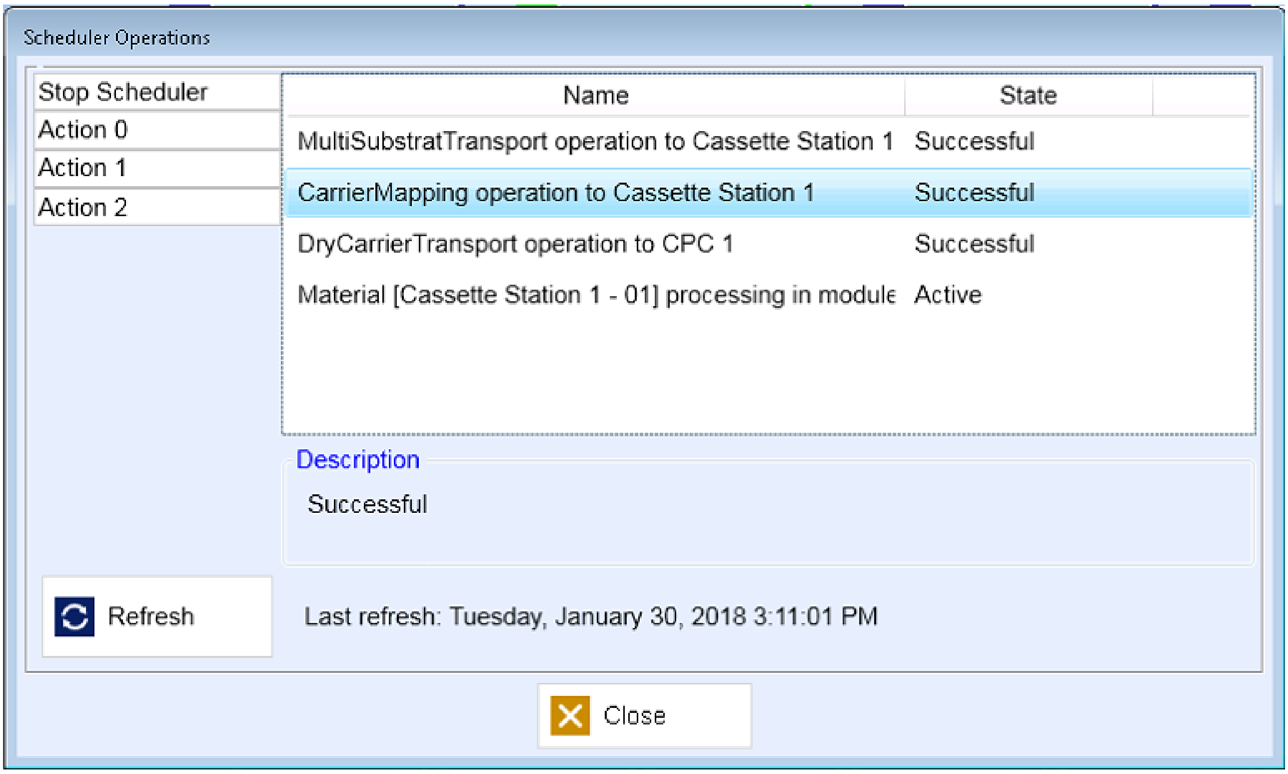


Figure 30 - Dialog Scheduled Operations

The list presents the name and the current state of an operation. The width of the list columns are variable and can be adapted.

Click on a list entry to select the operation. The selected operation is highlighted in a different color. Below the list, a further description of the selected operation is shown.

Specific recovery options are available for an operation. The possible recovery options are presented by the button bar at the left side. Press the required button to execute the function. The recovery options are dependent on the selected operation and the buttons vary correspondingly.

The list of operations is refreshed at once with opening the dialog. Pressing the button **Refresh** a reload can be triggered manually. The time stamp of the latest refresh is shown and comprises the date as well as the time.

Press the button **Close** to finish viewing of scheduler operations.

3.9 Interlocks

The production tree contains another element called Interlocks. Interlocks are available to prevent damages to humans or the equipment itself. They are shown in the production tree in three different places (in case they are available for that specific area):

First place is right below the root element (equipment) of the production tree.

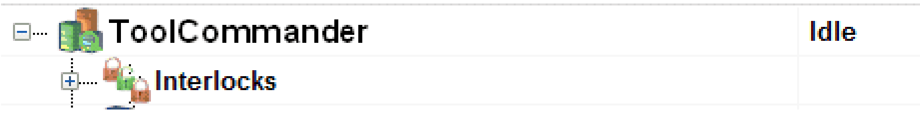


Figure 31 - Interlocks for equipment

A physical module may contain Interlocks.

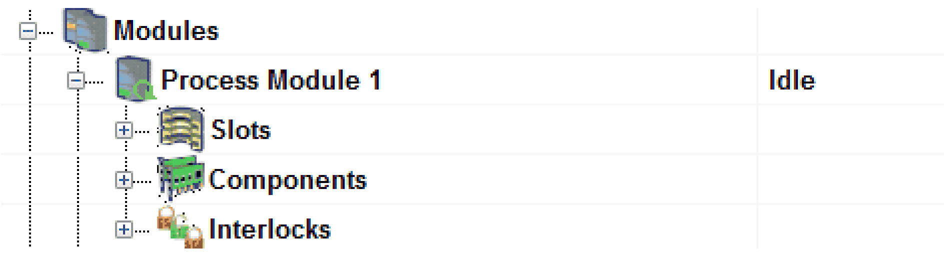


Figure 32 - Interlocks for modules

And as a third option module components may have associated Interlocks.

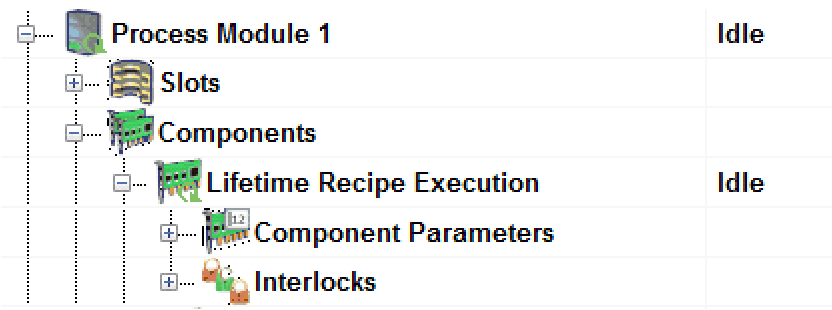


Figure 33 - Interlocks for module components

Each Interlocks section contains one or more interlocks that have different output states.

Value		Description
	Unlocked	Lock is currently not enabled.
	Locked	Lock is currently active.

Table 44 - Interlock states

Interlocks have one or more children that provide the base for locking or unlocking of a specific feature. The figure below shows that the *MainInterlock* contains two children *SubInterlock1* and *SubInterlock2*. It is possible that a child contains again children.

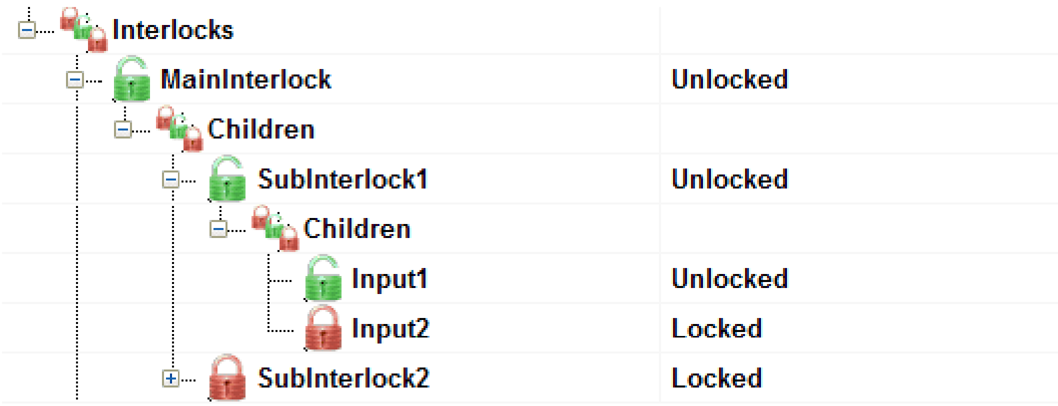


Figure 34 - Interlocks with children

Based on the children and the operation mode an interlock calculates its current state.

Operation Mode	Description
AND	All inputs must be true.
OR	One input must be true.
NAND	If all inputs are true the output is false otherwise it is true.
NOR	If all inputs are false the output is true.

Table 45 - Interlocks operation mode

The final decision of the output state of an interlock is finally based on the work mode.

Work Mode	Description
Lock	Basic work mode is lock something.
Release	Work mode is to release something.

Table 46 - Interlocks work mode

Interlocks have the following properties:

Property name	Description	Editable
Name	Name of the interlock.	
Description	More detailed description of that interlock.	
Work Mode	Working mode of the interlock	
Operation Mode	Operation mode of the interlock.	
Result	Result of all the inputs.	

Table 47 - Interlocks properties

4

Production Log

4 Production Log

The ToolCommander has a variety of logging functions. These functions take care of the recording and displaying of jobs, of events that take place during the production as well as the progression of chosen process values.

To take a look at the recorded data the user can access the Production Log screen. The data is displayed in a tree view and can be filtered and exported.

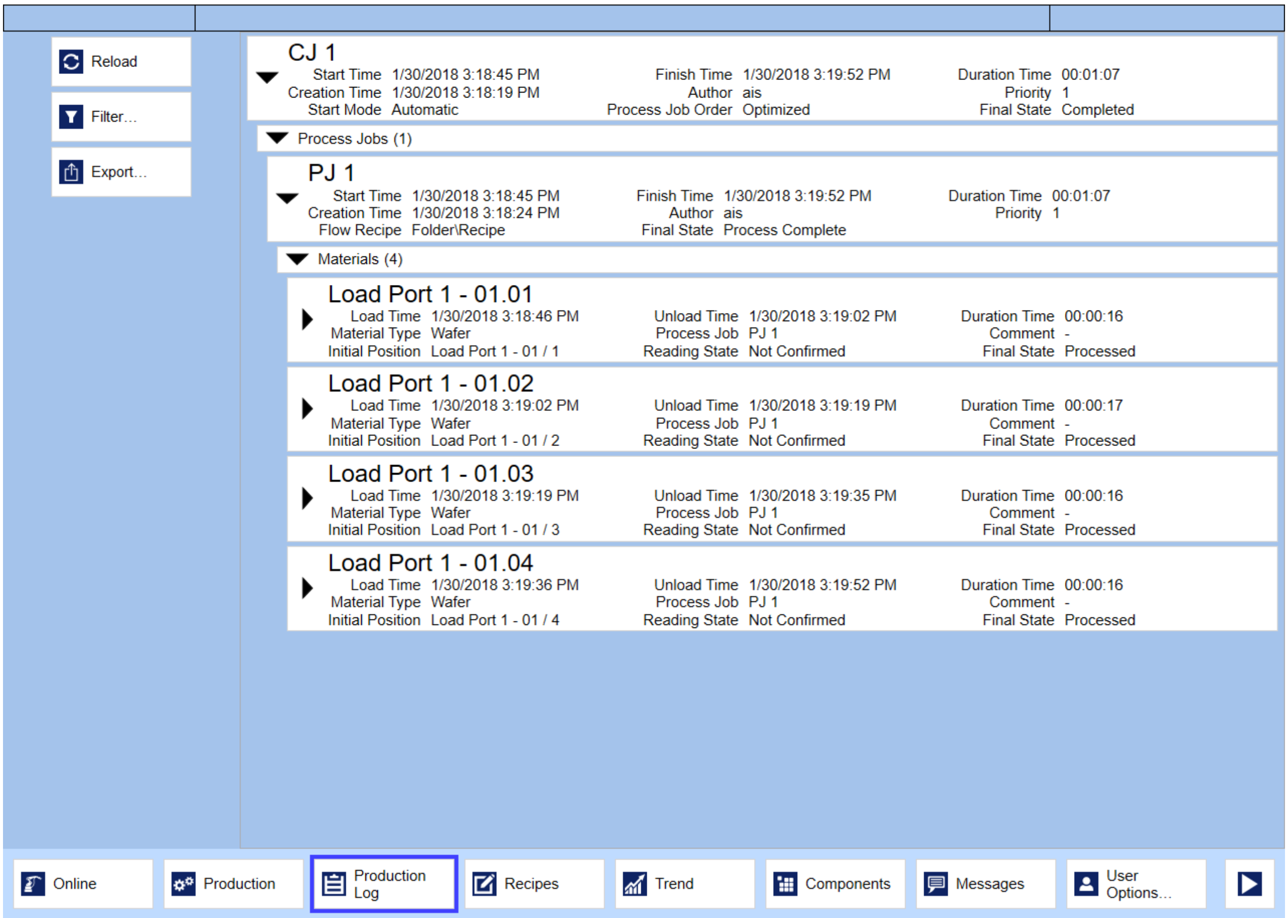


Figure 35 - Production Log

The screen is divided in two areas. In the menu there are buttons to control the log contents. The log can be seen next to it.

The following functionalities are available to control the log:

Function	Description
Reload	This button can be used to reload the list of production logs.

Function	Description
Filter	This button opens a dialog where a variety of filters can be defined, that will take effect on the currently displayed log.
Export	This button opens a dialog where the export path can be specified. After providing the path the log that is currently shown will be saved to this path as a Microsoft XPS-Document.

Table 48 - Production Log Functions

4.1 Log Tree

The visual representation of the log is a tree view. At the top most layer there are the different control jobs sorted by their corresponding process dates.

The control job is defined by a variety of properties, e.g. start time, end time or author. A double-click on the control job with the left mouse button will expand the control job. The sub elements of the control job – consisting of a list of events and process jobs – will be displayed. Any other control job will be hidden in the view.

The events will only be loaded when their corresponding control job is expanded. This means the expansion of a control job can take some time, depending on the amount of recorded data. To be able to view the results the events element needs to be double-clicked, which will in turn display the sub elements of the events list.

Other than the events there is also a list of process jobs for each control job. The behavior of the process job list is corresponding to the behavior of the event list.

While expanding a particular element within the production log using a double-click, all elements staying at the same level will be hidden. In order to unhide these elements, simply collapse the expanded element.

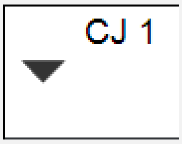
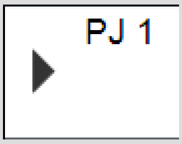
Visualization	Description
	The element is expanded; All sub elements will be displayed; Elements of the same level are hidden.
	The element is collapsed; All sub elements are hidden; Elements of the same level (if available) are displayed.
	This element has no sub elements; Elements of the same level (if available) are displayed.

Table 49 - Element Visualization in Production Log

All other elements of the Log tree behave mostly like the just described elements. The following is a complete overview of the Log tree structure:

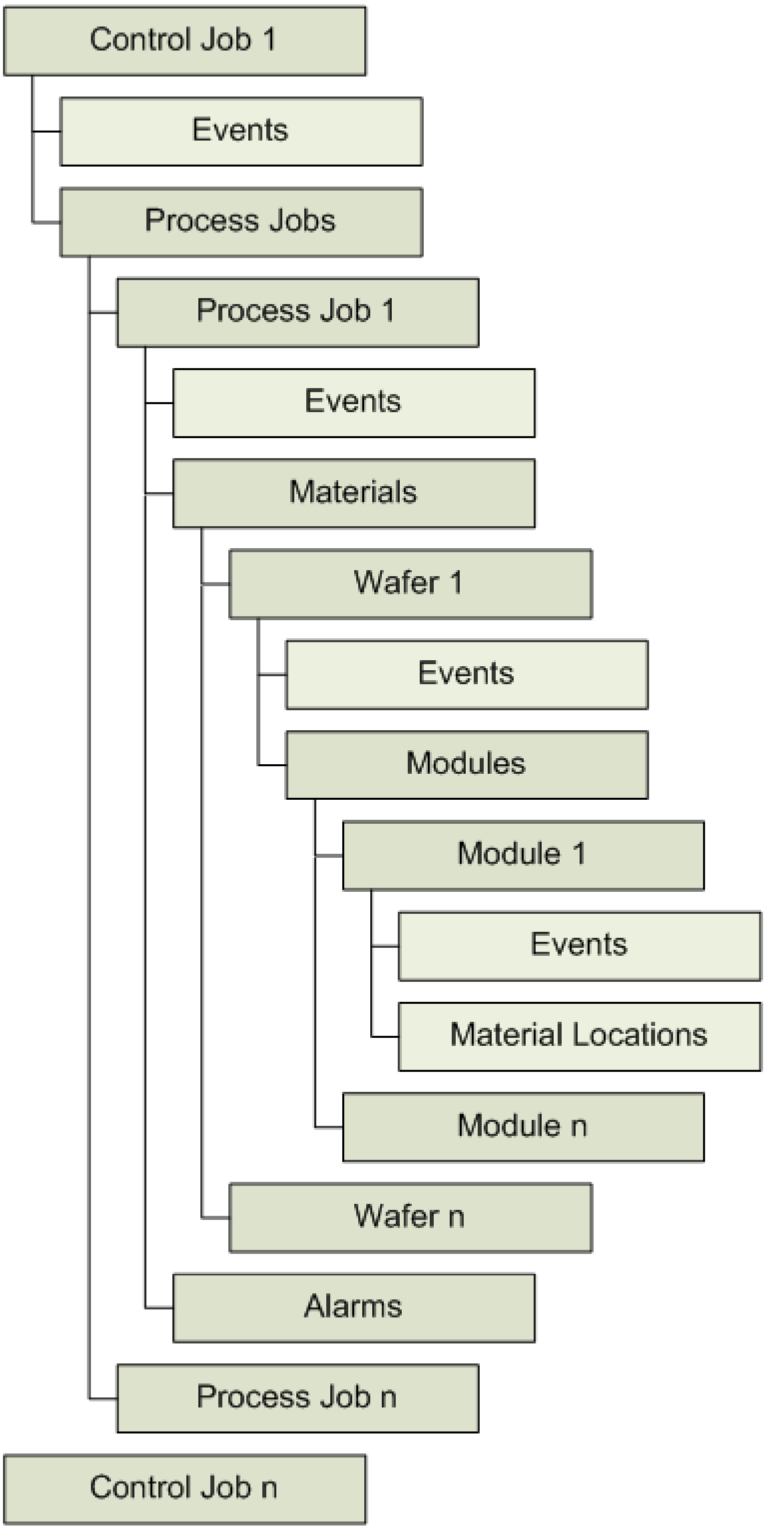


Figure 36 - Structure of production log

4.2 Filter

To be able to modify the range of displayed information in the log corresponding to specific criteria there is the possibility to create a filter.

After clicking the button **Filter...** the Filter dialog is opened. In the filter dialog different criteria can be selected for control jobs, process jobs and materials. These criteria can then be linked via a logical And or Or operation. The following filters are available:

- Control Job
  - Name (Free Text)
  - State (Selected, Waiting for Start, Executing, Paused, Completed)
  - Start Date (Range)
  - End Date (Range)
- Process Job
  - Name (Free Text)
  - State (Setting Up, Waiting for Start, Processing, Stopping, Stopped, Pausing, Paused, Aborting, Aborted, Process Complete)
  - Author (Free Text)
  - Flow Recipe (Free Text)
- Material
  - Name (Free Text)
  - State (In Process, Processed, Aborted, Stopped, Rejected, Lost, Skipped)

To activate a filter, the checkbox next to the filter needs to be activated. After that the filter is active and can be parameterized.

After confirming the selected filters by clicking on the button **Apply** the filters will be used on the available production log data.

4.3 Export

To permanently store the production log, all currently displayed control jobs and their sub elements can be exported. As format to store the log data, the Microsoft XPS-Format will be used.

To execute an export a folder and a file name has to be chosen with the button **Find…** as location for the log data. Then the export process can be started by clicking on the button **Start**. An active export process will be indicated by a corresponding text below the text field **Output field**.

Function	Description
Output file	This text field is used to insert and display the path, where the log data will be exported to.
Find ...	Press the button to open a standard dialog and select a folder and filename as target location for the log data that will be exported to.
Start	Press the button to start the export process.
Cancel	Press the button to abort the export and close the export dialog.

Table 50 - Export Dialog

5

Flow Recipes

5 Flow Recipes

Access authorization

Only users who have the right "Edit Flow Recipes" have the possibility to make changes to flow recipes. Users who have the right "View Flow Recipes" can only view the flow recipes.

Flow recipes describe how and in which order substrates are to be process inside the machine using different process modules and module recipes.

5.1 Properties of a Flow Recipe

A flow recipe consists of a number of steps which are executed sequentially. Steps can contain choices and choices can contain jumps.

Property name	Description	Editable
Name	The unique name of the Flow Recipe.	Yes
Author	The author of the Flow Recipe.	No
Comment	A comment for the flow recipe.	Yes
State	The actual flow recipe state; Possible values are Development, Archive and Production.	Yes
Version Name	The name of the version of the flow recipe.	Yes
Version Number	The version number of the flow recipe.	No
Modification Time	Date and time of last modification of the flow recipe.	No
Creation Time	Date and time of creation of the flow recipe.	No

Table 51 - Flow Recipe Properties

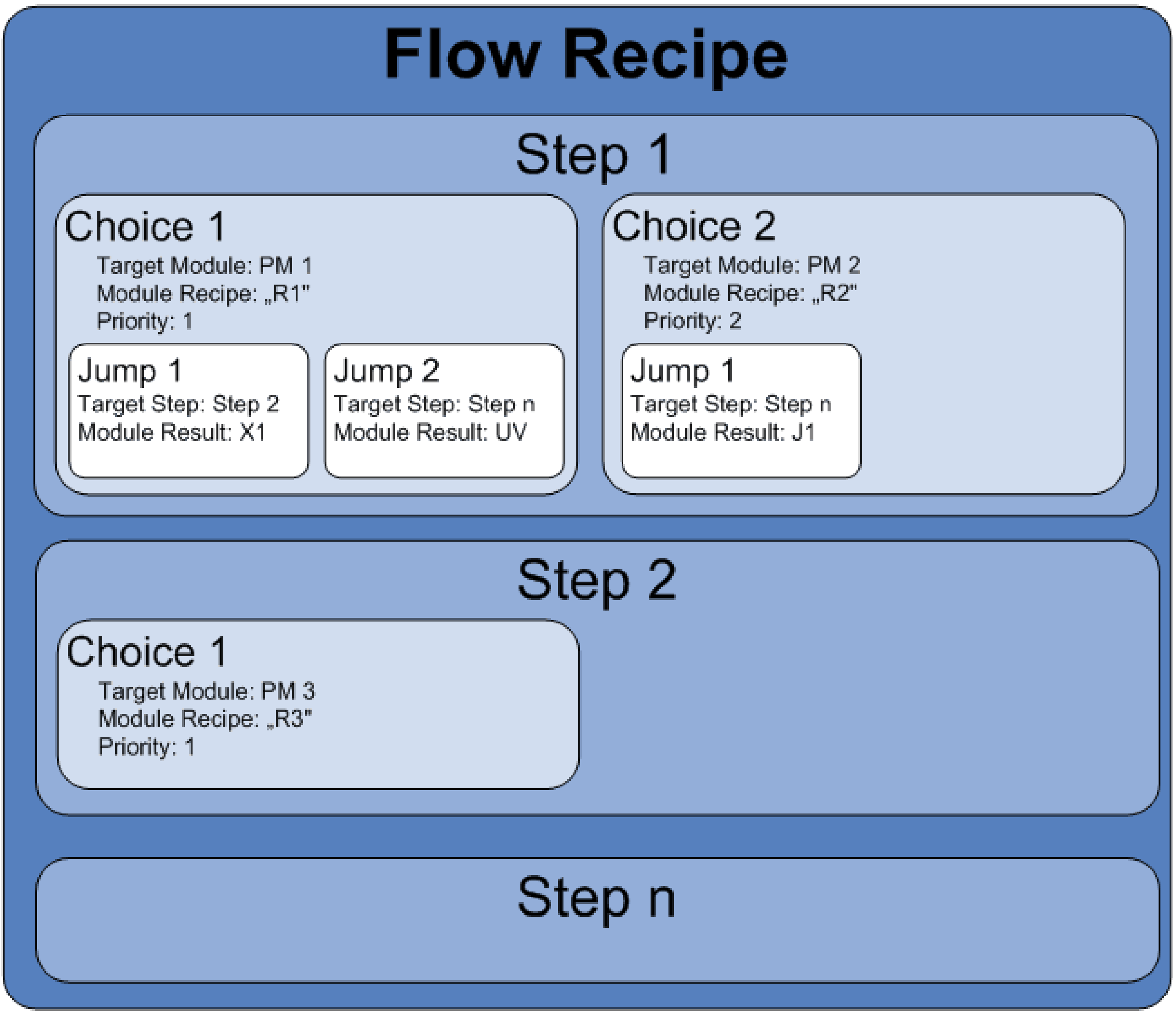


Figure 37 - Structure of Flow Recipe Containing Steps, Choices and Jumps

5.2 Steps

The steps of a flow recipe describe the transfer of materials in the tool. Within recipes the steps are executed sequentially.

Property name	Description	Editable
Name	The unique name of the step	Yes
Number	The number defines the order of the steps in the recipe. Change the order of the steps with the button **Move down** or **Move up** in the edit function area.	

Property name	Description	Editable
Type	The type of the step (see next table Flow Step Types). The step types to be used can be selected in the VCTC.ini	Yes
Sampling Rate (optional)	The percentage Sampling rate defines how often material is measured in the step. The Sampling rate is a percentage value. 100 means all material is measured, 50 means every other material is measured. The property is only available if **UseSamplingRate=1** is set in the VCTC.ini.	Yes
Comment	A comment for the step	Yes

Table 52 - Flow Step Properties

The available types of a step are described in the table below. Depending on the type of the step, steps can have various properties.

Property name	Description
Process	The step consists of the transport of the material to a target module and the processing with a selected module recipe. The module recipe describes the steps inside the module. The step must contain at least one choice. Choices can contain various jumps.
Unload (optional)	The step contains the transport of the material to a destination module for unloading it. The step must contain at least one choice. Choices of an unload step cannot contain jumps. The property is only available if **EnableFlowStepTypeUnload=1** is set in the VCTC.ini.
Loop (optional)	A Loop step contains a number of sub steps which are executed sequentially. The property Loop Count describes how often that sub-sequence should be executed. The Loop step may even contain sub steps of type loop. The property is only available if **EnableFlowStepTypeLoop=1** is set in the VCTC.ini.
Jump (optional)	The step contains no transport of material. It only contains a reference to another step (Parameter: destination step) where the execution of the recipe shall continue. Steps of this type cannot contain choices or sub-steps. The property is only available if **EnableFlowStepTypeJump=1** is set in the VCTC.ini.
Stop (optional)	The step contains no transport of material. It only contains the information that the sequence is to be stopped at this step. A Stop step in a recipe finishes the recipe. Steps of this type cannot contain choices or sub-steps. The property is only available if **EnableFlowStepTypeStop=1** is set in the VCTC.ini.

Table 53 - Flow Step Types

5.3 Choice

Property name	Description	Editable
Name	A unique name of the choice	Yes
Priority	The priority assigned to the choice. Change the priority of the choices with the button **Move down** or **Move up** in the edit function area.	
Target Module	The target of the transport of the material.	Yes
Target Module Slot (optional)	The slot in the target module. The property is only available if **UseTargetModuleSlots=1** is set in the VCTC.ini.	Yes
Module Recipe	The recipe to be executed in the target module. This property is only available if the assigned step is of type **Process**.	Yes
Default Jump (optional)	If a module returns a result which is not described by a jump the default jump will be executed. This property is only available if the assigned step is of type **Process**. The property is only available if **EnableFlowStepTypeLoop=1** is set in the VCTC.ini.	Yes
Alignment Angle (optional)	Angle for material alignment. Only useful if the machine supports alignment (aligner hardware is installed). The property is only available if the **UseAlignment=1** is set in the VCTC.ini.	Yes
Postconditioning Module Recipe (optional)	The module recipe to be executed after the actual module recipe finishes. The property is only available if **EnablePostconditioningModuleRecipe=1** is set in the VCTC.ini.	Yes
Preconditioning Module Recipe (optional)	The module recipe to be executed before the actual module recipe starts. The property is only available if **EnablePreconditioningModuleRecipe=1** is set in the VCTC.ini.	Yes
Comment	A comment for the choice.	Yes

Table 54 - Flow Choice Properties

Steps of type Unload and Process are divided into choices. Choices of a process step contain a target module and a corresponding module recipe. Choices of an unload step only contain a target module.

Based on their priority, the scheduler sorts the choices of a step and then tries to find an executable one. This function is used when the tool consists of several modules of the same type. If the module with the lowest priority number is active or not available the scheduler can try to find an alternative destination. Thus, parallel processes are supported.

An example will illustrate this: A flow recipe contains one step named "Step 1". The step contains 2 choices:

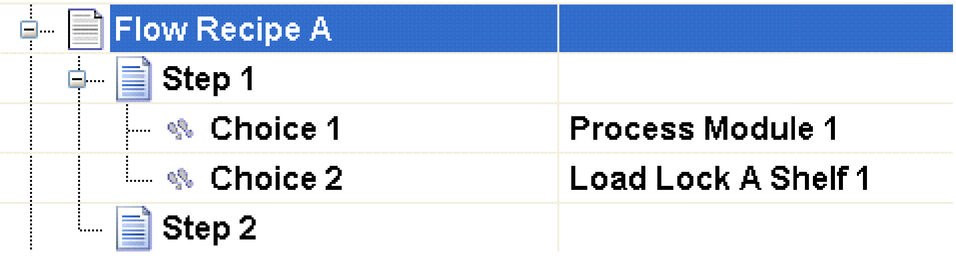


Figure 38 - Example of Priorities of Choices

In the example the priority of Choice 1 is 1 and the priority of Choice 2 is 2. At first, the scheduler checks whether the material can be transferred into Process Module 1 for processing the material. If this transfer is not possible, the scheduler checks whether the material can be transferred into Load Lock A Shelf 1.

The order of execution depends on the priorities of the choices. The priority 1 means the highest priority. In this example Choice 1 has a higher priority than Choice 2.

Within one step, each module can assigned to a choice only one time. So it isn't possible to define a module as Target Module in 2 different choices of one step.

5.4 Jump

If a step is of type Process, the choices can also contain a number of conditional jumps. After executing a module recipe each module returns a value to the ToolCommander. A jump provides functionality for selecting the target module depending on the module result. The number and meaning of such module results depend on the module implementation.

Property name	Description	Editable
Name	A unique name of the choice	Yes
Module Result	The return value of the module that is valid for this jump.	Yes
Target Step	The step on which the execution of the flow recipe continues if the module returns the value defined in Return Value.	Yes

Table 55 - Flow Jump Properties

Within a step choice, return values of modules can only be used once. So it is not possible to use a return value that is already used within that same choice.

5.5 Lifecycle

Flow recipes are structured into following lifecycles:

Property name	Description
Development	The recipe is still in development. The recipe can be edited.
Production	The recipe is ready for production and cannot be edited.
Archive	The recipe is no longer used. It is archived for later use. Archived recipes cannot be used within process jobs.

Table 56 - Flow Recipe Life Cycle

Basically, recipes can change their condition from one to another. However, there are the following constraints:

- State changes from Archive to Production are not possible
- For changing a flow recipe from Production to another condition, the flow recipe will not be assigned to a process job.
- The state change from Development to Production requires the following conditions:
  - The flow recipe requires at least one step
  - All steps of type Process and Unload require at least one step choice
  - All step choices have to contain a Target Module
  - All step choices have to have a Module Recipe assigned
  - All Module Recipes used by choices have to be in state Production.

5.6 Editor

Using the Flow Recipe Editor it is possible to create new flow recipes, edit previously created recipes or delete them. The editor can be opened by selecting Recipes from the menu and selecting the machine name (top-level node) in the navigation tree. The following figure presents the three areas of the Flow Recipe Editor.

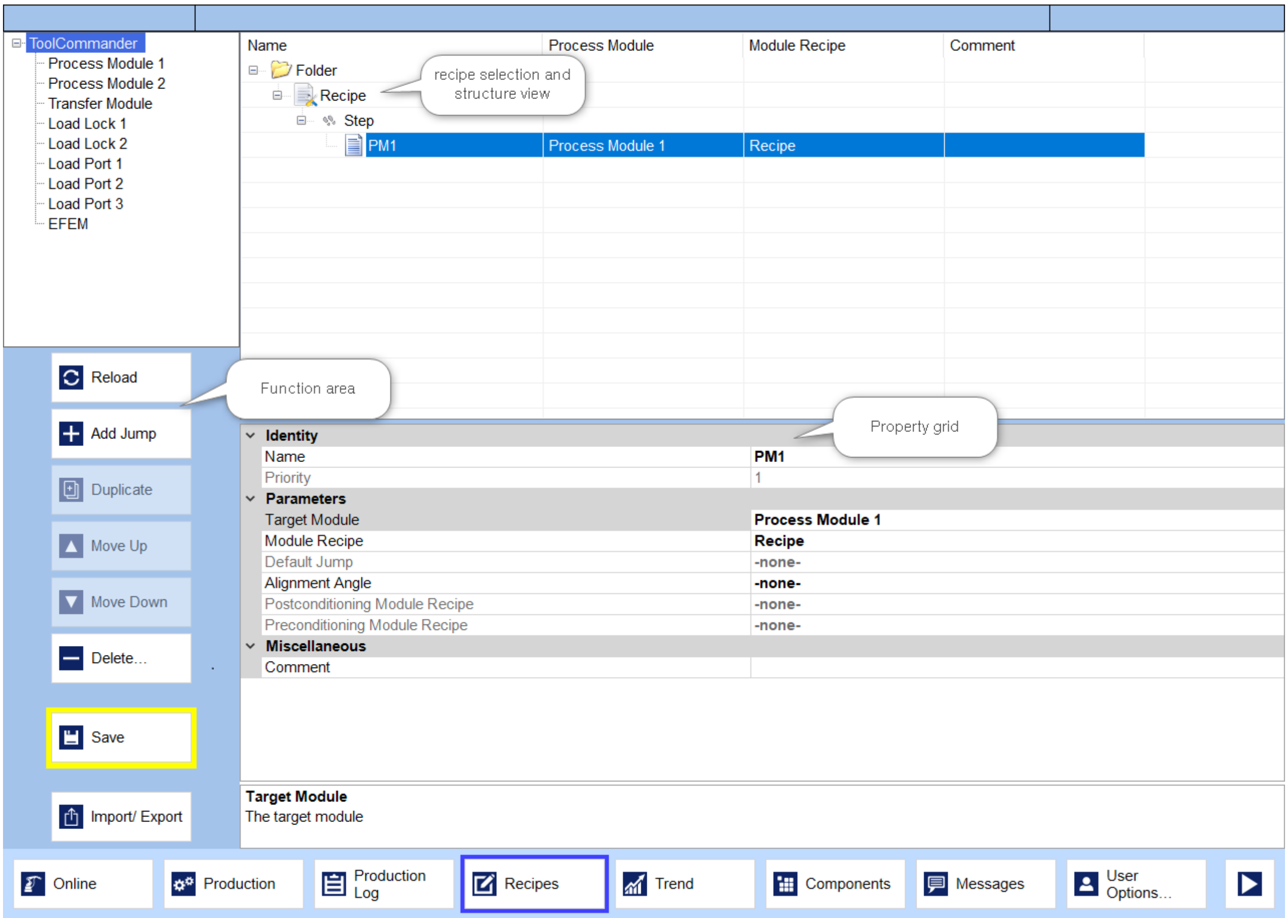


Figure 39 - Flow Recipe Editor

5.6.1 Flow Recipe View

The flow recipe editor provides a hierarchical visualization and modification of the data of flow recipes. The data of the flow recipes are displayed in a multi-column list in combination with a tree view. The actual position in the list will be highlighted.

Figure 40 - A Highlighted Flow Recipe

Depending on the present hardware, the selection can be changed by clicking the keys Page Up and Page Down (↑ und ↓), selecting by mouse or by tipping on the touch screen.

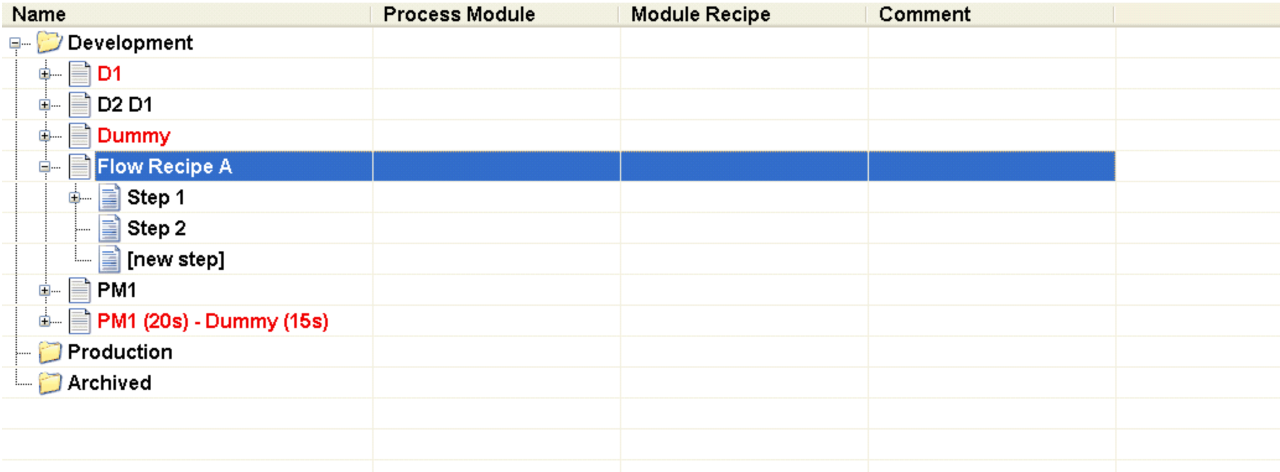


Figure 41 - List View of the Data of Flow Recipes

The tree view clarifies the hierarchical relationship between the components of the flow recipe.

For showing the sub elements of a tree node, click or press the key Cursor Right (→). For hiding the sub-elements click or press Cursor Left (←).

By default, the flow recipes are arranged in a folder structure you can define freely.
But, that property can be switched off by the VCTC.ini (configuration setting **EnableRecipeFolder=0**). Then, the flow recipes are organized depending on their state within the groups Development, Production and Archive.

The steps are sorted by their execution order (Property Number). The choices of a step are sorted by priority. The jumps are sorted alphabetically.

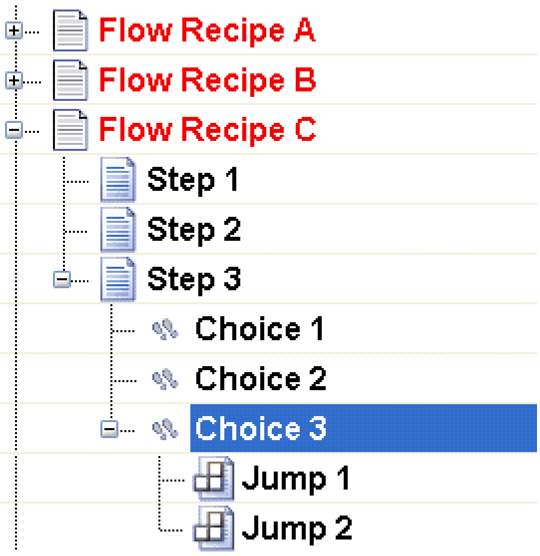


Figure 42 - Sorting Criteria

5.6.2 Function Area

The edit function area provides functionally for manipulating flow recipes in the recipe selection and structure view.

By clicking the button **Reload**, all flow recipes are loaded from the database again. After this reload the editor will be refreshed.

5.6.2.1 Add, Duplicate and Delete Elements

For adding new elements press the button Add Step. The type of the element depends on the currently selected element in the recipe selection and structure view.

Add Recipe: If the development node is selected, the newly added element is a flow recipe. This new flow recipe is always created as a sub-element of the development node. As a hint the button for adding the element will change its caption depending on the actual selection in recipe selection and structure view.

Add Step: If a flow recipe is selected the added element is a step. The processing sequence of the created step can be changed.

Add Choice: If a step is selected in the recipe selection and structure view, a new choice will be appended to the selected step.

Add Jump: If a choice is selected, a new jump will be added to it.

Add Recipe (disabled): If none of the above elements are selected, the button for adding elements is disabled.

Automatically, all newly added elements get a unique name. For better readability, it is recommended to change the name to a meaningful one. While changing the name of the elements it is important to give any element a unique name. For example: It is not allowed that two steps of the same flow recipe have equal names.

Duplicate: If the selected element is a flow recipe or a step of a flow recipe, the button Duplicate will copy the selected item with all its subordinate elements.

Duplicate (deactivated): Choices or jumps cannot be duplicated. If a duplicate action is not possible, this is pointed out by a deactivated button.

Name	Process Module	Module Recipe	Comment
Development			
Flow Recipe A			
Step 1			
Alternative 1	Load Lock A Shelf 1	[unknown]	
Alternative 2	[unknown]	[unknown]	
Step 2			
Step 3			
Flow Recipe A1			
Step 1			
Alternative 1	Load Lock A Shelf 1	[unknown]	
Alternative 2	[unknown]	[unknown]	
Step 2			
Step 3			
Production			
Archived			

Figure 43 - Duplicate of Flow Recipe A

Changes to the duplicated element will not change the original.

Press the button **Delete…** to remove the selected element. The button is disabled if no element is selected.

Before the final deletion of the element, a dialog will show in which the user has to acknowledge the deletion action.

After clicking Yes in the confirmation dialog the selected item will be deleted. Later access to the deleted element is not possible.

5.6.2.2 Organize Recipes in Folders

Next to the default organization of flow recipes into the 3 states 'development', 'production' and 'archive' there is the possibility to organize flow recipes in recipe folders. This functionality must be activated explicitly.

New recipe folders can be created on the root level or as sub folder in an existing recipe folder of the Flow Recipe View. The names of the recipes have to be unique within a recipe folder.

Press the button **Add folder** in the Function Area. The new recipe folder is created on the root level if you have not selected any element. If you have chosen an existing recipe folder, the new folder is created as sub folder of the selected folder. The button is activated if you have selected no element or a recipe folder.

The following figures presents exemplary a recipe folder structure with subordinated recipes.

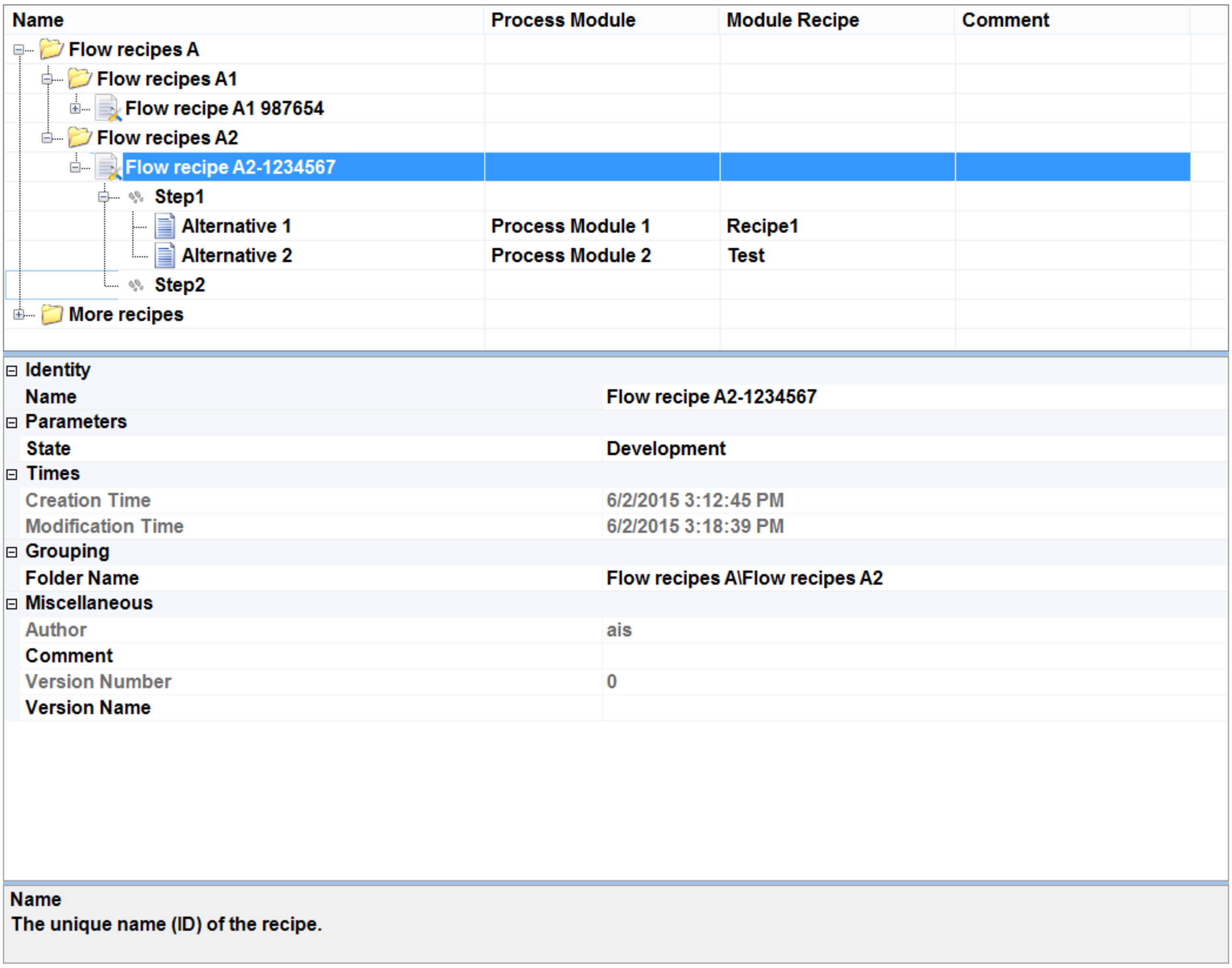


Figure 44 - Example of a recipe folder structure

Rename a Recipe folder

Select the recipe folder you want to rename. The property of the selected folder are presented below the Flow Recipe View. Enter the new name for the recipe folder in the input field **Name**. Press the button **Save** in the **Function Area** to take over the changes.

Move recipes in another recipe folder

Initially, select the recipe in the Flow Recipe View you want to move. The property of the selected recipe are presented above the Flow Recipe View. Open the combo box **Folder Name** in the **Grouping** group. The combo box presents the current recipe folder structure. Choose the required recipe folder. Press the button **Save** in the **Function Area** to take over the changes.

Structure with disabled Recipe folders

Organizing the recipe folders can be disable by the configuration of the **ToolCommander®**. In this case, the flow recipes are arranged by their current states in the groups Development, Production and Archive. The recipes cannot be moved. Additional folders cannot be created.

5.6.2.3 Changing the Step Sequence or the Priority of Choices

For changing the step sequence or the priority of choices use the buttons **Move up** and **Move Down**.

By clicking the buttons, the selected Step or Choice changes its position in the grid upwards or downwards. The position in the grid equals the position in the process sequence of a step.

Name	Process Module	Module Recipe	Comment
Development			
Flow Recipe A			
Step 2			
Step 1			
Alternative 1	Load Lock A Shelf 1	[unknown]	
Alternative 2	[unknown]	[unknown]	
Step 3			
Flow Recipe A1			
Production			
Archived			

Figure 45 - Change the Priority of Steps

For example: If the step named **Step 1** is selected and the user clicks Move Down, the selected step will be moved downwards and the step named **Step 2** has to finish before **Step 1** can start.

Changing the position of a choice means, that the priority of the choice is changed. If a choice is moved up in the tree, the priority of this choice is incremented. If the choice is moved downwards, its priority is decremented.

Name	Process Module	Module Recipe	Comment
Development			
Flow Recipe A			
Step 1			
Alternative 2	[unknown]	[unknown]	
Alternative 1	Load Lock A Shelf 1	[unknown]	
Step 2			
Step 3			
Flow Recipe A1			
Production			
Archived			

Figure 46 - Change Step Sequence

If a movement of a step or a choice isn't allowed or there is only one step or choice, the buttons for moving are disabled.

5.6.2.4 Import and Export of Flow Recipes

To import and export flow recipes the edit function area contains the button **Import/ Export**.

The import and export is carried out in or from a file. After clicking the button Import/ Export a dialog will be shown on which the import or export can be specified.

By clicking the button Import or Export the dialog changes its mode between save flow recipe to a file and load flow recipe from a file. The current mode is displayed by a blue salience around the button.

If the dialog is in export mode, it contains a list of flow recipes that exist in the flow recipe editor. If the dialog is in import mode, the dialog contains a list of all flow recipes that were found in the selected file. The imported or exported file can be selected by entering its path directly into the text field or by using the open file dialog which is displayed after that button **Find ...** was clicked.

Each recipe in the import/ export dialog has its own checkbox for selecting it.

For selecting all recipes or none of the recipes use the buttons **Select all** or **Select none**. This method of selecting and deselecting recipes is useful if the dialog contains many recipes.

After selecting the import/ export file and selecting the recipes, the import or export action starts when you click the button **Start**. If the export/import action completes, the dialog closes automatically.

For canceling the import/ export, simply press **Close**.

The import/ export functionality if only available if a correct file name was selected and at least one flow recipe is selected.

5.6.3 Object Property Grid

The object property grid allows users to view and modify the properties of each flow recipe, step, choice or jump. The object property grids location is at the bottom of the screen.

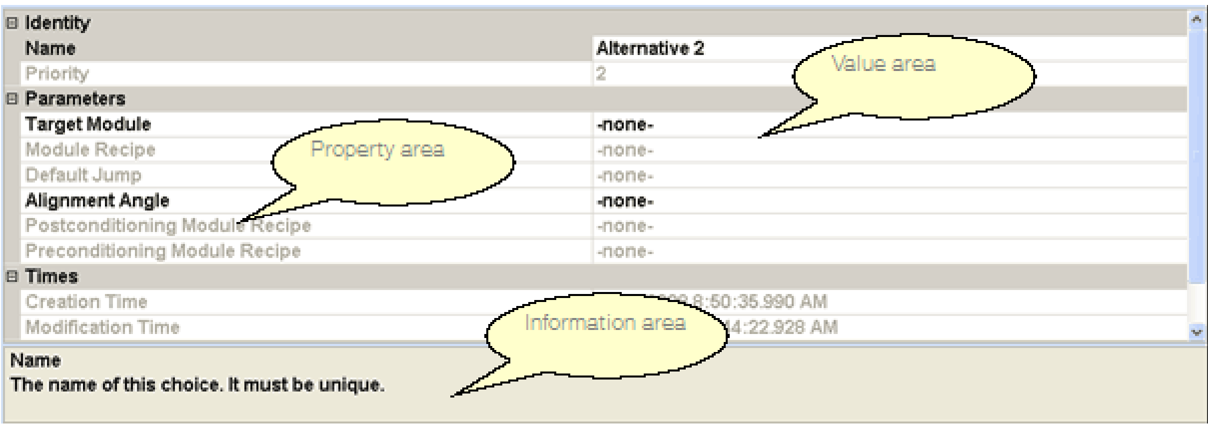


Figure 47 - Object Property Grid

The object property grid is divided into three sections:

- Property area
- Value area
- Information area

The property area contains the properties of the currently selected item in the recipe selection and structure view. The value area contains the appropriate values of the properties. The information area at the bottom of the screen provides helpful information about the currently selected property. In most cases, the information area contains the description of the currently selected property. Furthermore, if the state property of a recipe is selected, the information area can provide information about the reason why the recipe state cannot be changed from Development to Production.

Figure 48 - Example for Additional Information in Information Area

Properties displayed in gray cannot be changed.

The input options of the changeable properties differ. Properties like the name of flow recipes, steps, choices or jumps can be entered directly via keyboard. Properties like the target module of a choice give the user a list of possible values on which the user can select an item (see Figure 5-24: Combo box for Selecting the Target Module).

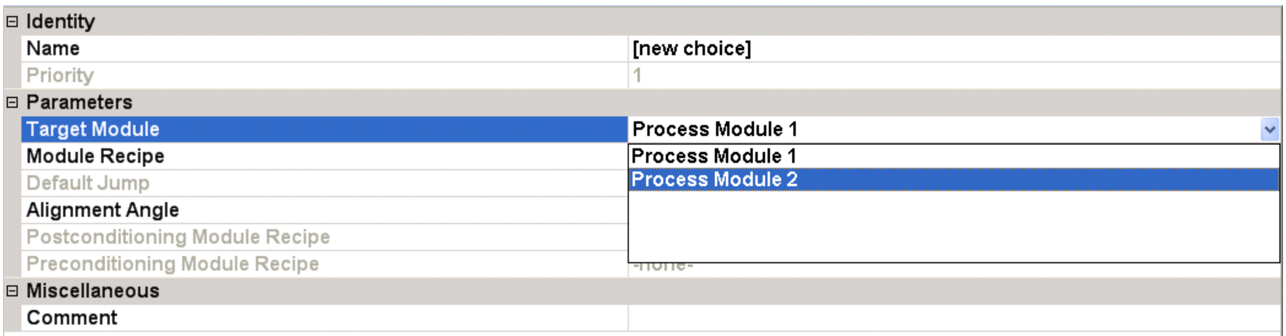


Figure 49 - Combobox for Selecting the Target Module

The third input method is the Tree box. The Tree box looks like a Combo box but the selection is rather a hierarchical tree:

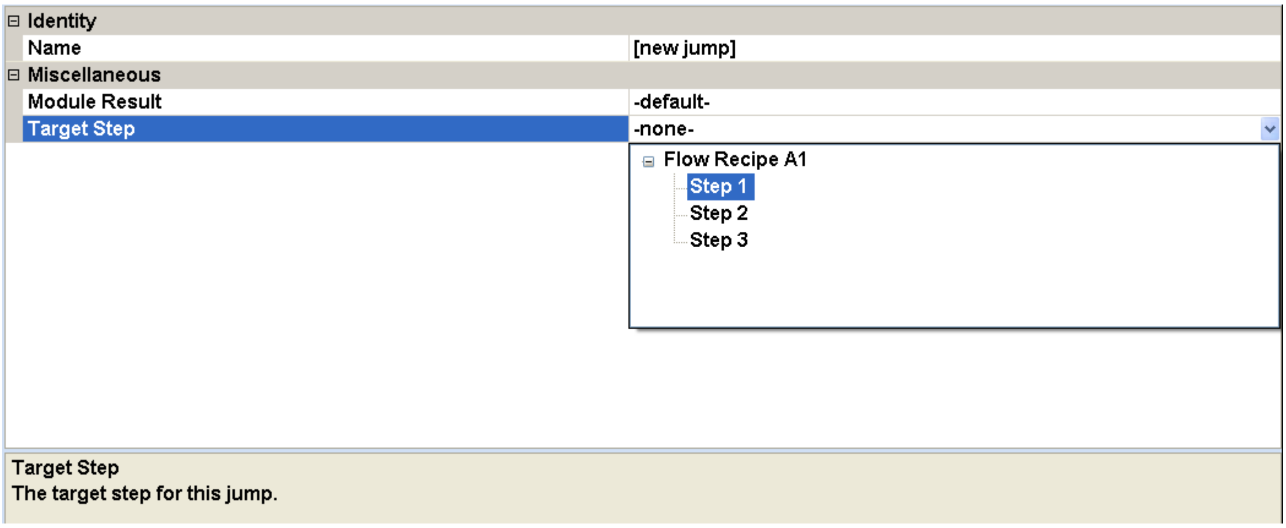


Figure 50 - Treebox for Selecting the Target Step

By clicking the button ⊞ in the Tree box the sub-element of an item can be displayed. Clicking the button ⊟ symbol hides its sub-elements. For selecting an element in the Tree box, click on it.

A fourth input option is a list box with filter functionality. Properties such as recipes offered on this type available. The selection can be made directly via the keyboard (arrow keys) or mouse. Also may a input of a character string be possible by a separate text box according to which the list is filtered. It checks whether the string (case-sensitive) is found in the specified list items.

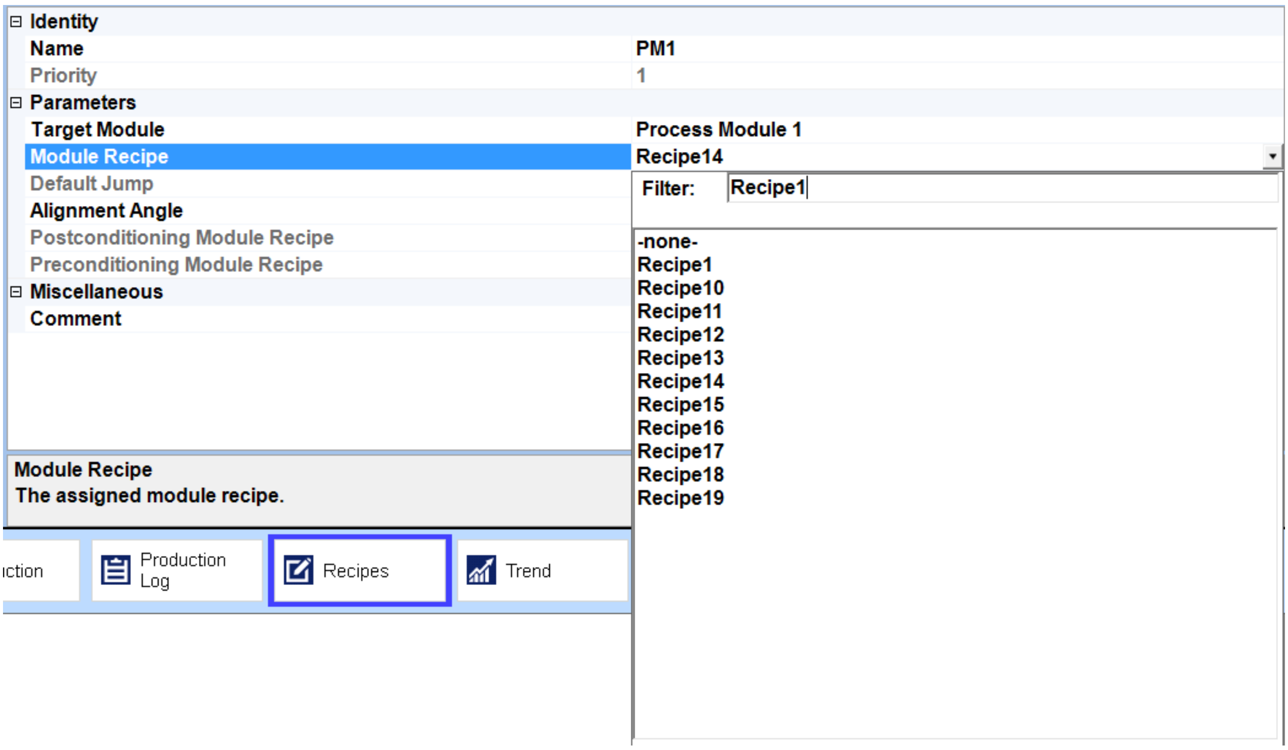


Figure 51 - Combo box for Selection the Recipe

For a better readability the properties are grouped. Possible groups are Identity, Parameters, Times and Miscellaneous. Depending on the selected item some groups may be missing.

6

Trend

6 Trend

The Trend module offers you to present the Online trend of several process values (channels). A user has to be logged on to use the Trend module.

Group several channels to a graph to compare them with each other. The Trend module provides a convenient graphical interface to create and configure new graphs in only a few steps. The graph has a **Diagram View** and a **Table View** you can switch between them anytime.

A graph is assigned to a schema. You can assign more than one graph to a schema. A schema illustrates for example a defined characteristics of the equipment or simplifies the presentation for a specific user groups. A new schema with one graph is created in a few steps. Changes can be saved automatically or can be discarded. The following figure illustrates the structure of the Trend module.

Figure 52 - Detail view of a schema with four graphs

ToolCommander Application

The upper area (application menu, header and message line) as well as the lower menu bar of the **ToolCommander®** are always visible. See Screen Layout for detailed information.

Schema management

The upper area presents the available schemas and the currently selected schema. In the lower area, you can create, save, configure and delete schemas by the action buttons. See Schema Management for detailed information.

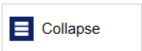
Trend views

The major presentation of the Trend module is the trend viewer. The Trend viewer shows the **Detail View** of the selected graph while all other graphs are shown as thumbnails. The thumbnails show the current values of the channels. The **Compact View** shows all graphs in equal size and provides all information. You can edit the selected graph and switch between **Diagram View** and **Table View**. See Trend Views for detailed information.

6.1 Schema Management

Graphs are saved in schemas. Therefore, a schema is a set of graphs. Use the schema to present the characteristics of a module.

Expand the schema management with the button  to select a schema. In this state, the upper area of the schema management lists all available schema and the lower area provides the action buttons with labels.

If you have already selected a schema, you can collapse the schema management with the button 

to save place on the screen. In this state, the upper area presents the currently selected schema and the lower area provides the action buttons without labels.

The following figure shows the collapsed and the expanded schema management of the Trend module.

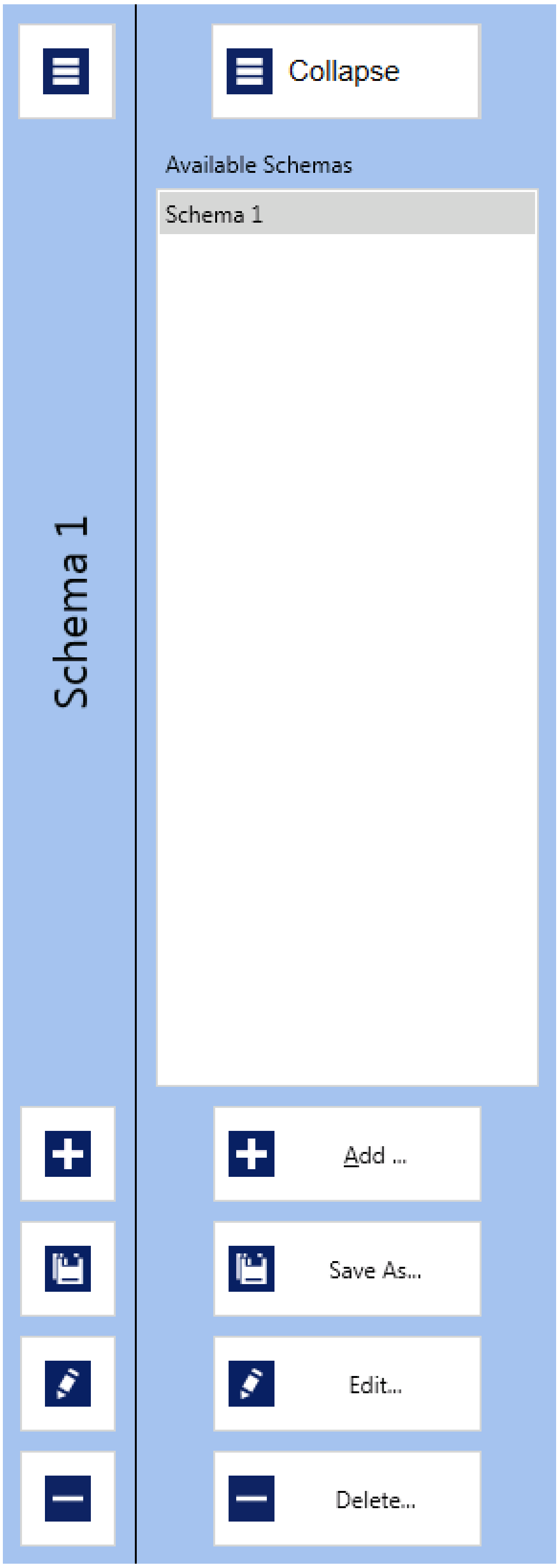


Figure 53 - The schema management collapsed and expanded

Schema selection

Expand the schema management with the button . The list shows all available schemas. The currently selected schema is marked.
Select the required schema in the list. The **ToolCommander®** loads the schema. An indicator is shown if the loading takes long. Then you can cancel the loading. Press the button **Cancel** of the indicator.

Action buttons

The lower part of the schema management presents the actions buttons. The buttons can be used in the collapsed and in the expanded view of the schema management.

Press the button **Add...** to create a new schema. See Create a Schema for detailed information.

Press the button **Save as...** to save a copy of the selected schema under a new name. The button is only enables if a schema is selected. The dialog **Save Schema as** will be shown. Enter the required name in the text box **Schema name** for the copy of the schema. Press the button **Save as...** to save the schema with the new name. Press the button **Cancel** to do not save the schema. Both buttons close the dialog.

Press the button **Edit...** to change the selected schema. The dialog **Edit Schema** will be shown. Edit the schema and the related graphs in the dialog. The button is only activated if a schema is selected. See Edit a Schema for detailed information.

Press the button **Delete...** to delete the selected schema. The button is only enabled if a schema is selected. You have to confirm the action in a separate dialog. Press the button **Delete** to delete the schema. Press the button **Cancel** to abort the action. The confirmation dialog will be closed. After deleting a schema no schema is selected.

6.1.1 Create a Schema

Create a new schema in a few steps. The new schema contains a graph with the selected channels and one Y axis in the diagram view. After creating, you can always edit the schema and add more graphs. Press the button **Add...** in the schema management. The dialog **New Schema** will be opened. The following figure shows the dialog **New Schema**.

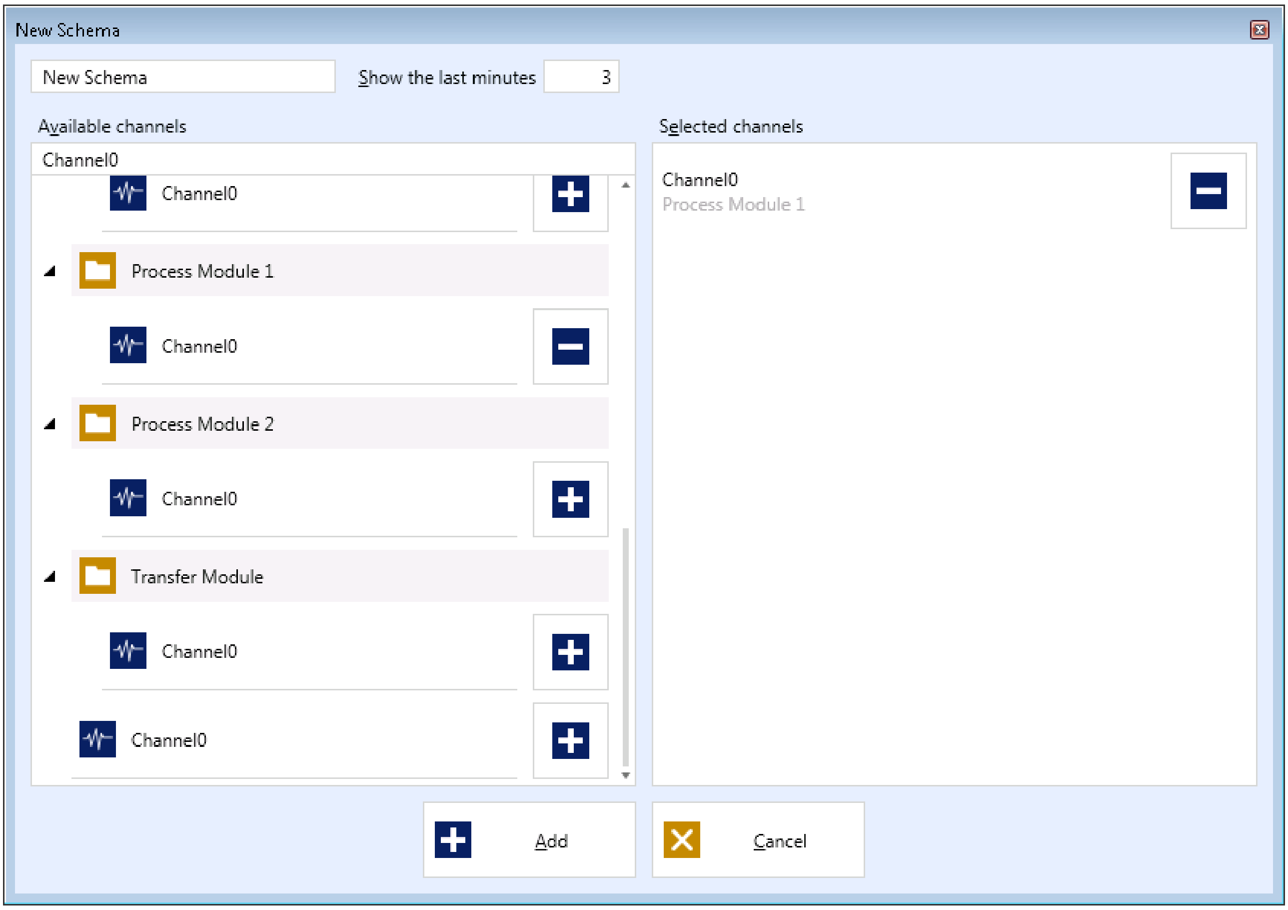


Figure 54 - Dialog for creating a new schema

Specify a name for the new schema or leave the default name **New Schema xx** (x is a consecutive number). Please consider, the name of the schema is required and the schema name has to be unique. The text box indicates the error if the schema name does not fulfill the requirements. The button **Add** is deactivated. Please input a valid schema name.

The input field **Show the last minutes** contains the time span for the Online Data. The Online Data of the graph will be set from the current time to the selected time in the past. Modify the number of minutes if desired. You can change the value individually while the graph is shown.

Select the required channel in the left list **Available Channels.** The channels are arranged in an hierarchy below the modules. Globally defined channels are not assigned to a module. They are shown in the root level of the channel tree. Modules are labeled by the symbol . Channels are labeled by the symbol . Search for a channel when there are to many channels. Enter any part of the channel name in the search box to filter the list.

Press the button to assign the selected channel to the graph. Repeat the selection to add further channels.

The right list **Selected Channels** shows the channels you have already chosen for the graph. Press the button to remove a channel from the list.

Press the button **Add** to add a new schema. The button is deactivated if you have not selected a channel or the name of the schema is not unique.
Press the button **Cancel** to do not create a new schema. The dialog **New Schema** will be closed.

6.1.2 Edit a Schema

Press the button to edit the selected schema. The dialog **Edit Schema** will be open as presented in the following figure.

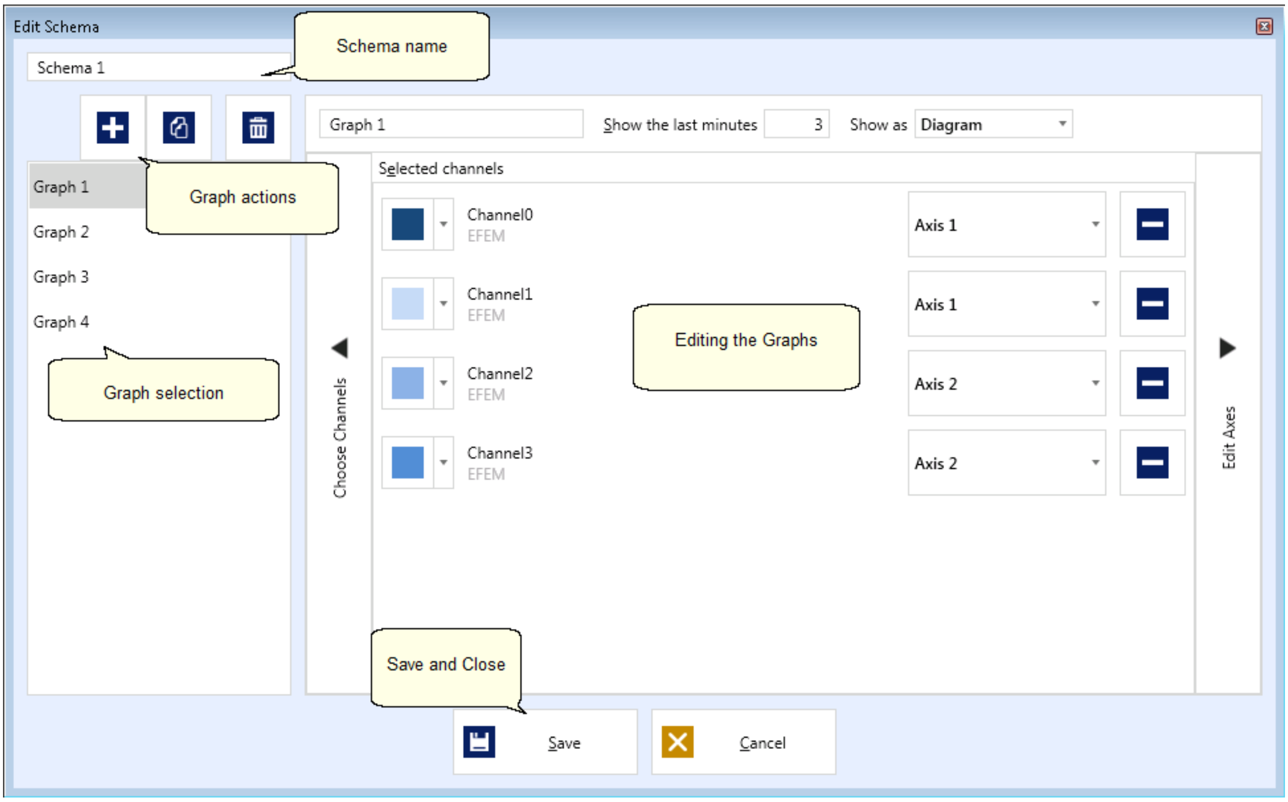


Figure 55 - Edit Schema

Edit schema name

Enter a new schema name in the input field. Please consider, the name of the schema is required and the schema name has to be unique. If the schema name does not fulfill the requirements, the text box indicates the error. Please input a valid schema name.

Graph actions

You manage the graphs by the action buttons.

Press the button **Add Graph** to add a new graph to the schema. The new graph will be added to the list of graphs with a default name. Open the graph to edit it.

Press the button **Copy Graph** to add a copy of the selected graph. The button is only enabled if a graph is selected in the list. The copied graph is added to the list of graphs by a default name. Open the graph to edit it.

Press the button **Delete Graph** to delete the selected graph. The button is only enabled if a graph is selected in the list. Please consider, no confirmation is required to delete a graph.

Selection of a graph

The graphs are arranged in a list. The selected graph is marked and presented at the right side.

Edit a graph

You edit individual graphs, assign channels and configure the related axes in this area. See Edit a Trend for detailed information about editing graphs.

Save and Cancel

Press the button **Save** to save pending changes. The button is only activated if the name of the schema as well as the name of the graph are valid.

Press the button **Cancel** to discard all pending changes.

6.2 Trend Views

Graph

You can display each graph as diagram and table. Both trend views are presented in tabs. Therefore, you can easily switch between the views.
The **Diagram View** visualizes each channel in a trend curve. Show to the Online Data of the trend curves or select a defined time. Compare the trend curves by show or hide the trend curves. Additionally, the diagram view provides the mapping of the trend curves to different Y-axes. Similar channels can also be combined to one Y-axis. See Diagram View for detailed information.

The **Table View** of the graphs shows the recorded data of the selected time span of the individual channels in columns. When you switch over to the diagram view, the currently selected time span will be used. You can select a time span or you can go to a defined time to put the data of the channels into the table. Furthermore, you can sort the data of the table and also export it to a file. See Table View for detailed information.

Schema

The Trend module is able to present the graphs of a schema at the same time. Therefore, two views are available.
The **Compact View** shows you all graphs in the same size. Keep track of by this view. You can represent the individual graphs as diagram as well as table and you can use all functions of the graphs. The following figures presents an example for the compact view.

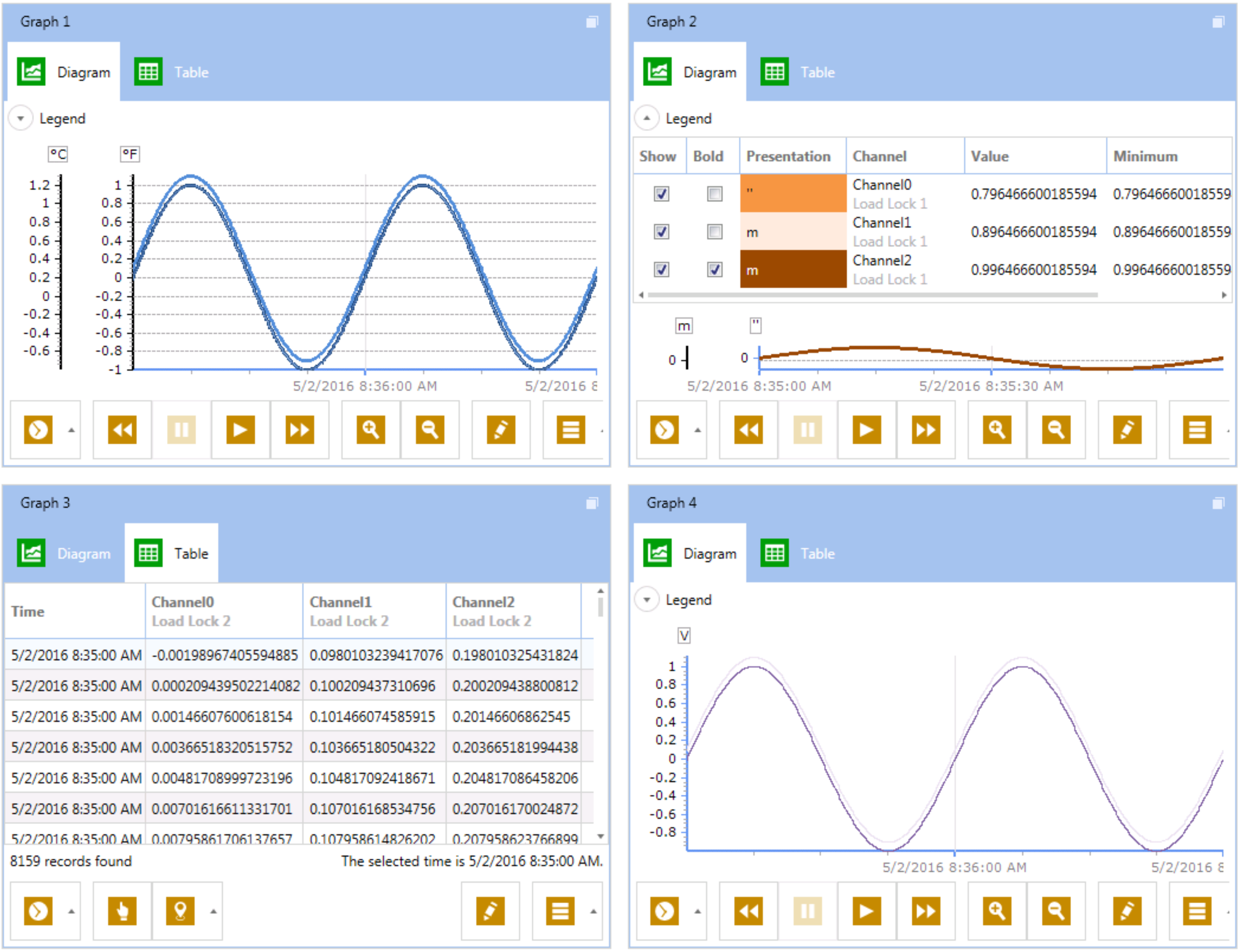


Figure 56 - Compact view of a schema with four graphs

Switch over to the **Detail view** of the schema to enlarge one of the graphs. Click on the symbol in the right upper corner of the required graph. You can use the selected graph both in Diagram view and in Table view. All other graphs will be represented as thumbnails and display the current process values of the channels. Click on the symbol in the right upper corner of the selected graph to switch over to the compact view. The following figure illustrates the Detail view by an example.

Figure 57 - Detail view of a schema with four graphs

6.2.1 Diagram View

The **Diagram View** shows the representation of the individual channels in an Online trend. The following figures shows an example.

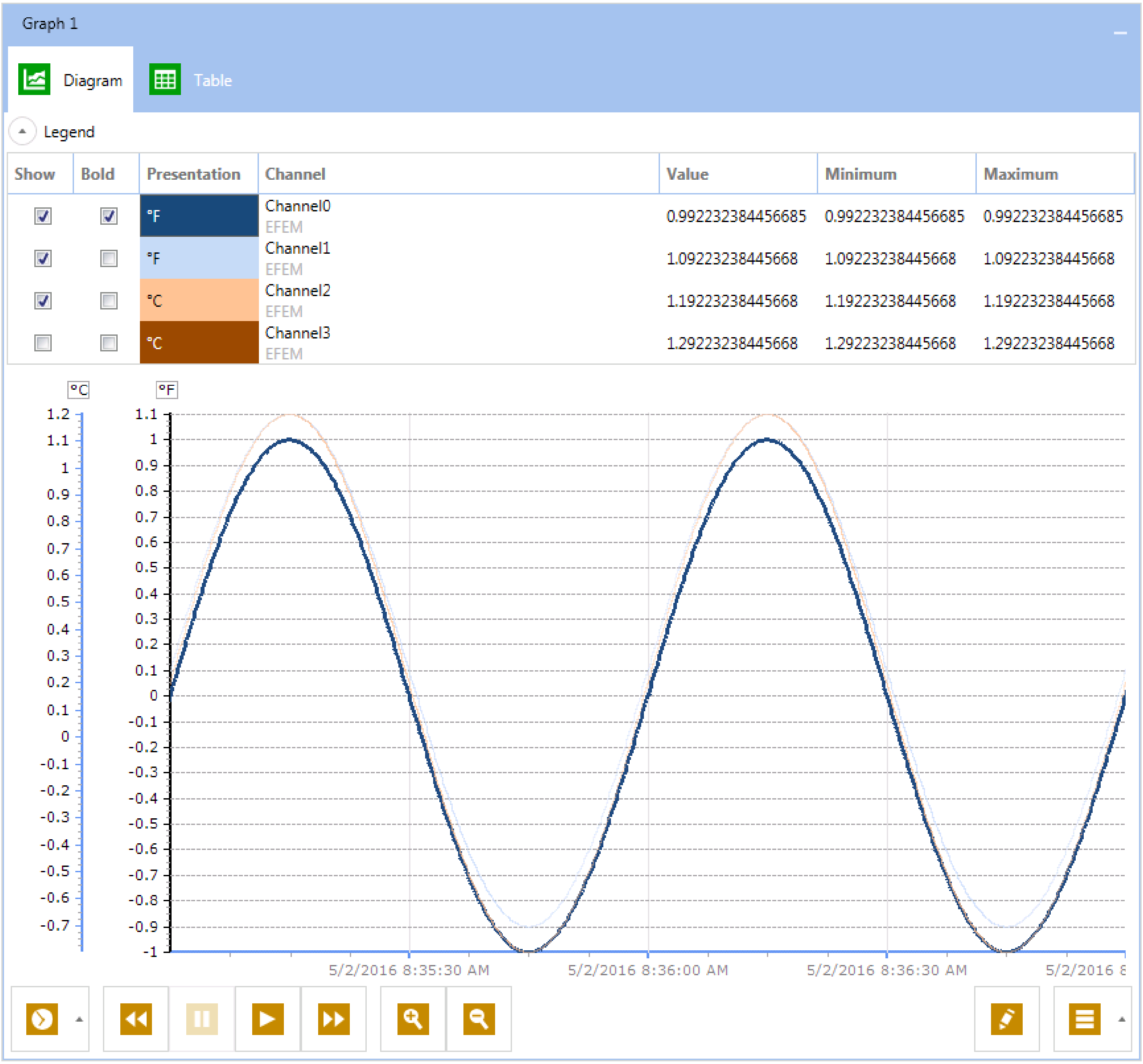


Figure 58 - Diagram view

Legend

You can adapt the visualization as well as get important information about the current process values by the legend in the upper area of the Diagram View. Press the symbol in the title bar of the legend to open the legend. Press the symbol to collapse the opened legend to create a space.

The legend contains a row for each channel assigned to the graph with the following information.

- Activate the check box of the column **Show** to display the channel in the diagram. Deactivate the check box to hide the channel.

- Activate the check box **Bold** to highlight the curve of the channel in the diagram. Furthermore, the horizontal grid lines will be shown.
- The column **Presentation** shows the colors and the y axis assigned to the curves in the diagram.
- The column **Channel** shows the name of the channel. You see the name of the module below the channel name if the channel is assigned to a module.
- The column **Value** displays the current process value of the channel.
- The columns **Minimum** and **Maximum** contains the minimum value or the maximum value of the process value in the selected range.

Diagram

The diagram represents the process values of the individual channels in a temporary trend. You see the assignment of the curves to the channels in the legend. All axes are drawn which are required for the shown channels. If different channels are assigned to an axis then the axis is shown while one of the channels is presented as a curve.

The shown area matches the presetting for the graph. By default, you see the Online Data, whereas the current value is shown on the right side. You can move the curves to the left or to the right to see the elapsed time. For it, press and hold the left mouse button at the graph. Move the mouse to the right or to the left to select the required time span and release the mouse button. The Online Data of the diagram is deactivated. Move the curve to the right to activate the Online Data again or press the button **Start** of the function bar.

Function bar

Adapt the presentation of the curves to your requirements by the function bar below the diagram.

Set time span

Press the button to set the shown time span. Select the required start time and end time. Use the calender tool for an easier selection or enter date and time directly. By default, the presented time span is selected. press the button **Select** to confirm.

Navigating in time

Navigate within the diagram by the button group Trend **Navigation** .

- Press the button **Rewind** to go backwards in the trend. For it, the Online Data are stopped. One rewind click corresponds to 50% of the presented time span.
- Press the button **Pause** to stop the Online Data. The button is only available if the Online Data is activated.

- Press the button **Start** to activate the Online Data. The button is only available if the Online Data is deactivated.

- Press the button **Fast forward** to go forward in the trend. The button is deactivated if the Online Data is activated. When you reach the current time, the button will be deactivated and the Online Data will be activated.

Zoom graph

Enlarge or reduce the time span of the graph by the button group **Zoom** .

Edit graph

Press the button **Edit** to open the edit dialog of the graph. See Edit a Trend for detailed information.

Save screenshot

Press the button **Options** and select **Screenshot...** to save a screenshot of the current presentation. A Wizard will open to guide you through the process.

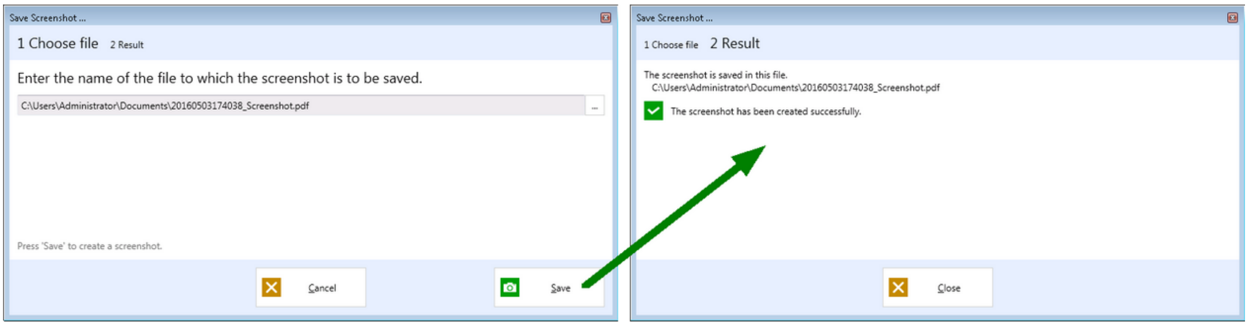


Figure 59 - Wizard for saving screenshots

1. Enter the file name for the screenshot. Use the directory selection to choose a path and define the file name. By default, the user directory for documents is chosen. Press the button **Save** to archive the screenshot. Press the button **Cancel** to abort the Wizard.

2. You get information about the screenshot was successfully saved. If necessary, you get an error message. Press the button **Close** to quit the wizard.

Export trend data

You can export only Trend data when the Online Data is deactivated. Select a time span the Trend Data will be exported for. Press the button **Pause** if necessary to deactivate the Online Data.

Press the button **Options**  and select **Export...** to export the Trend Data. The Wizard is open to guide you through the process.

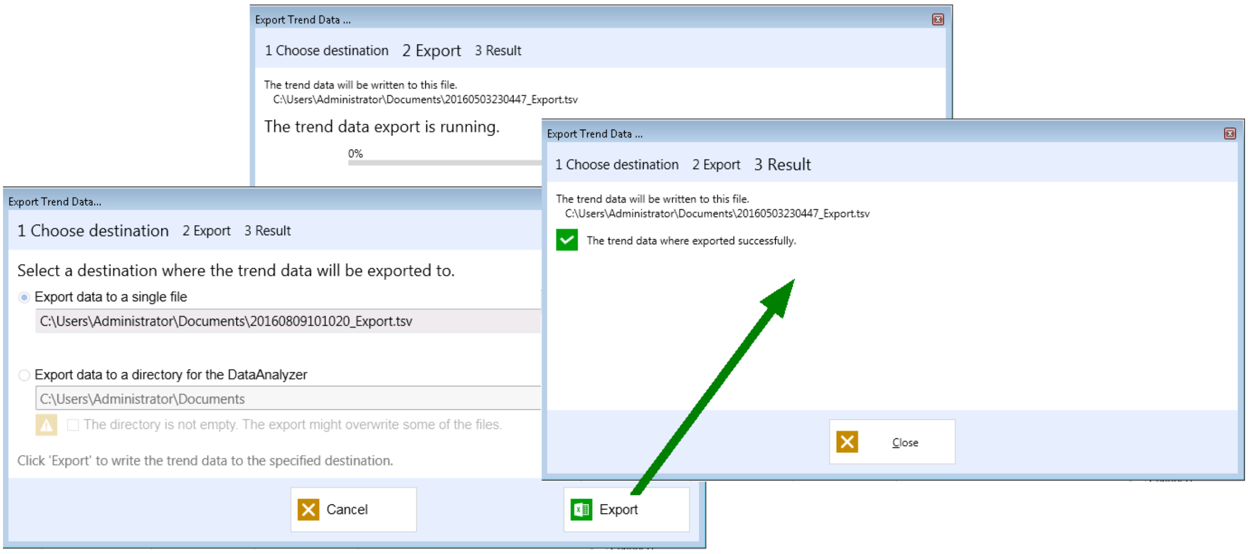


Figure 60 - Wizard for exporting the Trend Data

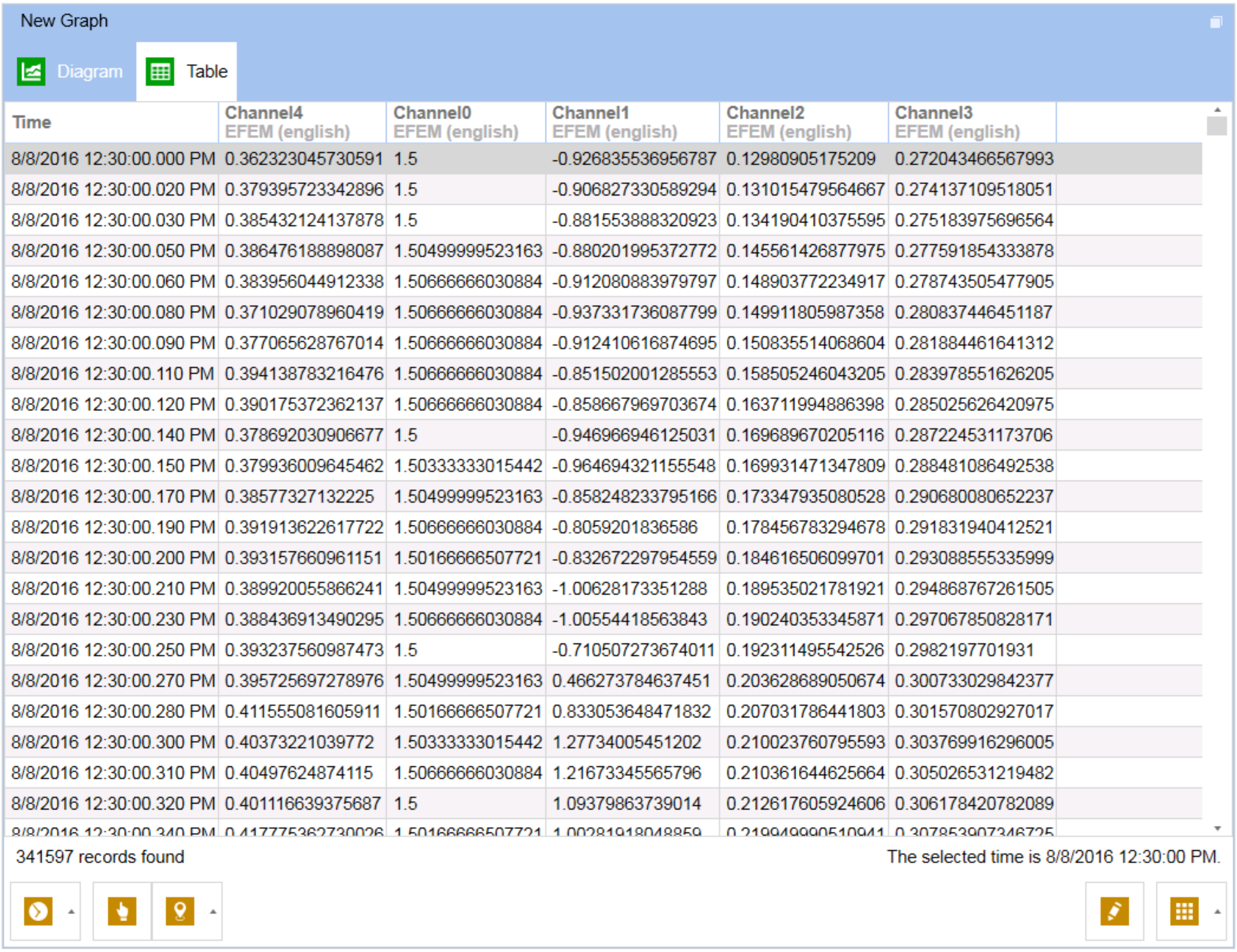
1. Select if you want to export the trend data to a single file (for example to process the data in a spreadsheet) or in a directory for the DataAnalyzer.
   Activate the option **Export to a single file** to export the trend data in a TSV file. Select the directory and define the name of file. By default, the user directory for documents is chosen.
   Activate the option **Export data to a directory for the DataAanalyzer** to save the data for the DataAnalyzer. Select the directory. A directory tree is created for each year and each month in the selected directory. The data of one day will be exported in a text file and arranged in the directory tree. The text file is named as <Module name> <YYYY-MM-DD>.txt. If the directory is not empty, be sure to do not overwrite existing data. If you are sure, activate the check box below.
   Press the button **Export** to go to the next step. Press the button **Cancel** to quit the wizard and do not save the trend data.

2. The data will be exported automatically. Press the button **Cancel** to abort the export. The data are not saved and the wizard is closed.

3. You get information about the trend data was successfully exported. If necessary you get an error message. Press the button **Close** to quit the wizard.

6.2.2 Table View

The **Table View** shows the process data of the channels of the graph in a table. The following figure shows the Table View by an example.



Time	Channel4 EFEM (english)	Channel0 EFEM (english)	Channel1 EFEM (english)	Channel2 EFEM (english)	Channel3 EFEM (english)
8/8/2016 12:30:00.000 PM	0.362323045730591	1.5	-0.926835536956787	0.12980905175209	0.272043466567993
8/8/2016 12:30:00.020 PM	0.379395723342896	1.5	-0.906827330589294	0.131015479564667	0.274137109518051
8/8/2016 12:30:00.030 PM	0.385432124137878	1.5	-0.881553888320923	0.134190410375595	0.275183975696564
8/8/2016 12:30:00.050 PM	0.386476188898087	1.50499999523163	-0.880201995372772	0.145561426877975	0.277591854333878
8/8/2016 12:30:00.060 PM	0.383956044912338	1.50666666030884	-0.912080883979797	0.148903772234917	0.278743505477905
8/8/2016 12:30:00.080 PM	0.371029078960419	1.50666666030884	-0.937331736087799	0.149911805987358	0.280837446451187
8/8/2016 12:30:00.090 PM	0.377065628767014	1.50666666030884	-0.912410616874695	0.150835514068604	0.281884461641312
8/8/2016 12:30:00.110 PM	0.394138783216476	1.50666666030884	-0.851502001285553	0.158505246043205	0.283978551626205
8/8/2016 12:30:00.120 PM	0.390175372362137	1.50666666030884	-0.858667969703674	0.163711994886398	0.285025626420975
8/8/2016 12:30:00.140 PM	0.378692030906677	1.5	-0.946966946125031	0.169689670205116	0.287224531173706
8/8/2016 12:30:00.150 PM	0.379936009645462	1.50333333015442	-0.964694321155548	0.169931471347809	0.288481086492538
8/8/2016 12:30:00.170 PM	0.38577327132225	1.50499999523163	-0.858248233795166	0.173347935080528	0.290680080652237
8/8/2016 12:30:00.190 PM	0.391913622617722	1.50666666030884	-0.8059201836586	0.178456783294678	0.291831940412521
8/8/2016 12:30:00.200 PM	0.393157660961151	1.50166666507721	-0.832672297954559	0.184616506099701	0.293088555335999
8/8/2016 12:30:00.210 PM	0.389920055866241	1.50499999523163	-1.00628173351288	0.189535021781921	0.294868767261505
8/8/2016 12:30:00.230 PM	0.388436913490295	1.50666666030884	-1.00554418563843	0.190240353345871	0.297067850828171
8/8/2016 12:30:00.250 PM	0.393237560987473	1.5	-0.710507273674011	0.192311495542526	0.2982197701931
8/8/2016 12:30:00.270 PM	0.395725697278976	1.50499999523163	0.466273784637451	0.203628689050674	0.300733029842377
8/8/2016 12:30:00.280 PM	0.411555081605911	1.50166666507721	0.833053648471832	0.207031786441803	0.301570802927017
8/8/2016 12:30:00.300 PM	0.40373221039772	1.50333333015442	1.27734005451202	0.210023760795593	0.303769916296005
8/8/2016 12:30:00.310 PM	0.40497624874115	1.50666666030884	1.21673345565796	0.210361644625664	0.305026531219482
8/8/2016 12:30:00.320 PM	0.401116639375687	1.5	1.09379863739014	0.212617605924606	0.306178420782089
8/8/2016 12:30:00.340 PM	0.417775362730026	1.50166666507721	1.00281918048859	0.219949990510941	0.307853907346725

Figure 61 - Table View

Table

When you switch from the **Diagram View** to the **Table View**, the time span presented in the Diagram View is mapped to the Table View. The table shows the process value of the individual channels for each measurement time. The time as well as the individual channels are arranged in columns. If two entries differ only by the measurement time, the entry with the first measurement time will be presented in the table. Therefore, one process value at least has to be changed since the previous measurement time to add a another row in the table.

You can sort the table by a column. For it, click on the header of the required column. A small arrow illustrates the sort order. Click again on the header to change the sort order.

Status bar

The **Status Bar** presents the number of data sets in the selected time span on the left side. The right side of the status bar shows the time of the currently selected data set.

Function bar

Adapt the table to your requirements by the function bar below the table.

Set time span

Press the button to set the shown time span. Select the required start time and end time. Use the calender tool for an easier selection or enter date and time directly. By default, the presented time span is selected. press the button **Select** to confirm.

Goto a time

Press the button to jump to a time in a table. Use the calender tool for an easier selection or enter date and time directly. By default, the presented time span is selected. Press the button **Goto** to confirm the selection.

Edit graph

Press the button **Edit** to open the edit dialog of the graph. See Edit a Trend for detailed information.

Export trend data

You can export only Trend data when the Online Data is deactivated. Select a time span the Trend Data will be exported for. Press the button **Pause** if necessary to deactivate the Online Data.

Press the button **Options** and select **Export...** to export the Trend Data. The Wizard will open to guide you through the process.

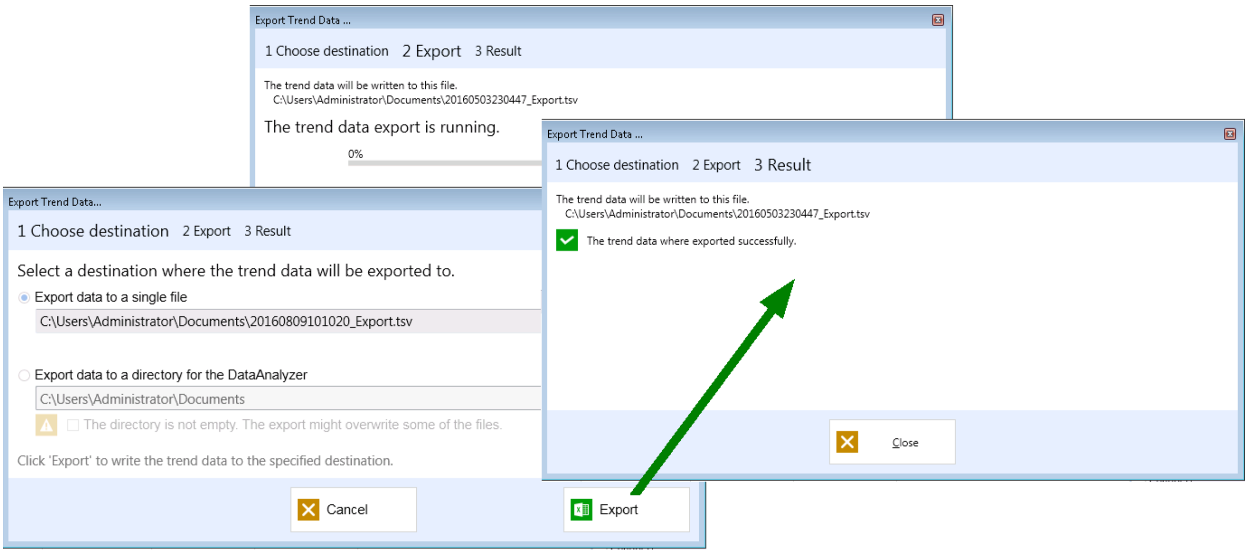


Figure 62 - Wizard for exporting the Trend Data

1. Select if you want to export the trend data to a single file (for example to process the data in a spreadsheet) or in a directory for the DataAnalyzer.
   Activate the option **Export to a single file** to export the trend data in a TSV file. Select the directory and define the name of file. By default, the user directory for documents is chosen.
   Activate the option **Export data to a directory for the DataAanalyzer** to save the data for the DataAnalyzer. Select the directory. A directory tree is created for each year and each month in the selected directory. The data of one day will be exported in a text file and arranged in the directory tree. The text file is named as <Module name> <YYYY-MM-DD>.txt. If the directory is not empty, be sure to do not overwrite existing data. If you are sure, activate the check box below.
   Press the button **Export** to go to the next step. Press the button **Cancel** to quit the wizard and do not save the trend data.

2. The data will be exported automatically. You can finish the export by the button **Cancel**. The data are not saved and the wizard is closed. Please note that the exported files contains all measurement times even if the process values of the selected channels are unchanged.

3. You get information about the trend data was successfully exported. If necessary you get an error message. Press the button **Close** to quit the wizard.

6.3 Edit a Trend

Press the button in the schema management and select the graph you want to edit.

Press the button in the Diagram View or in the Table View to edit the currently shown graph.

The configuration of the graph is shown in the following figure.

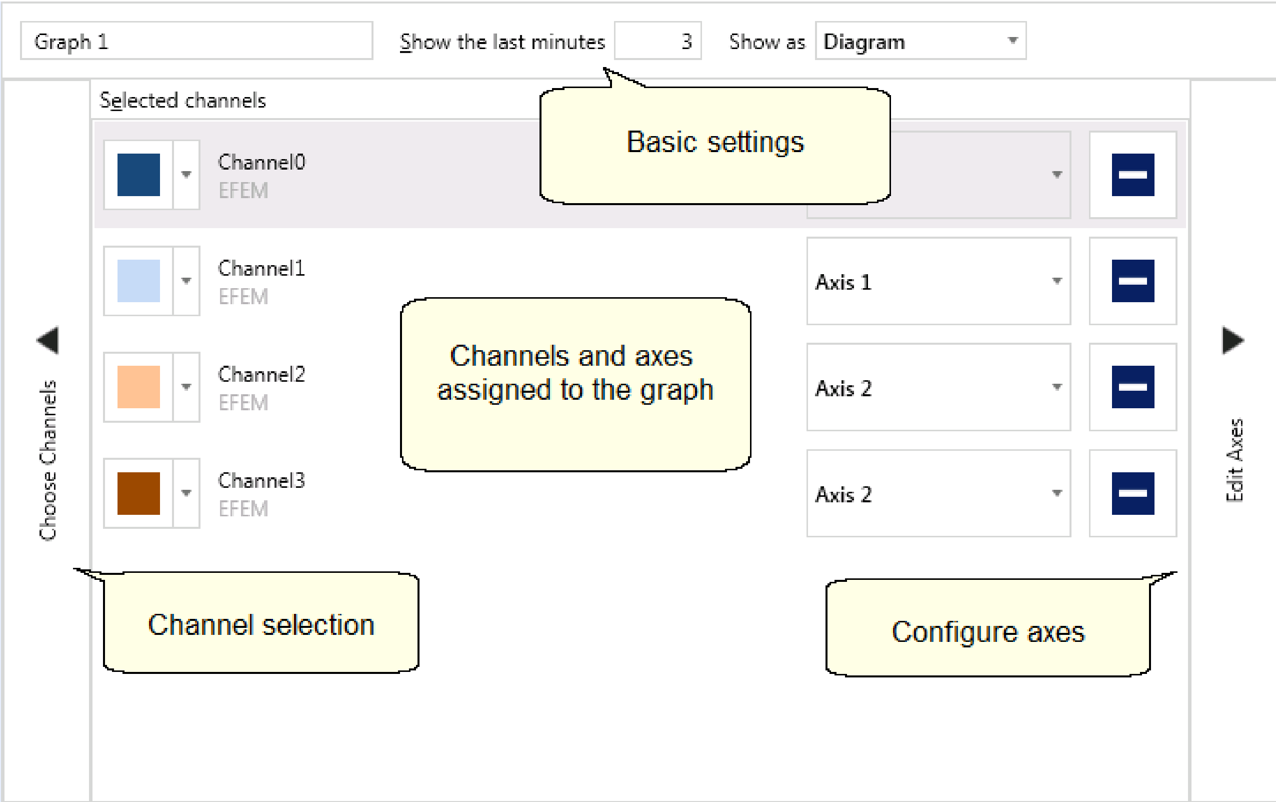


Figure 63 - Edit graph

Change basic settings

Change the name of a graph by entering the required name in the input field. Consider, the name of the graph is necessary and has to be unique. If the name does not fulfill the requirement, the input field shows an error message. Please enter an other name for the diagram.

The input field **Show the last minutes** contains the time span for the Online Data. The Online Data of the graph will be set from the current time to the selected time in the past. Modify the number of minutes if desired. You can change the value individually while the graph is shown.

Select in the combo box **Show as** the view the graph will be open with. The Table View and the Diagram View are available. You can switch between them anytime.

Configure selected channels and assign axes

Configure the channels assigned to the graph by the list. Each row contains a channel. The name of the channel will be provided by the system and cannot be changed.

Change the color of the trend curve: Open the color selection to change the color of the trend curve in the diagram view.

Select assignment of an axis: Open the combo box of the axis name to assign the channel to another axis. All available axes of the chart are shown. Select the required axes.

Remove channels: Press the button to remove channels of the graph.

Select channels

Press the expander **Select channels**. The List of selected channels is moved to the right and the list of available channels will be open. See Add Channels for detailed information.

Edit axes

Press the expander **Edit axes** to edit the Y-axes of the schema. The list of the selected channels is moved to the left and the axes configuration will be opened. See Configure Axes for detailed information.

6.3.1 Add Channels

Press the Expander **Choose Channels** to add channels to the graph or to delete already assigned channels. The list of the selected channels is moved to the right and the list of the available channels will be open as shown in the following figure.

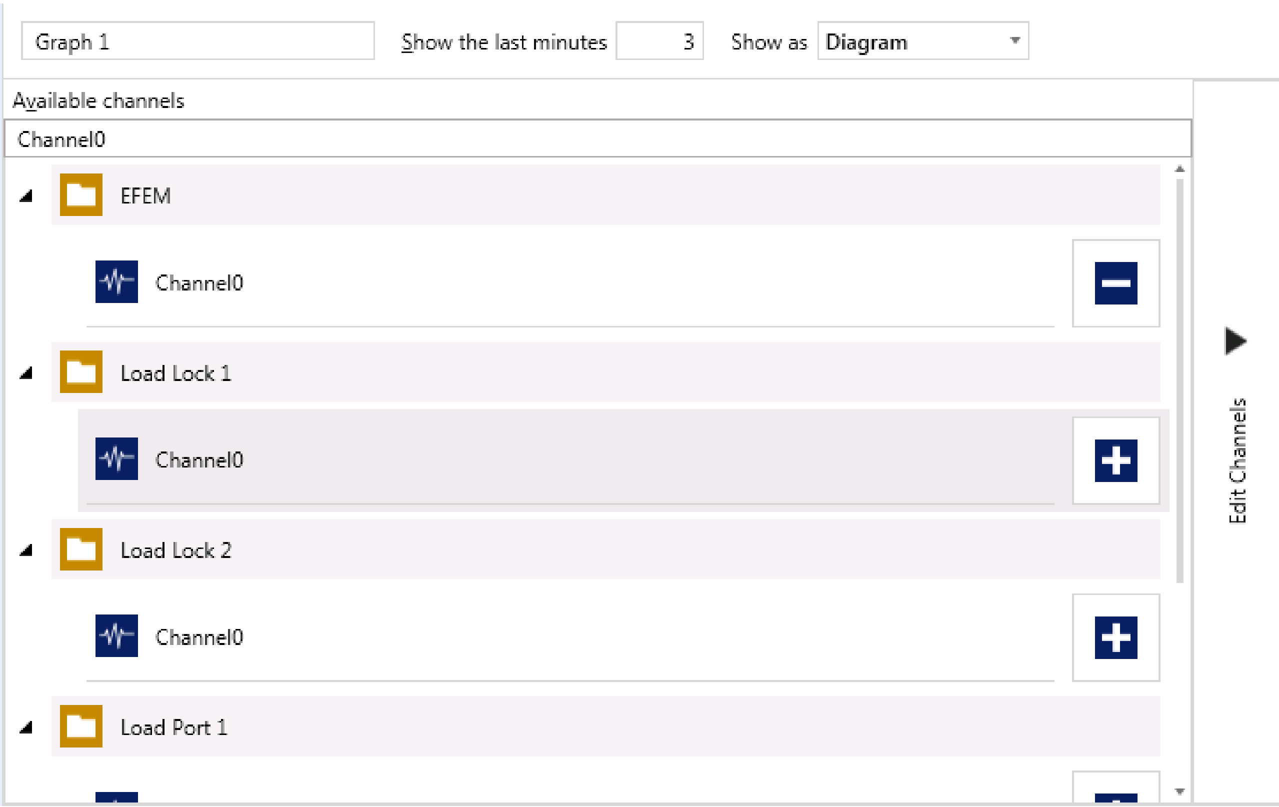


Figure 64 - List of available channels.

The channels are arranged in the hierarchy below their modules. Global defined channels are not assigned to a module. Modules are labeled by the symbol . Channels are labeled by the symbol . Search for a channel when there are to much channels. Enter any part of the channel name in the search box to filter the list.

Each channel has a button on the right side to assign the channel to the graph or remove the channel of the graph.

Press the button to add the selected channel to the graph. The button is not available if the channel is already assigned to the graph.

Press the button to remove the channel of the graph. The button is only available if the channel is already assigned to the graph.

Press the expander **Edit channels** to switch back to the view of the selected channels.

6.3.2 Configure Axes

Press the Expander **Edit Axes** to configure the axes of the graph. The list of available channels is moved to the left side and the axes configuration is opened as shown in the following figure.

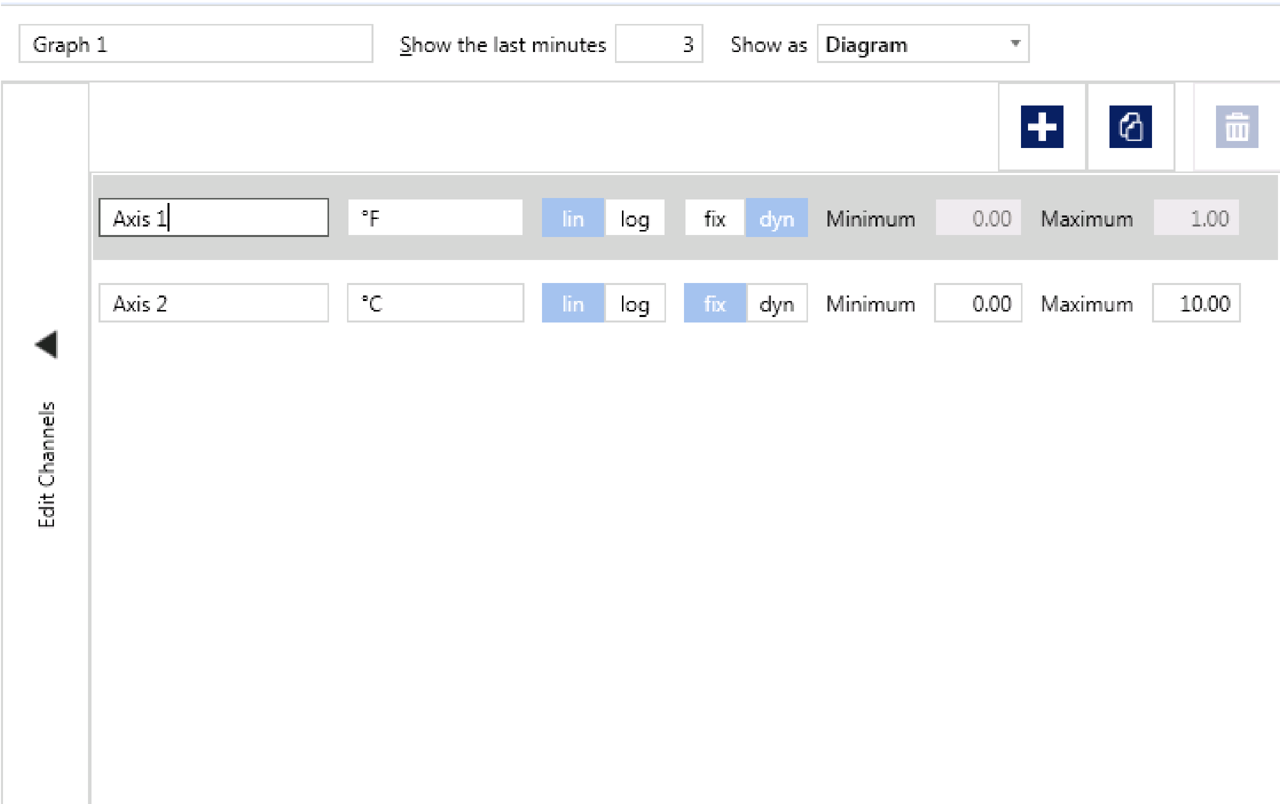


Figure 65 - Edit axes

The axes configuration shows all axes in a list configured in the graph. Press the corresponding action button to add, copy or delete axes.

Press the expander **Edit channels** to return to the view of selected channels.

Manage axes

Press the button **Add axis** to add a new axis to the graph. The new axis is added to the list of available axes with a default name.

Press the button **Copy axis** to add a copy of the selected axis. The button is only activated if an axis is selected in the list. The copied axis is added to the list of available axes with a default name.

Press the button **Delete axis** to remove the selected axis. The button is only activated if an axis is selected in the list and the axis is not used anymore. Please consider no confirmation is necessary to delete the axis.

Configure axes

Name of axis: Enter a new name in the input field. Please consider the name of the axis is required and has to be unique. The input field shows an error message if the requirement is not fulfilled. Please enter an other name for the axis.

Unit: Enter the required unit in the input field.

Scaling of the axis: Select the scaling for the axis. Press the switch **lin** for a linear scaling. Press the switch **log** for a logarithmic scaling of the axis.

Range of values of the axis: Press the switch **dyn** for a dynamic range of values. The limits of the range corresponds to the lowest and highest of the values to show. The input fields for the minimum and maximum are deactivated.
Press the switch **fix** for a fix range of values for the axis. Enter a minimum and a maximum into the corresponding input fields to define the limits of the range.

7

Components

7 Components

The control system of the machine is hierarchically structured into modules. Each module can include several components. Usually, a component has its own device parameters and manual operations. Some components offer an own online trend window.

7.1 Components screen

The component screen displays all components of the machine in a tree structure. It allows opening manual operation and configuration dialogs of each component. Choose between two available views on the Components screen.

- The *Structure view* displays components by the module assigned to.
- The *Type view* displays components sorted by their component type.

A table with the parameters of the selected component is shown below the component tree if activated.

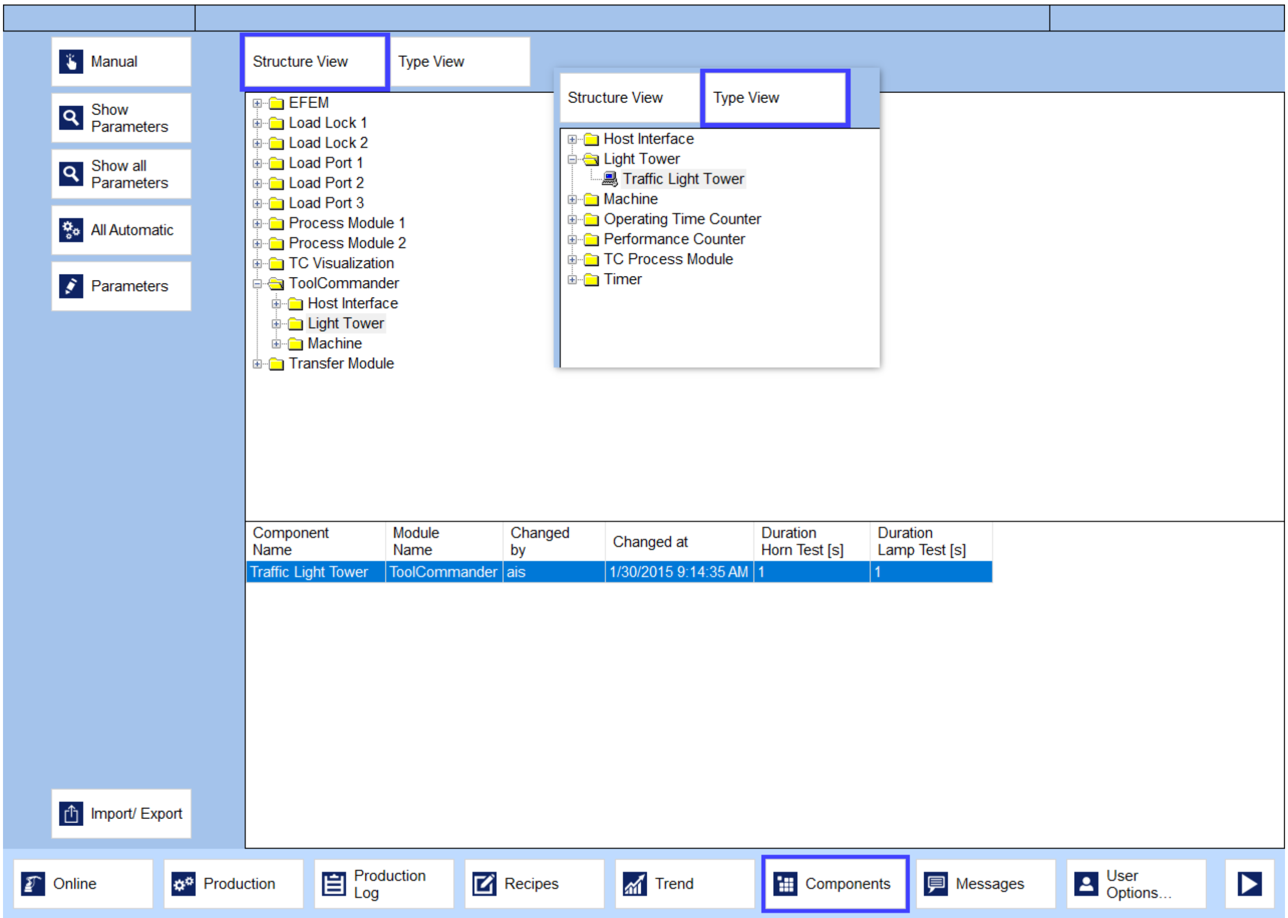


Figure 66 - Components – Structure View

The Components screen provides the following functions on the left.

Press the button **Open** to open the manual operation dialog of the selected component. The button is disabled if no component is selected. For further information see Manual Operations Dialog.

Press the button **Show params** to open an overview of the parameters for the selected component/ component type/ module. For further information see Device parameters. You can print the view if a printer is installed.

Press the button **Show all params** to open an overview of all the parameters for the entire machine. For further information see Device parameters. You can print the view if a printer is installed.

Press the button **All auto** to set all components to automatic mode.

Press the button **Parameters** to open the parameter dialog of the selected component. For further information see Device parameters.

7.1.1 Manual Operations Dialog

Open the operation dialog by one of the two possible ways:

- Press the button **Manual** in the component screen.
- Select the symbol of a component in the online visualization. Press the left mouse button in the status of the component to open the operation dialog.

Operation modes

A component works always in one of the following component modes.

Production: The **ToolCommander** (by the module sequencer) controls the component. Recipe parameters and recipe commands operate the component. You cannot use the manual operation dialog in the mode **Production**.

Maintenance: You control the component manually in the mode **Maintenance**. Recipe parameters and recipe commands do not influence the component.
You can switch the component to **Maintenance** mode only if the assigned module is always in manual mode. This mechanism avoids problems between manual mode and running recipes. The system shows a warning message if the component is switching to manual mode.

Disabled: The component is switched off and can neither be controlled by the **ToolCommander** nor the user.
You can switch the component to **Disabled** mode only if the assigned module is always in **Maintenance** mode. This mechanism avoids problems between manual mode and running recipes. The system shows a warning message if the component is switching to off mode.

The operation mode of a component is represented by one of the following symbols in component tree.

Production   Maintenance   Disabled

Usage

The following figure shows an example of a manual operation dialog.

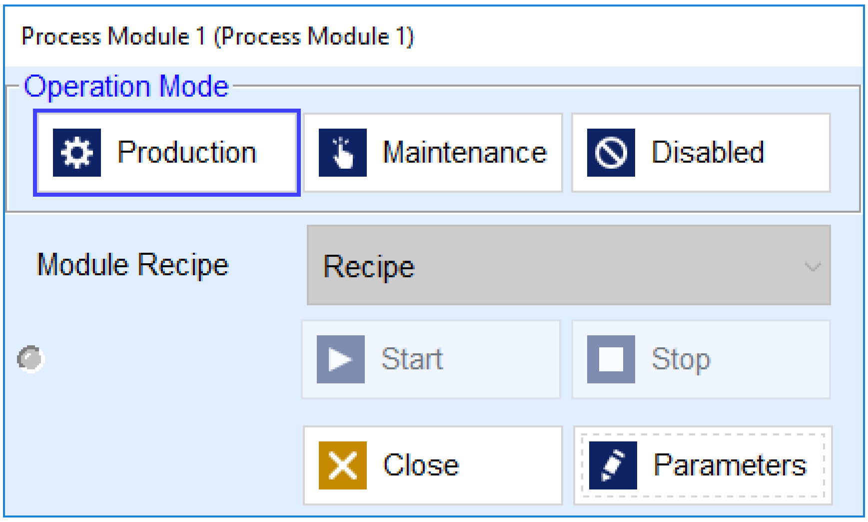


Figure 67 - Example of a manual operation Mode

Switching the operation mode

Press the button **Production**, **Maintenance** or **Disabled** to switch directly to the required operation mode. The current operation mode is blue highlighted. The message line shows the switching of operation mode of the component if the operation mode is now **Maintenance** or **Disabled**.

Manual mode

You can operate the component functions and set points manually in **Maintenance** mode. The functions depend on the selected component. The functions can be used only if the operation mode is set to **Maintenance**.

Access authorization

You require the permission Manual Operation.

Parameters

Press the button **Parameters** to display the parameters of the selected component. If the component does not have parameters, the button is disabled. The parameter are shown on an additional dialog. For further information see Device parameters.

Close

Press the button **Close** to finish the Manual operations dialog. The system notifies you if the component is running in **Maintenance** or in **Disabled** mode. Press the button **Yes** to leave the mode and close the Manual operations dialog. Press the button **No** to return to the Manual operations dialog.

7.1.2 Parameter overview

Exported configuration files (Params.xml or Params.txt in the Database folder) can be used to compare the actual configuration with saved configuration files.

Use cases

- Comparison of components of system A to components of system B
- Comparison of components of process module 1 to components of process module 2

Press the button **Back** to return to the component screen.

7.1.3 Device parameters

Access authorization

Device parameters can only be changed by users who have the right "Change parameters".

Device parameters are a set of values defined once for a component defined and always valid. Components operates by the values independently of the current running recipe.

Note

The other parameter type, recipe parameters, is defined for each module recipe individually.

Device parameters influence the properties of a component. On the one hand there are specific settings of the hardware like connection names or features. On the other hand the parameter define the control behavior of a component like trigger values and timeouts.

NOTICE

The parameters in the parameter window must correspond to the hardware installed in the system.
Wrong parameter values can cause damage of units and hardware.

The components have different configuration dialogs. The following figure shows an example.

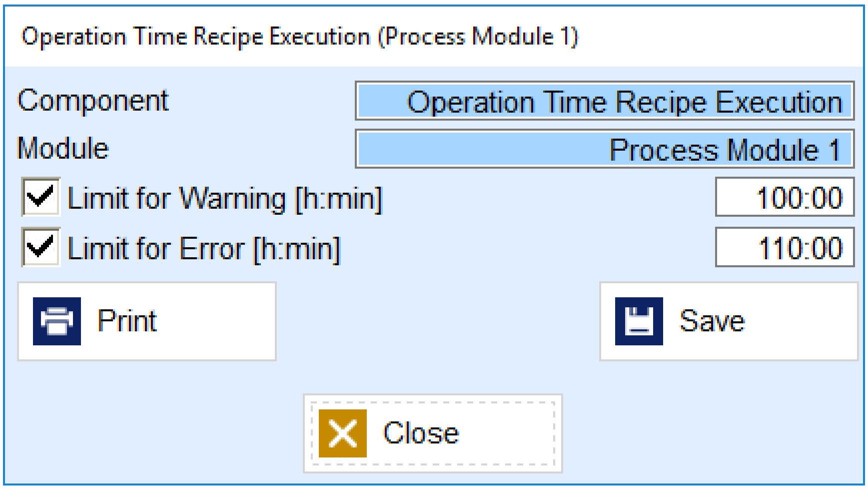


Figure 68 - Example of a Parameter Dialog

Press the button **Save** to save changes and to apply.
Press the button **Close** to finish the dialog. The system notifies you if there are pending changes. Press the button **Yes** to acknowledge, discard changes and close the dialog. Otherwise, press the button **No** to return to the configuration dialog.

Press the button **Print** to print an hard copy of the parameters dialog. The button is only visible if the printing is activated or a printer is installed.

7.2 Performance Counter

This component is used for watching the free disk space, CPU usage and free memory on a PC.

7.2.1 Parameters

Select the component **Performance Counter**. Open the Parameters dialog. For it, press the button **Parameters** on the Component screen. You can also open the Parameters dialog directly in the manual operation dialog of the component. Press the button **Parameters** in the manual operation dialog.

Following parameter settings are available:

Enable: Activate the check box to activate the component to monitor. Deactivate the check box to deactivate the component to monitor.

System

The text box **Machine Name / IP Address** (string [0...255])) displays the name of the machine to be monitored. By default, the character . (dot) is used, indicating the local PC. Further valid values are computer name, localhost and 127.0.0.1.

The setting **Interval [min:s]** defines the cycle time for checking disk space, CPU usage and free memory on the monitored system.

Disk drive

You can configure up to 2 disk drives to be monitored.

The text box **Disk drive** (single character) shows the physical or logical drive letter of the monitored disk drive. If you do not enter a value into this text box, the disc monitoring is disabled.

The data field **Warning Limit [MB]** (integer) shows the warning limit of the available disk space. When the free disk space falls below the warning limit, a warning message is created.

The data field **Error Limit [MB]** (integer) shows the error limit for the available disk space. When the available disk space falls below the error limit, the system creates an error message.

Memory

The data field **Warning Limit [MB]** (integer) shows the warning limit for the available memory space. When the free memory space falls below the warning limit, a warning message is created.

The data field **Error Limit [MB]** (integer) shows the error limit for the available memory space. When the available memory space falls below the error limit, the system creates an error message.

CPU

The data field **Warning Limit [%]** (Range [0...100]) shows the warning limit for the CPU usage. When CPU usage exceeds the warning limit for longer than the tolerance time a warning message is created.

The data field **Error Limit [%]** (Range [0...100]) shows the error limit for the CPU usage. When CPU usage exceeds the error limit for longer than the tolerance time, the system creates an error message.

The data field **Tolerance Time [min:s]** defines the tolerable time for exceeding the warning and error limits of the monitored CPU.

Use the drop down list **Monitoring Type** to define if the total CPU usage or the CPU usage of the single CPU cores shall be monitored.

Print

Press the button **Print** to print an hard copy of the parameters dialog. The button is only visible if the printing is activated or a printer is installed.

Save

Press the button **Save** to save changes of the parameter settings. The button is disabled if no changes are pending.

Close

Press the button **Close** to finish the dialog. The system notifies you if there are pending changes. Press the button **Yes** to acknowledge, discard changes and close the dialog. Otherwise, press the button **No** to return to the configuration dialog.

7.2.2 Manual Operation

Select the component Performance Counter. Press the button **Manual** on the component screen to open the manual operation dialog.

The manual dialog shows only the actual values. There are no manual operations possible.

Disk Drive

The field **Empty [MB]** shows the current available disk space.

Memory

The field **Empty [MB]** shows the current available memory.

CPU

The field **Usage** shows the current CPU usage. The value Total represents the total CPU usage and the values CPU 1 … CPU n for the usage of a single CPU cores.

Press the button **Parameters** to open the parameter dialog of the component in an additional dialog.

Press the button **Close** to finish the manual operation dialog.

7.2.3 Messages

Alarm	Type	Description
Error during read/ write of database [SQL Server message]	Error	An error occurred while accessing the database.
No parameters found	Warning	For this component still no parameter values are configured.
The free disk space of the drive [drive letter] dropped below the error limit.	Error	The free disk space of the given drive is below the configured error limit.
The free disk space of the drive [drive letter] dropped below the warning limit.	Warning	The free disk space of the given drive is below the configured warning limit.

Alarm	Type	Description
The free memory dropped below the error limit.	Error	The free main memory is below the configured error limit.
The free memory dropped below the warning limit.	Warning	The free main memory is below the configured warning limit..
The CPU usage is above the error limit.	Error	The total CPU usage is longer as configured tolerance time above the error limit.
The CPU usage is above the warning limit.	Warning	The total CPU usage is longer as configured tolerance time above the warning limit.
The CPU usage for the core [number] is above the error limit.	Error	The CPU usage of the given core is longer as configured tolerance time above the error limit.
The CPU usage for the core [number] is above the warning limit.	Warning	The CPU usage of the given core is longer as configured tolerance time above the warning limit.

Table 57 - Messages of the Component Performance Counter

7.3 Stop Watch

The following pages describe the component **Stop Watch**.

7.3.1 Parameters

The component does not have any parameters.

7.3.2 Manual Operation

Select the component **Stop Watch**. Press the button **Manual** in the component screen to open the manual operations dialog of the component Stop Watch.

Operation Mode

Select the required operation mode for the component by pressing the corresponding button. The available operation modes are Production, Maintenance and Disabled. The selected operation mode is blue highlighted.

Operations

The functions are only available if the component is in Operation mode **Maintenance**.

Press the button **Start** to begin the time measurement. The button is disabled if the stop watch is always running.

Press the button **Stop** to pause the time measurement. The button is disabled if the stop watch does not running.

Press the button **Reset** to set the time of the stop watch back to 00:00 and to clear the table of intermediate times.

Actual Time

The field **Current Time** [h:min:s] shows the current time.

Intermediate times

The table **Intermediate times** shows the stored times. The column **Number** indicates the consecutive number of the intermediate times. The column **Time [s]** contains the intermediate time.

Press the button **Store** to write the actual time into the table Intermediate times. The time measurement does not stop. The button is disabled if the stop watch does not running.

Close

Press the button **Close** to finish the manual operation dialog. The system notifies you if the component is running in **Maintenance** or in **Disabled** mode. Press the button **Yes** to leave the mode and close the manual operations dialog. Press the button **No** to return to the manual operations dialog.

7.3.3 Alarms and Recovery

The following information messages are provided by this component:

Exception	Reaction, recovery
Intermediate time is saved.	none
The measurement was stopped.	none

Table 58 - Alarms and Recovery for Stop Watch

7.3.4 Interlocks

None.

8

Message System

8 Message System

The integrated message system provides the capability of recovery actions for messages. Every message has its own related recovery actions. One of these recovery actions can be selected by user. The selected action will be executed if the message is acknowledged.

Depending on the light tower configuration, alarm and warning states influence the light tower operation.

As long as there are unacknowledged or active alarms, the header line background will be displayed in red color.

If the background of the Messages button is yellow or red, a warning (yellow) or an alarm (red) is active or unacknowledged.

The button **Alarms** activates the alarm view; see chapter Alarms View. This view shows messages that are active or unacknowledged.

The button **Archive** activates the message archive view; see chapter Message Archive View.

8.1 Message Line

The message line extends across the full monitor width on top of the screen area. It always displays the latest active message.

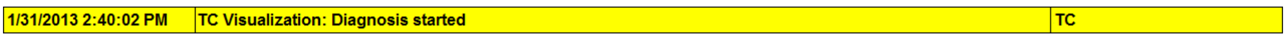


Figure 69 - Message Line

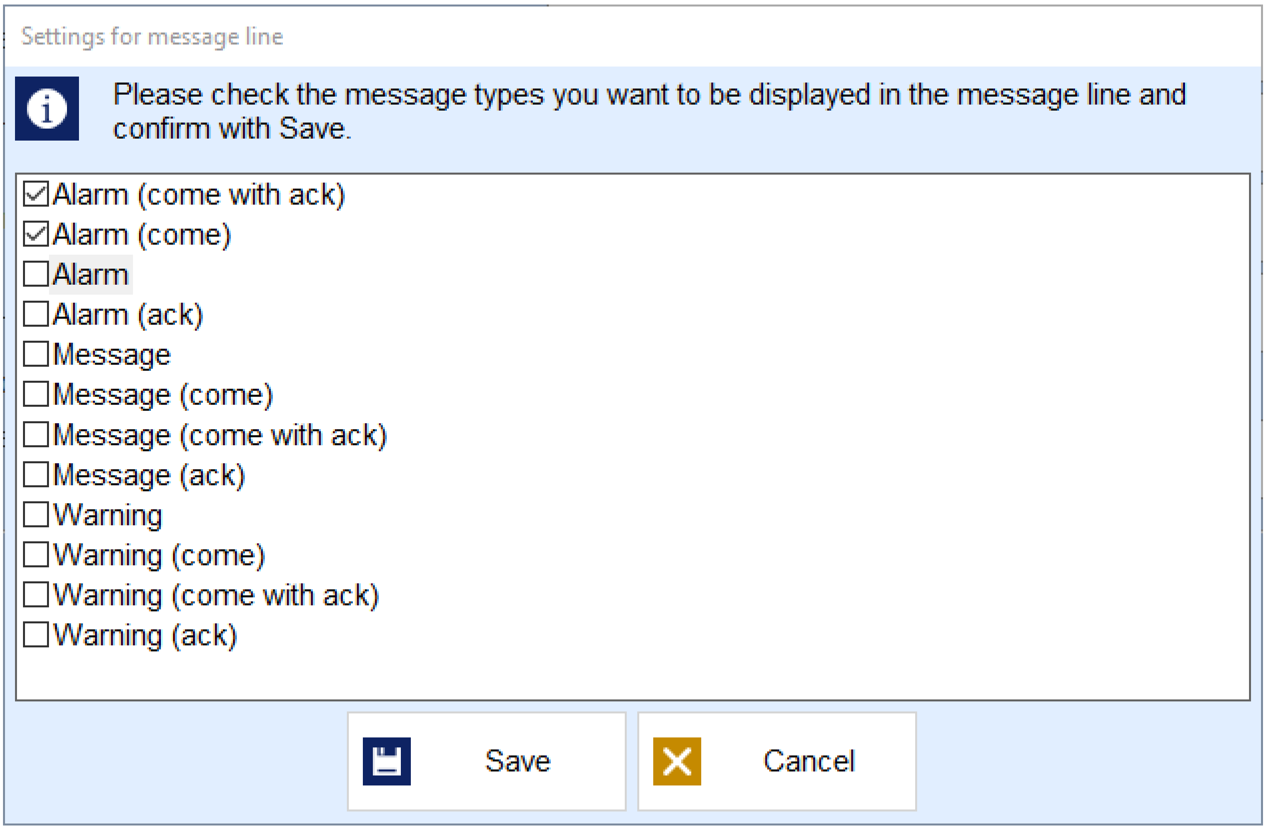


Figure 70 - Message Line Settings

On this dialog the user can set up which types of messages are being displayed in the message line. Messages of unchecked types will not be displayed in the message line. These settings don't affect the alarm list.

8.2 Alarms View

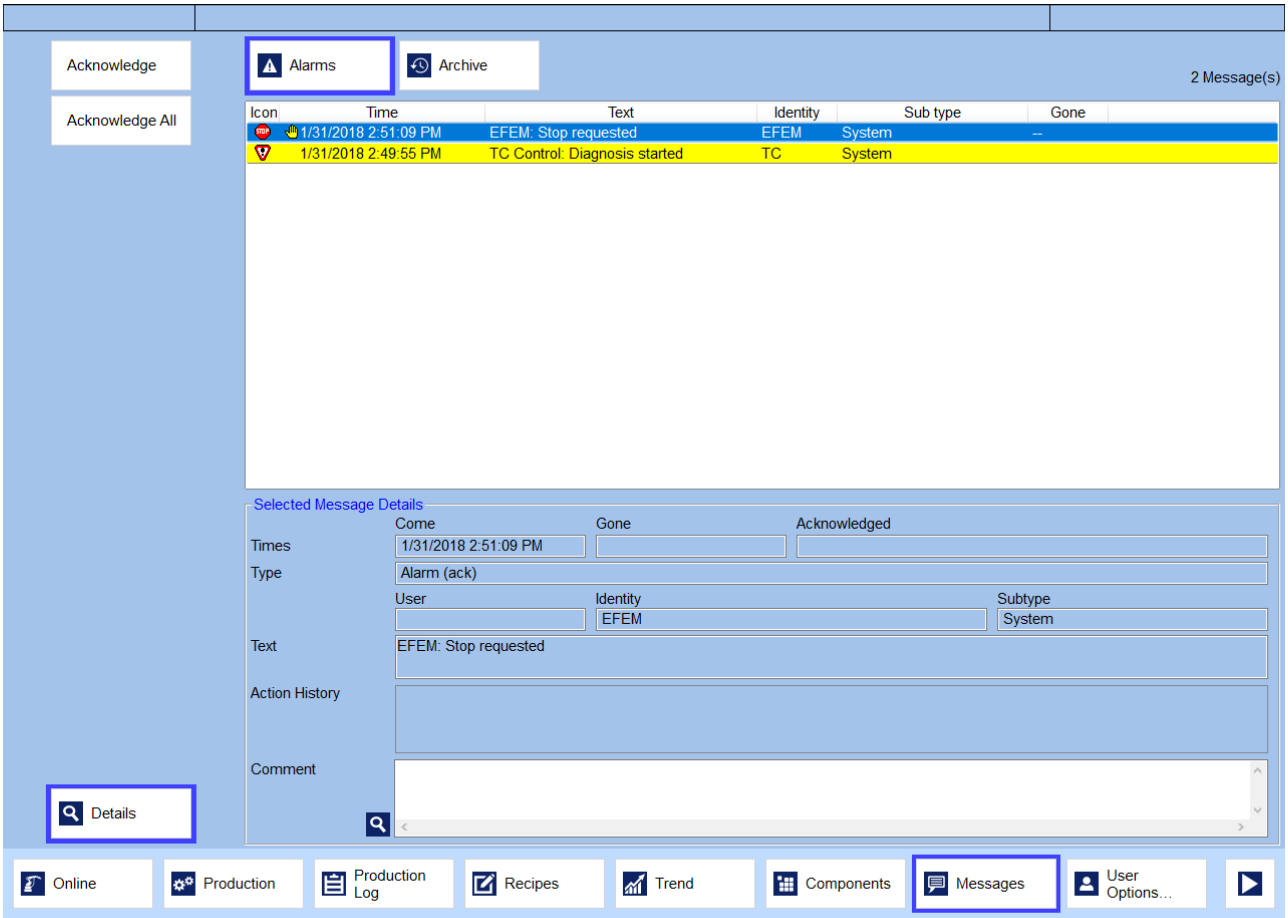


Figure 71 - Alarm List

This view delivers information about start (Time) and end (Gone) date/ time of the message, the associated error text, component type and component location. Depending on the message selected buttons will be displayed to acknowledge it and select a recovery action. If the message has no recovery action attached only *Acknowledge All* will be available.

Messages that are not active any longer and are acknowledged are automatically moved to the message archive.

Press the button **Acknowledge** to acknowledge the currently selected message.

Press the button **Acknowledge All** to acknowledge all messages.

Press the button **Details** to activate or deactivate the detail view.

8.3 Message Archive View

The system archives all messages occurring in the system (alarms, warnings, info messages) and displays them in the message archive. If a message has recovery actions, these are displayed in the detail view of the selected message.

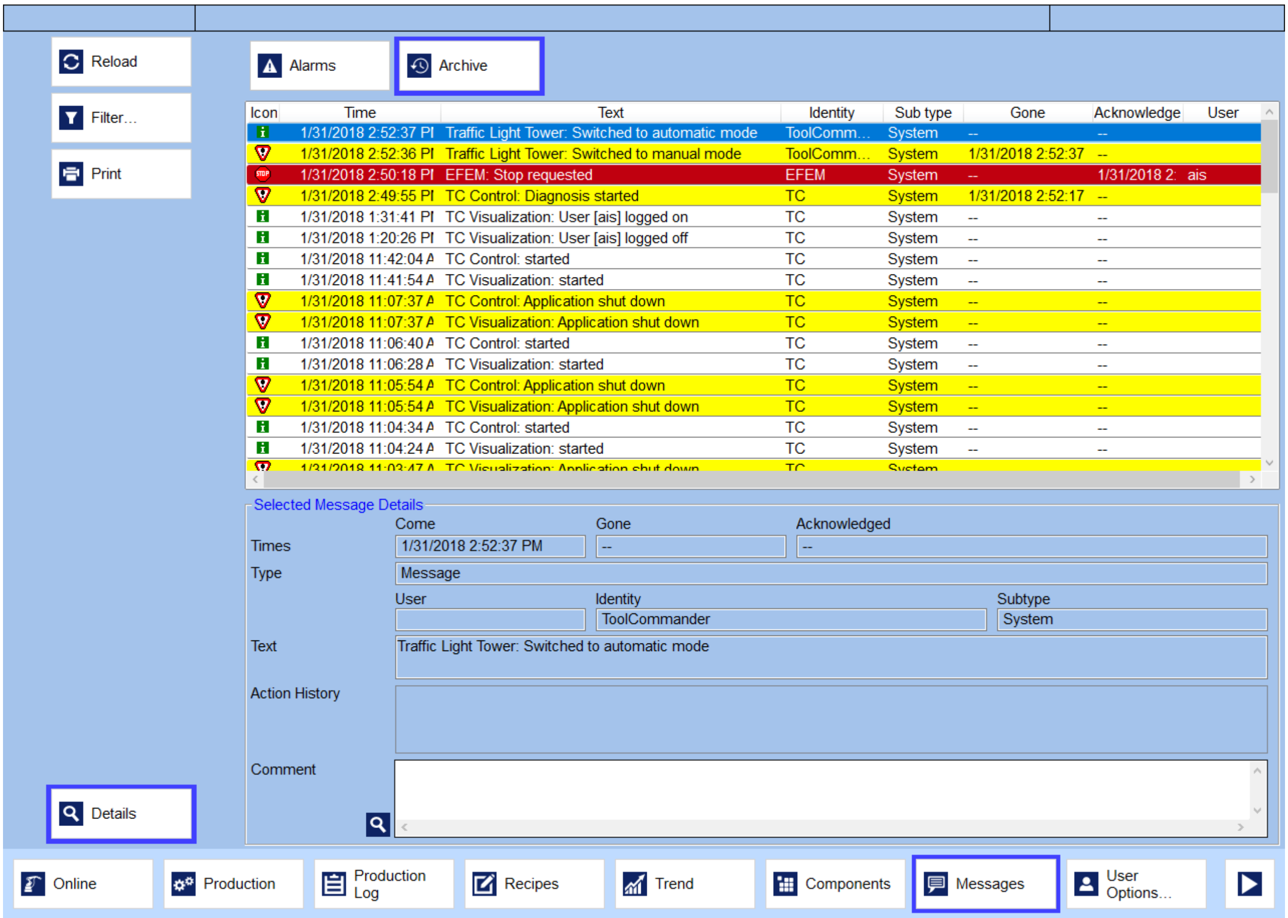


Figure 72 - Message Archive

Press the button **Reload** to reload the data according to the filter settings.

Press the button **Print** to print the list of messages. The button is hidden if no printer is installed.

The button **Details** activates or deactivates the detail view.

Press the button **Filter...** to open the dialog with the filter setting.

In this dialog you can define filter criteria to display messages in archive. By selecting *Message Types*, *Identifier* and *Sub Types* options you are able to limit the list to special kind of events. Additionally, time limits can be set to limit the number of shown messages.

8.4 Alarm List Popup

The Alarm List Popup window displays all active alarms. It displays the same information that is shown in the alarm list of the message view (see chapter Alarms View). It can be opened manually using the application menu or, if set up in the application settings (see chapter Application Settings), automatically whenever a new alarm is raised.

The benefit of this window is that it can be opened from any screen without needing to change the currently active view.

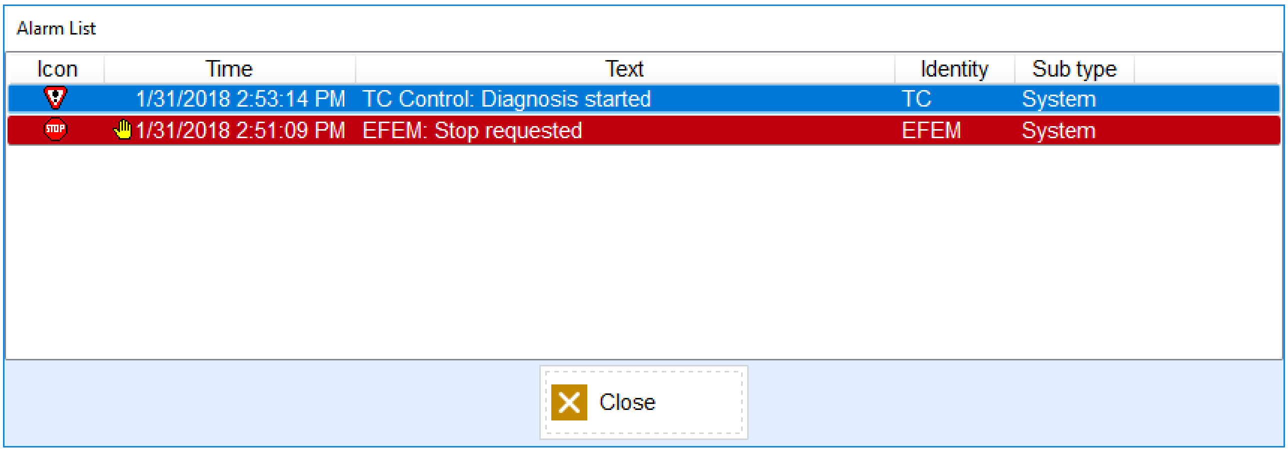


Figure 73 - Alarm List Popup

9

I/O Diagnosis

9 I/O Diagnosis

Access authorization

Only users who have the right I/O Diagnosis can access this function.

NOTICE

Using I/O diagnosis, it is possible to create dangerous machine conditions. Therefore, only authorized persons are allowed to use it.

The I/O diagnosis can be used for following tasks:

- viewing state of binary input channels, overwrite the input state with a user defined state; the control program works then as if the user defined state would be the real state.
- viewing values of analog input channels, overwrite the input value with a user defined value; the control program works then as if the user defined value would be the real value.
- viewing state of binary output channels, overwrite the output state with a user defined state; the state will be used independently of the original program state then.
- viewing values of analog output channels, overwrite the output value with a user defined value; the value will be used independently of the original program value then.

The I/O Diagnosis is a layer between I/O and the control system. This layer can be used to simulate both sides of the system.

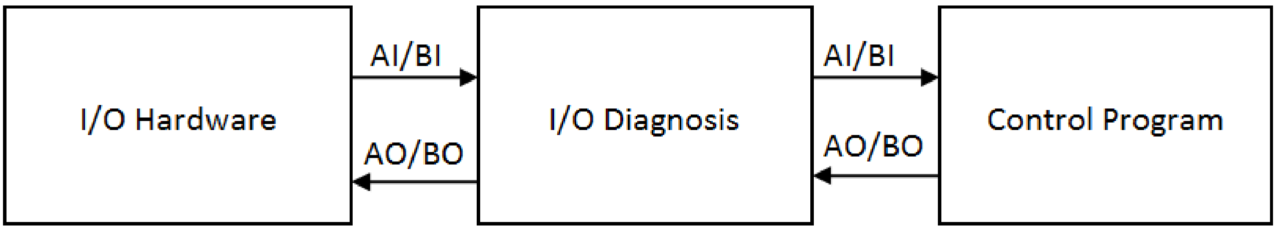


Figure 74 - I/O Diagnosis Layer

9.1 Overview

Following figure shows the I/O Diagnosis user screen.

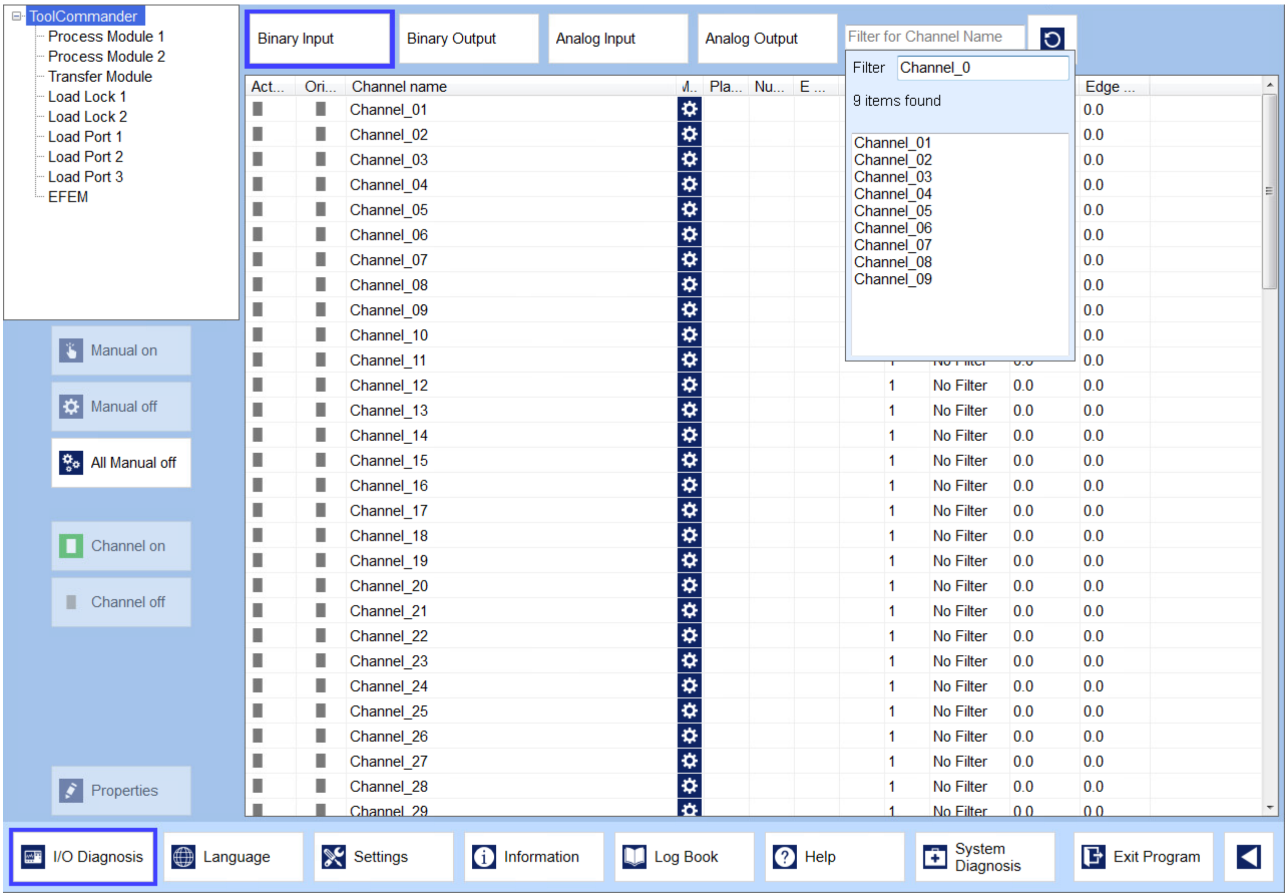


Figure 75 - I/O Diagnosis

Channels are shown in tables that are separated between the modules and the channel type (Binary Input, Binary Output, Analog Input, and Analog Output). Select a module in the module tree and the desired channel type to select a channel table.

The channels presented in the selected channel table can be filtered.
For it, key in three characters at least into the search field. The filter does not distinguish between upper and lower case (is not case sensitive).
The number of found channels as well as the name of the matching channels are shown in the expanded list below. If there are no matching channels, the list is empty and the number of found channels is set to zero.
Select a channel name in the list below by the mouse. Use the cursor keys to navigate in the list and select another channel. The currently selected channel is marked. Press the key **ENTER** to take over the channel and show it in the channel table.
if you are searching for all channels matching the search string, press the key **ENTER** after you have keyed the search string (three characters at least). The found channels are shown in the channel table.
Press the key **ESC** to leave the search field. The current filtering of the channel table is kept up.
Press the button **Reset** on the right side of the search field to remove the filter. Filtering of the channel table is revoked and the search string is removed.

Only channels of modules with activated maintenance mode can be switched to manual mode. Channels of modules in production mode are view-only.

As long as channels are in manual mode, it is not possible to switch the corresponding module from maintenance to production mode.

9.2 Binary Channels

Actual	Original	Channel name	Mode (graph.)	Place	Number	E Title	Comment	DP Address	Signal Processing Mode	Edge Up [s]	Edge Down [s]
		Test Channel						1	No Filter	0.0	0.0

Figure 76 - Binary Channels

Binary channels provide following information.

Column	Meaning	
Actual	BI: state that is reported to the control system BO: state that is transmitted to the I/O unit	green (on)
Original	BI: state that is received from the I/O unit BO: state that is generated by the control system	gray (off)
Channel name	name of the I/O channel	
Mode	diagnosis mode of the channel automatic (no simulation) manual (channel is simulated to be on or off)	
Place	user defined information	
Number	user defined information	
E Title	user defined information	
Comment	user defined information	
DP Address	user defined information: DP address of the channel; changing the value has no influence to the control system	
Signal Processing Mode	The defined signal processing mode	

Column	Meaning
Edge Up [s]	The time span of the signal processing mode for edge up in seconds
Edge Down [s]	The time span of the signal processing mode for edge down in seconds

Table 59 - Binary Channel Information

To change the user defined properties of a channel, select the channel and press **Properties**.

Access authorization

You requires the right Change Parameters to change the channel properties.

In the **signal processing** group you can determine whether the values of a channel are filtered. Press the **Debounce** button to debounce the channel values. The channel value is immediately accepted. From this time onwards any further edge is ignored for a defined time span. This filter can only used for input channels. Press the **Delay** button to delay the channel values. The channel value is accepted after a defined time span. Value changes are ignored within this time. Press the **No Filter** button to output the channel value directly. Enter a time span in seconds for the **edge up** or **edge down** to configure the selected filter.

To activate simulation mode for a selected channel press *Manual on*. Thereafter the buttons *Channel on* resp. *Channel off* can be used to change the state of the channel. The function has to be used with care. Absolutely no software interlock checks will be active on ToolCommander side when using the I/O diagnosis function.

9.3 Analog Channels

Figure 77 - Analog Channels

Analog channels provide following information.

Column	Meaning
Actual	AI: value that is reported to the control system AO: value that is transmitted to the I/O unit (ET200)
Original	AI: value that is received from the I/O unit (ET200) AO: value that is generated by the control system
Unit	unit of the I/O value

Column	Meaning
Channel name	name of the I/O channel
Mode	diagnosis mode of the channel automatic (no simulation) manual (channel value is simulated)
Place	user defined information
Number	user defined information
E Title	user defined information
Comment	user defined information
DP Address	information on DP address of the channel; changing the value has no influence to the control system
Input Minimum	value scaling: smallest value at control program side
Input Maximum	value scaling: largest value at control program side
Output Minimum	value scaling: smallest value at I/O hardware side
Output Maximum	value scaling: largest value at I/O hardware side
Accuracy	accuracy of the channel to be processed

Table 60 - Analog Channel Information

To change the user defined properties of a channel, select the channel and press *Properties*.

Access authorization

You requires the right Change Parameters to change the channel properties.

The information is almost the same as the binary channel information. There is only one important difference. The values Input Minimum, Input Maximum, Output Minimum, Output Maximum and Accuracy influence the control system. They define how to scale the hardware input value to the corresponding process value and how exactly the control system must respond to changes of the value (accuracy).

To activate simulation mode for a selected channel press *Manual on*. Thereafter the frame *Manual Value* can be used to change the diagnosis value. The simulated value must be activated using the button **Set Value**. Use this function with care. No software interlock checks will be made when using the I/O diagnosis function.

10

User and Rights Management

10 User and Rights Management

The dialog **User Options** dialog allows user log on and log off as well as user administration.

Press the button **Log On** to open the Log on dialog. See Log On/ Log Off for details.

Press the button **Log Off** to log off the current user. See Log On/ Log Off for details.

Press the button **Change Password** to change your password. See Change Password for details.

Press the button **Administrate Users** to open a dialog where users can be administrated (add or change users and user groups and the rights assignment). See User Manager for details.

10.1 Log On/ Log Off

Pressing *Log On* or clicking the header button User a window where user name and password can be entered or the user currently logged in may log off.

At one time only one user can be logged in. When using the log on function while still a user is logged in, this user will be logged off automatically.

The name of the current user is shown in the header.

Pressing the button *Log Off* causes the log off of the current user.

10.2 Change Password

Press the button **Change Password** to change the password of the current user.

The user needs to enter his/ her current password. The new password then has to be entered in **New Password** and **New Password (repeated)** to make sure there is no typo when changing the password. By clicking **Save** the new password will be stored (if the new password in both edit-frames is equal; otherwise an error message will be displayed); clicking **Cancel** closes the window without changing the users password.

10.3 User Manager

Access authorization

Only users with the right „Administrate“ or "User administration" can access user management functions.

The user manager can be opened by pressing button *Administrate Users*.

In this window you can set access rights for user groups within the system. Every user is assigned to exactly one group.

Creating new user accounts is done at run time with the visualization system. All users of a user group have identical rights; this means if an administrator changes the user rights of any group, all users within this group will get changed user rights as well.

Users and groups can be changed and deleted without any limitations. There is only one exception: administrators can be deleted only if there will be one administrator left. That means, in each system at least one user exists: an administrator.

Administrators must handle user rights with care. Creating a user with extensive user rights will make him/ her possible to cause serious damage.

User Groups

The list shows all defined user groups.

Press the button **Add...** to open the dialog Create User Group.

Press the button **Delete** to delete the selected user group.

Press the button **Rename...** to open the dialog Rename User Group.

Users in selected Group

The list shows all defined users of the selected group.

Press the button **Add...** to open the dialog Add User.

Press the button **Delete** to delete the selected user.

Press the button **Edit...** to open the dialog Edit User.

Group rights

Rights are assigned to the User Groups, not to a single user. The available rights are listed in chapter User Rights.

10.3.1 User Group

Create a user group

The field **Group Name** defines the name of the new group. It must be unique.

Press the button **Add** to close the dialog and create the new group in the group list.

Press the button **Cancel** to close the dialog. No user group is created.

Rename a user group

The field **Group Name** defines the new group name. It must be unique.

Press the button **Save** to close the dialog and rename the selected group in the group list. The changes are not stored in the user database.

Press the button **Cancel** to closes the dialog. The user group is not renamed.

10.3.2 User

Add User

The field **User Name** defines the new user name. It must be unique.

The field **Password** defines the password. It can be empty.

You have to confirm the new password in the field **Password confirmation**.

Press the button **Add** to close the dialog and create the new user entry in the user list of the selected group.

Press the button **Cancel** to close the dialog. No user will be created.

Edit User

Enter the new user name into the field **User Name**. It must be unique.

The field Password defines the password. It can be empty.

You have to confirm the new password in the field **Password confirmation**.

Press the button **Save** to close the dialog and apply the changes to the selected user entry.

Press the button **Cancel** to close the dialog without changing the user settings.

10.3.3 User Rights

The following user rights are available:

Right	Description
Administrate	Allows to create, to change and to delete system user accounts and user groups
Application Settings	Allows changing the general application settings

Right	Description
Assign Operator Rights for Module Recipes	This right might not be available for all ToolCommander based machine control systems: Allows to change the settings for module recipes which define which properties of a module recipe can be changed by an operator (who does not have the right to change module recipes)
Automatic Mode Operation	Allows to control the production functions of the system
Change Parameters	Allows changing device parameters of components and also import and export them
Delete Production Log	Allows to delete production logs
Edit Flow Recipes	Allows to edit flow recipes
View Flow Recipes	Allows to view flow recipes
Edit Module Recipes	Allows to create and to modify module recipes. If the right **Edit Module Recipes (Expert Mode)** is also supported, a user needs both rights, **Edit Module Recipes** and **Edit Module Recipes (Expert Mode)**, for creating and editing module recipes.
Edit Module Recipes (Expert Mode)	This right might not be available for all ToolCommander based machine control systems: Allows to create and modify module recipes in case the user has the right **Edit Module Recipes,** too.
I/O Diagnosis	Allows to use the I/O diagnosis functions
Manual Operation	Grants access to manual operation dialogs of components
Manually control Sequencer	Allows the manual control the execution of module recipes

Right	Description
Quit Application	Allows to exit the program and to shut down the computer(s)
User Administration	Allows to administrate users/ user groups that have the same user rights assigned as the user who gets this right
VAC Diagnosis	Allows to use the VAC Diagnosis tool
View Parameters	On the components screen allows the user to display the parameters of components and to print them
View Production Log	Allows viewing production logs
Windows Access	Allows access to the operating system Windows TM

Table 61 - User Rights

11

Host Interface Integration

11 Host Interface Integration

ToolCommander is prepared for the communication to a host system. In case the host is connected and the control state is set to *Online Remote* different functions on the GUI will be disabled for the user, e.g. creation of jobs.

For a detailed description of the host interface a separate manual is provided if your machine has it.

11.1 Carrier Management

The operator can control the automatic transport from or to the Load Port by the Carrier Management. The integration of an Host Interface in the equipment control is required. Additionally, the equipment control has to support standardized Load Ports which are used to hold carriers.

Press the button **Carrier Management** in the online visualization of the appropriate Load Port to open the dialog of the Carrier Management. Depending on the current equipment status the buttons are enabled or disabled. A blue salience around a button represents an already active state. The following table gives an overview about the functions.

Button	Function
Proceed with Carrier	Acknowledge a previously read Carrier ID or Carrier SlotMap.
Cancel Carrier	Cancel an carrier standing on the Load Port. The carrier is released and can be removed either manually by the operator or by the automatic transport system (AMHS).
Carrier Recreate	Recreate an released carrier standing already on the Load Port on the part of software. Following, the carrier will be treated as new and unknown carrier.
Out of Service	Deactivate the transfers temporary from or to the Load Port. The property influences automatic transports as well as manual transports from or to the Load Port.
In Service	Reactivate the transfers from or to the Load Port. The property influences the automatic transports as well as manual transports from or to the Load Port.
Manual	Activate the manual material transfer from or to the Load Port by the operator only. Automatic transports by AMHS will be declined.
Automatic	Activate the automatic material transport from or to the Load Port by the automatic transport system (AMS), only. Manual transports from or to the Load Port will be declined.

Table 62 - Carrier Management functions

12

Auxiliary Functions

12 Auxiliary Functions

12.1 Language Selection

The ToolCommander supports online language switching. Click the button **Language** in the menu bar to open the Language Selection dialog.

By selecting the desired language and confirming the selection with *Save* the language of the application will be switched. The selection can also be changed by using the *Up* and *Down* buttons.

12.2 Log Book

The Log Book allows writing and reading notes.

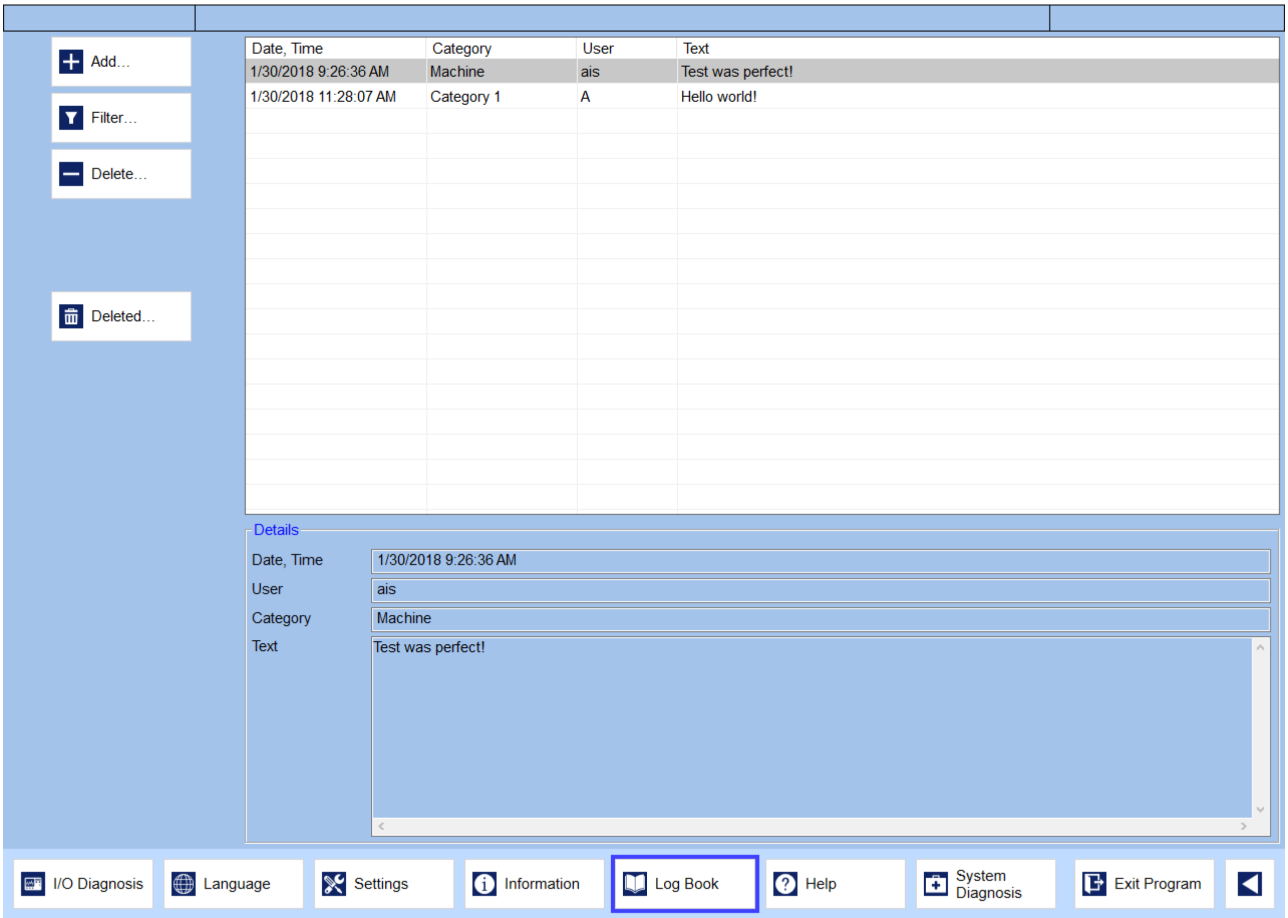


Figure 78 - Log Book

Log book entries are always assigned to a category.

Following functions are available:

Add new log book entry

To insert a log book entry, press **Add...**, thereafter select an already existing category using the button **Select...** or enter a new category name. Enter a text and accept the entry using the button **Save**.

Log book filter settings

To filter the displayed log book entries select the button **Filter...** The dialog Log Book Filter Settings is opened. Following filter criteria are available

- time span of creation
- author (user)
- category of the log entry

After selecting the desired filters accept the filter settings using the button *Apply*.

Delete log entries, restore deleted log entries

To delete a log book entry, select the corresponding line and press Delete.

Confirm the user request with *Yes* to delete the log entry. To restore deleted entries, select *Deleted…*, choose a table entry and press *Restore*.

12.3 VAC Diagnosis

Access authorization

Only users who have the right „VAC Diagnosis" can access this function.

The VAC Diagnosis which is opened after the button System Diagnosis was selected from the menu is a diagnostic tool for ToolCommander software developers only. It is not to be used by the application user since its usage might cause the application to crash or not behave in the supposed in manner.

Depending on application settings the button might be hidden. The button is disabled if the user currently logged on does not have the user right *VAC Diagnosis*.

12.4 Exit Program

Access authorization

Only users who have the right „Exit Application" can access this function.

To exit the application click the button Exit Program from the second menu. That will open the following window:

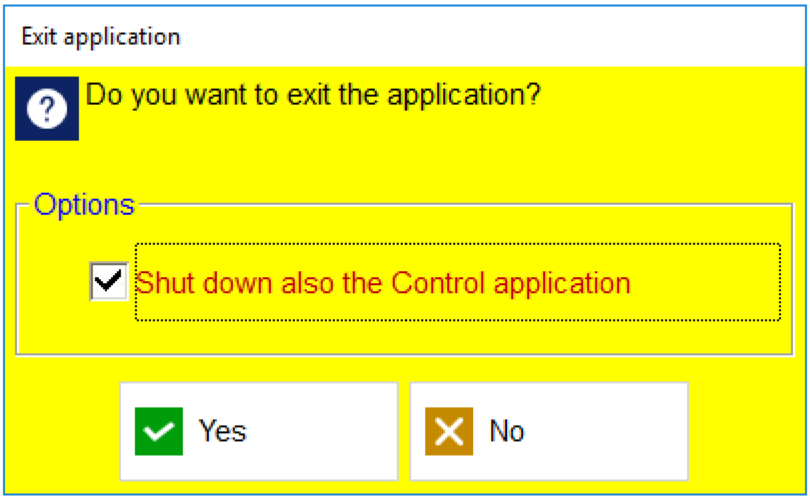


Figure 79 - Exit Application

Checking the option *Shut down Control application, too* (only visible if the application contains a separate control application and if this application is connected to the visualization application) the control system will also be exited after the user confirmed the shut-down by clicking Yes.

13

Application Settings

13 Application Settings

This dialog allows adjusting common settings. The information is stored in the file *VCTC.ini* which must not be write-protected if changes in this dialog are to be saved.

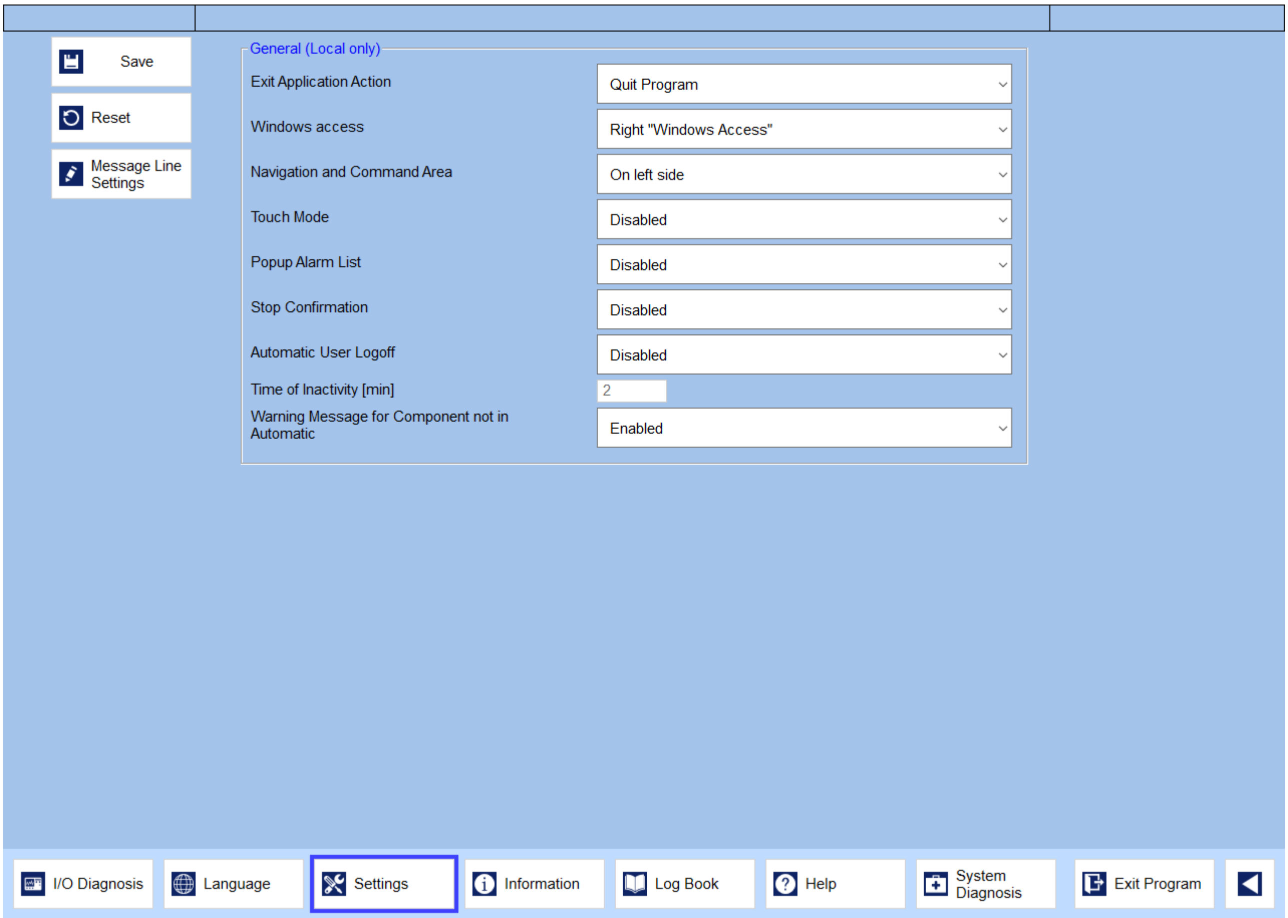


Figure 80 - Settings Dialog

General	
Exit application action	
	Following configurations are possible: • quit program • log off from Windows • shut down PC
Windows Access (enables Alt-Tab etc.)	
	Following configurations are possible:

General	
	• enable or disable always • only for supervisor • according to right Windows Access Note: To change this setting option the user needs also the user right 'Windows access.'
Application Menus	
	Following configurations are possible: • pane on left • pane on right side
Touch Mode (activates/deactivates touch screen keyboard)	
	Following configurations are possible: • enabled • disabled
Alarm List Popup (incoming messages are automatically shown in a popup window)	
	Following configurations are possible: • enabled • disabled
Stop Confirmation (technological stop requests must be confirmed by the user)	
	Following configurations are possible: • enabled • disabled
Automatic User Logoff (If activated the user automatically is logged off from the system if he does not do any action for a defined time.)	

General	
	Following configurations are possible: • enabled • disabled
Time of Inactivity [min]	Range [1..300]
	This value defines the time of inactivity before automatic user logoff.
Warning Message for Component not in Automatic	
	Defines whether or not the application shall display a warning message when the user tries to close a component dialog for a component that is not in Automatic (Production) operation mode. Following configurations are possible: • enabled • disabled
Commands	
Apply	This button saves the settings.
Reset	This button reloads the last saved settings.
Message Line Settings	This button opens the message line settings; see Message Line.

Table 63 - Application Settings

14

Control Application

14 Control Application

The control application of the ToolCommander handles all logical operations of the system to any of the components (e.g. Scheduler), contains all interfaces (e.g. I/O) and acts as a server to the GUI application(s) which only displays information it receives from control.

The control application at runtime presents this screen:

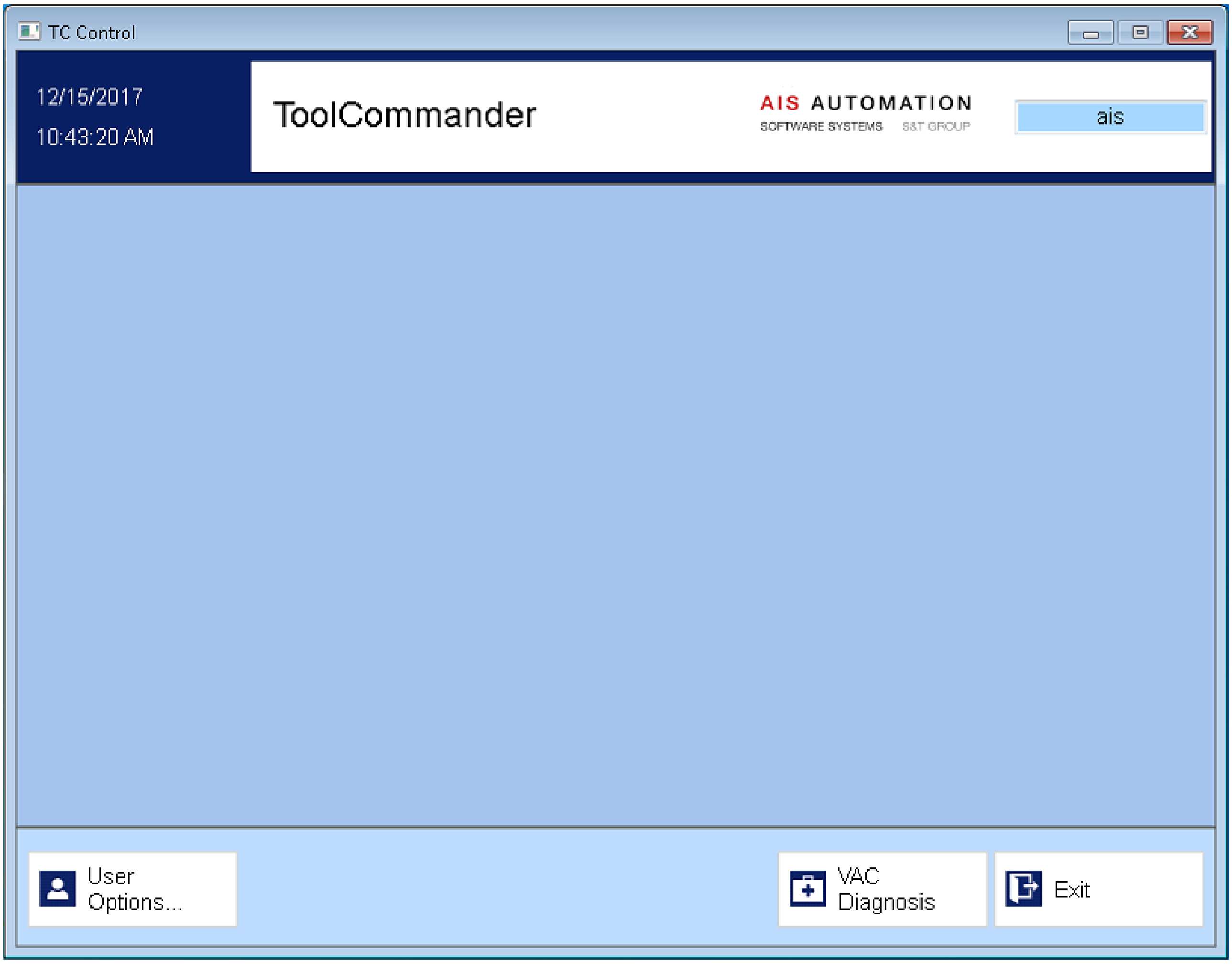


Figure 81 - Control Application

The menu on the bottom of the application window contains the following three buttons:

Commands	
User Options...	Opens the user options window as described in chapter User and Rights Management.
VAC Diagnosis	Opens the VAC Diagnosis window as described in chapter VAC Diagnosis. This button only is enabled if the user currently logged on to the control application has the right.

Commands	
Exit	Opens a window in which the user has to confirm to exit the control application. This button only is enabled if the user currently logged on to the control application has the right *Quit Application*.

Table 64 - Control Application Menu

Note

The control application can also be shut down from the GUI application when the user selects the according check-box in the confirmation window that is displayed when trying to shut down the GUI application (see chapter “Exit Program”).

15

Data Logging and Analysis

15 Data Logging and Analysis

ToolCommander logs different types of data.

15.1 Trace Data

The system will delete or move old log files according to parameter values set in machine component. The system keeps log files at least for 24 hours. Log files will be stored in a directory structure of [project directory]\Logging\[year]\[month]\ on the control PC.

Alarm Log	This file contains entries describing posted alarms.
Application trace log	ToolCommander stores debugging/ tracing information in svclog-Files which can be displayed using Microsoft Service Trace Viewer.

Table 65 - Data Logging

15.2 Actual Values

All actual values (analog and digital) which are displayed in one of the module trends will be automatically logged to the database.

16

Version history

16 Version history

Beginning with current version of the document the history contains statements to all modifications.

Version	Release	Author	Date	
1.06		Team Tool Automation	02-06-2018	Updated figures and removed not necessary figures.
1.05		Team Tool Automation	01-09-2018	Description regarding rights Edit Module Recipes and Edit Module Recipes (Expert Mode) improved respectively added.
1.04		Team Tool Automation	07-12-2017	Added note if material types are editable
1.03		Team Tool Automation	03-14-2017	Added right to view flow recipes
1.02		Team Tool Automation	02-01-2017	Added signal processing for digital I/O channels
1.01		Team Tool Automation	10-19-2016	Configure flow recipe steps using the VCTC.ini
1.00		Team Tool Automation	06-28-2016	Initial version

Version: Version of this document formatted <major_version>.<minor_version>

Release: Release date of the version of the document - empty if the version is not released.

Author: If there are several authors, the ones are named here that has inserted the modifications of a version into the document. If there is only one editor the field can remain free.

Date: Date of the modification.

Modification: Short description of the all modifications for this release.

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